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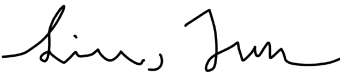
Building upon Bradley-Terry and
Plackett-Luce: some methods for modeling
paired comparison and rank order data

presented by Sanqian Zhang

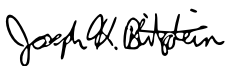
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Building upon Bradley-Terry and Plackett-Luce: some methods for modeling paired comparison and rank order data

A DISSERTATION PRESENTED
BY
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TO
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ABSTRACT

A common model used for analyzing paired comparison data is the Bradley-Terry model. An implicit assumption embedded within the Bradley-Terry framework is stochastic transitivity. This implies that if A is preferred to B, and B is preferred to C, then A must be preferred to C. Chapter 1 proposes two Bayesian hierarchical paired comparison models that extend the Bradley-Terry model by introducing a stability parameter for each player. This additional dimension on player characteristics permits the violation of stochastic transitivity among players. This can improve paired comparison inferences fit and out-of-sample prediction accuracy when there is evidence of violation of stochastic transitivity in the data. Chapter 2 extends the model developed in Chapter 1 by developing a rating system that characterizes athletes through strength and stability parameters. We find that the new rating system is capable of identifying athletes with high level of instability.

Chapter 3 evaluates a common practice of analyzing rank order data (e.g., from multi-competitor races) by converting them into paired comparison data and using Bradley-Terry models to make inferences and predictions. We establish this practice within existing statistical frameworks, assuming the rank order data can be analyzed via the conventionally used Plackett-Luce model, and prescribe the necessary adjustments to the covariance matrix of the Bradley-Terry estimators. We also describe the loss of efficiency in doing so. The second part of Chapter 3 draws on the formulation of rank order data as paired comparisons to propose a new model diagnostic tool for the Plackett-Luce model. In a simulation study, this new model diagnostic has good power in detecting departures from Plackett-Luce models.

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I DEDICATE THIS THESIS TO MY PARENTS AND MY GRANDPARENTS.

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Introduction

The field of quantitative analysis in sports has become substantially more established over the last two decades. One intrinsic source of data from sports event is the observed outcome of sports competitions. We can divide these sports into two broad categories: head-to-head sports and multi-competitor sports.

In head-to-head sports, exactly two players or teams directly face each other during the game. Examples of head-to-head sports include team sports such as baseball, soccer and basketball, individual

sports such as tennis and badminton, mind sports such as chess and e-sports such as League of Legends and Counter-Strike. Outcomes of these games can be reported as which player (team) won the game. The results of head-to-head games are examples of paired comparison data. In some sports such as tennis and volleyball, the outcome of the game is always binary. The rules of these sports dictates that each game must finish with an unequivocal winner. In other sports such as chess, games can finish with a winner or in a tie.

In multi-competitor sports, players (or teams) perform a specific task, and are compared against each other based on some performance measure. For example in sports such as swimming, running and skiing, athletes' performance are measured by the time taken to finish a particular distance. In sports such as diving, ice skating and gymnastics, athletes are judged by a common panel by how well they performed a routine. These competitions can be summarized by listing players' performance from best to worst. Such records are examples of rank order data.

A common model used for analyzing paired comparison data is the Bradley-Terry model (Bradley and Terry, 1952), while rank order data is often modeled using the Plackett-Luce model (Plackett, 1975; Luce, 1959). Both models represent individual athletes' average skills using a numerical value, and prescribe the distribution of outcomes of the game as a function of these athletes' average skills. Using these models, we can obtain estimates of athletes' relative strength and make predictions regarding unobserved or future outcomes.

The first two chapters of this dissertation focus on the analysis of paired comparison data. An implicit assumption embedded within the Bradley-Terry framework is stochastic transitivity. In the context of modeling athletes' abilities, stochastic transitivity implies that if player A is stronger than player B, and player B is stronger than player C, then player A must be stronger than player C. This can be a restrictive assumption. In Chapter 1, to accommodate the violation of stochastic transitivity, we propose two variants of a Bayesian hierarchical paired comparison model. The hierarchical models we propose extends the Bradley-Terry model by introducing a stability parameter for each player. This

additional dimension on player characteristics permits the violation of stochastic transitivity among players. Our models embed the Bradley-Terry model as a limiting case when the stability parameters are exactly 0. Using simulation and empirical examples, we demonstrate that our hierarchical models can outperform the Bradley-Terry model when the data deviates from Bradley-Terry assumptions.

Chapter 2 extends the concept of characterizing athletes using average strength and stability. We propose a dynamic paired comparison model with time-varying strength and stability parameters. Paralleling the development of the Glicko (Glickman, 1999) and Glicko-2 (Glickman, 2001) rating systems, we adapt the dynamic paired comparison model into a new rating system. Given new observations from a fixed time period, the rating system updates the estimates of players' strength and stability. We demonstrate this rating system using data from professional Go.

The first two chapters introduce methodologies for a new modeling framework for outcomes from head-to-head sports. They accommodate the natural phenomenon of stochastic intransitivity. The new framework quantifies each athletes' ability through strength and stability. This introduces additional insights on athletes that is not present in Bradley-Terry models and can improve the predictive capabilities of the models. The methods introduced in the first two chapters can also be extended to analyses of paired comparison data in contexts other than sports. In Chapter 1, we demonstrate the application of our model to a citation data set.

The third chapter of this dissertation examines modeling of rank order data. We are aware of instances where practitioners in the field of sports analytics choose to analyze rank order data by converting the data into sets of paired comparisons. Instead of using the Plackett-Luce model, the Bradley-Terry model is applied instead to the formed paired comparisons. We would like to help these practitioners appreciate what they are doing from a statistical perspective. The first part of Chapter 3 establishes the validity of this approach. We also prescribe the necessary adjustments to the covariance matrix of the Bradley-Terry estimators, and describe the loss of efficiency in doing so. The second part of Chapter 3 draws on the formulation of rank order data as paired comparisons to propose a

new model diagnostic tool for the Plackett-Luce model. We demonstrate the behavior of this new model diagnostic through simulation studies. Using a simulation study, we show that this model diagnostic can detect departures from Plackett-Luce model and has good power in doing so. In short, the third chapter affirms an observed practice among sports analytics professionals in dealing with the analysis of rank order data. The chapter also introduces a new tool that can be used for checking the appropriateness of the Plackett-Luce model when modeling outcomes from multi-competitor sports.

1

Hierarchical Models for Modeling Violation of Stochastic Transitivity in Paired Comparison Data

1.1 INTRODUCTION

Paired comparison data arise out of observing the direct competition between two players, objects, or options. Common examples can be observed in head-to-head sports competitions, with each game involving two players facing off against each other. Examples include common sports like soccer, baseball, basketball and chess. Many other problems can also be analyzed under this framework. For example, analysis of observed data on consumer preferences between two brands can be seen as a competition between these two brands.

A classic and popular family of models for such problems is the family of linear paired comparison models (David, 1988). This family of models, and extensions based on these models, are extensively used in analyzing head-to-head sports competitions. It has also been extended to other context such as analyzing citation networks (Stigler, 1994; Varin et al., 2016), understanding decision-making in starting a doctoral program (Dittrich et al., 2004) and modeling organisms’ preferences in Ecology (Müller et al., 2015; De’ath and Moran, 1998; Larsen et al., 2008). The framework of paired comparisons and the application of linear paired comparison models can be found in a wide range of academic disciplines.

An implicit assumption of this family of linear paired comparison models is stochastic transitivity. We describe stochastic transitivity in the context paired competition among sports players. For any arbitrary player triplet A, B and C, if player A is stronger than player B, and player B is stronger than player C, stochastic transitivity implies that player A must be stronger than player C. We argue that this might be a restrictive assumption, and would like to contribute a way of allowing for violations of stochastic transitivity while modeling paired comparisons.

Linear paired comparison models characterize each player by a strength parameter. The main idea behind our work is to introduce an additional parameter at the player level which we refer to as the stability parameter. With this additional parameter, players can differ in strength and/or stability.

This extension allows violations of stochastic transitivity to exist. A simple analogy can be compared to the problem of stochastic dominance between random variables that are Normally distributed. If we fix the variance parameter, then the random variable with a higher mean parameter must stochastically dominate the other. However, when we allow the variance parameter to change, the stochastic dominance relationship between variables becomes inconclusive.

The remainder of this chapter is organized as follows. Section 1.2 reviews linear paired comparison models and current work on accommodating intransitivity. To account for intransitivity, we introduce two variants of a Bayesian hierarchical paired comparison model in Sections 1.3 and 1.4. For each of our models, we discuss interpretations of model parameters and describe computational details for drawing posterior samples. We demonstrate the new models in Section 1.5 and 1.6 using simulations and empirical data, respectively. In Section 1.5, the new models are compared against the Bradley-Terry model by evaluating the models' ability to recover true parameters based on simulated data. In Section 1.6, the new models are compared against the Bradley-Terry model by conducting out-of-sample predictions. The chapter concludes in Section 1.7.

1.2 STOCHASTIC TRANSITIVITY AND PAIRED COMPARISON MODELS

For simplicity, we assume that a comparison between two players can only result in one of two outcomes, a win or a loss. With a total of N players, given n_{ij} games played between player i and player j , we observe Y_{ij} cases where player i wins, for $i = 1, \dots, N - 1$; $j = i + 1, \dots, N$. The choice of $i < j$ is purely arbitrary for accounting purposes to avoid double counting player pairs. We define $Y_{ij} = n_{ij} - Y_{ji}$ for $i > j$ where needed.

We assume that every outcome resulting from games played between player i and player j are independently and identically distributed, and games across different player pairs are independent.

We can write this as

$$Y_{ij} \sim \text{Binom}(n_{ij}, p_{ij})$$

independently.

1.2.1 LINEAR PAIRED COMPARISON MODELS

The family of linear paired comparison models assumes that p_{ij} , the probability of player i defeating player j , can be described as

$$p_{ij} = F(\theta_i - \theta_j)$$

where F is the cumulative distribution function (CDF) of a continuous random variable with a density that is symmetric around 0. The zero-symmetric condition ensures that $p_{ij} = 1 - p_{ji}$. Setting F as the CDF of the standard Logistic distribution gives the classic Bradley-Terry model for paired comparison (Bradley and Terry, 1952), which can be expressed as

$$\text{logit}(p_{ij}) = \theta_i - \theta_j.$$

Another conventional choice, the CDF of the standard Normal distribution, gives the Thurstone-Mosteller model (Thurstone, 1927; Mosteller, 1951). We focus on the Bradley-Terry model in this chapter.

Frequentist inference on the Bradley-Terry model is typically done using maximum likelihood estimation. The problem of solving for the maximum likelihood estimator(MLE) of the Bradley-Terry model has been discussed by Hunter (2004). The Bradley-Terry model is also equivalent to a logistic regression with a special design matrix. Hence, the Bradley-Terry model can also be analyzed using standard software packages that perform logistic regression.

Consider paired comparisons among three distinct individuals i, j and k . Suppose that we know $p_{ij} \geq 0.5$ and $p_{jk} \geq 0.5$. Under weak stochastic transitivity, this implies $p_{ik} \geq 0.5$. Under strong stochastic transitivity, we have $p_{ik} \geq \max(p_{ij}, p_{jk})$ (Luce and Suppes, 1965). In this chapter we will focus on weak stochastic transitivity. Hereafter, stochastic intransitivity will refer to the existence of player triplets such that $p_{ij} \geq 0.5$, $p_{jk} \geq 0.5$ and $p_{ki} > 0.5$, that is, a violation of weak stochastic transitivity. Under the family of linear paired comparison models, we can see that

$$F^{-1}(p_{ik}) = F^{-1}(p_{ij}) + F^{-1}(p_{jk}).$$

When $p_{ij} \geq 0.5$ and $p_{jk} \geq 0.5$, the zero symmetry condition on cdf F guarantees that $F^{-1}(p_{ij}) \geq 0$ and $F^{-1}(p_{jk}) \geq 0$. Hence,

$$F^{-1}(p_{ik}) \geq \max(F^{-1}(p_{ij}), F^{-1}(p_{jk})) \geq 0.$$

Since F as a CDF is monotonically increasing, we see that the family of linear paired comparison models implicitly assumes strong and weak stochastic transitivity in all paired comparisons.

We argue that this is a strong assumption in thinking about paired comparison. For example, consider a student deciding to accept admission to colleges H, Y and P. Perhaps the student has a preference for college H when compared against Y because of their location, a preference for Y over P because of campus design but also a preference for P over H because of her parents' personal opinion on colleges P and H (this opinion may be non-existent on comparisons against Y). Here we arrived at a set of intransitive pairwise preferences such that $H \succ Y$, $Y \succ P$ and $P \succ H$, where " $\succ$ " means "is preferred to", because the pairwise decisions are not made based on the same considerations.

An empirical example of violation of stochastic transitivity is exhibited in data taken from an article on citations in Statistics journals (Varin et al., 2016). Stigler (1994) proposed to model the

log-odds of journal i being cited by j as the difference between a measure of each journal’s academic strength in influencing other journals. This set-up for the analysis of citations across journals can be considered a paired comparison problem. Conditioning on number of citations exchanged between two journals, a citation of journal i by journal j can be viewed as a win for journal i in its ability to academically influence journal j . Table 1.1 gives the empirical probability of journal i being cited by journal j , for four selected Statistics journals. From the table, we can see that $\hat{p}_{\text{CSDA},\text{JMA}} \geq 0.5$ and $\hat{p}_{\text{JMA},\text{StMed}} \geq 0.5$, but $\hat{p}_{\text{CSDA},\text{StMed}}$ is only 0.34. This illustrates violation of stochastic transitivity being observed.

	CSDA	JMA	JSPI	StMed
CSDA	-	0.54	0.36	0.34
JMA	0.46	-	0.47	0.58
JSPI	0.64	0.53	-	0.45
StMed	0.66	0.42	0.55	-

Table 1.1: Empirical probability of journals being cited by other journals from [Varin et al. \(2016\)](#).

Motivated by conceptual reasoning and empirical examples, we would like to examine how to extend linear paired comparison models, in particular, the Bradley-Terry model, to allow for violations of stochastic transitivity.

1.2.2 EXISTING WORK ON MODELLING INTRANSITIVITY

The Bradley-Terry model can be extended by assuming a set of $\binom{N}{2}$ parameters θ_{ij} and modeling the probability p_{ij} as

$$\text{logit}(p_{ij}) = \theta_i - \theta_j + \theta_{ij}. \quad (1.1)$$

Without any structure on θ_{ij} , the MLE of the p_{ij} for this model are the sample proportions $\frac{y_{ij}}{n_{ij}}$. To avoid overfitting, we can consider penalizing the log-likelihood using LASSO or ridge penalties. However, the θ_{ij} cannot be estimated if we do not have any observed data between two players. In other words, this model is unable to make inferences about player pairs who have not competed before without invoking additional assumptions.

Despite stochastic intransitivity being a such a natural phenomenon, the literature on modeling intransitivity is scarce. We recognize that one of the fundamental difficulties in developing a model is the prevalence of transitivity in mathematical operations. Early work by economist [May \(1954\)](#) discussed transitivity in the context of preferences and utility functions. He pointed out that a utility function built on a single dimension input must be transitive. Along these lines, several authors have attempted to allow for stochastic transitivity by extending scalar representations of player ability to multiple-parameter representations. The main tools in previous work has relied on vector operations such as measuring Euclidean distances and computing inner products.

[Causeur and Husson \(2005\)](#) proposed the model

$$\text{logit}(p_{ij}) = \sigma_{ij} \sqrt{(\lambda_{i,1} - \lambda_{j,1})^2 + (\lambda_{i,2} - \lambda_{j,2})^2},$$

where $\sigma_{ij} = -\sigma_{ji}$ is a sign parameter which takes value ± 1 . The vectors $\boldsymbol{\lambda}_i = (\lambda_{i,1}, \lambda_{i,2})'$, $i = 1, \dots, N$ must also satisfy identifiability constraints

$$\sum_{i=1}^N \lambda_{i,1} = \sum_{i=1}^N \lambda_{i,2} = 0 \text{ and } \sum_{i=1}^2 \lambda_{i,1} \lambda_{i,2} = 0.$$

To summarize their approach, the Euclidean distance of the ability vectors $\boldsymbol{\lambda}_i$ and $\boldsymbol{\lambda}_j$ determines the gap between players i and j , while a separate sign parameter is used to indicate which player is stronger. This model does not provide structure to infer the sign parameter for players that have not played

against each other.

Chen and Joachims (2016) also used vector representation of player ability, where each player is represented by two d -dimensional vectors $\mathbf{i}_{\text{blade}}$ and $\mathbf{i}_{\text{chest}}$. Then, the authors propose several ways to represent

$$\text{logit}(p_{ij}) = M(i, j)$$

such that violations of stochastic transitivity are allowed. Possible choices of the function $M(i, j)$ include

- $M(i, j) = \|\mathbf{j}_{\text{blade}} - \mathbf{i}_{\text{chest}}\|_2^2 - \|\mathbf{i}_{\text{blade}} - \mathbf{j}_{\text{chest}}\|_2^2$
- $M(i, j) = \mathbf{i}_{\text{blade}} \cdot \mathbf{j}_{\text{chest}} - \mathbf{j}_{\text{blade}} \cdot \mathbf{i}_{\text{chest}}$

The authors argued that these models with $\mathbf{i}_{\text{blade}}$ and $\mathbf{i}_{\text{chest}}$ can fully represent any structure in pairwise comparisons when the vectors are of sufficient length. The proof is based on vectors of length N and $N+1$ for the inner product and Euclidean distance models respectively. A drawback of their approach is that their models were proposed as a pure prediction problem, and it is unclear how to extend their approach to account for statistical uncertainty and obtain parameter inferences.

1.2.3 MOVING AWAY FROM VECTOR REPRESENTATION

Rather than adopt an approach to modeling intransitivity which relies on operating on vector parameters, our proposed approach starts with a single scalar parameter θ_i to describe player i 's average ability, and then introduces a second parameter, δ_i , at the player level, so that player i is associated with a set of parameters (θ_i, δ_i) , in a way that will accommodate stochastic intransitivity. The role of δ_i will be distinct from θ_i . We give a brief overview of the broad idea below.

We choose to focus on representing θ_{ij} in Equation (1.1) in a structured approach. We separate

θ_{ij} as

$$\theta_{ij} = \Delta_{ij} - \Delta_{ji}$$

so that

$$\text{logit}(p_{ij}) = \theta_i - \theta_j + \Delta_{ij} - \Delta_{ji}.$$

We represent this deviation from a basic Bradley-Terry model into contributions by player i and player j respectively. Naively, this extension creates a model with unidentifiable parameters. However, we will be imposing a hierarchical structure on the parameters. We let $\Delta_{ij}, j = 1, \dots, N; j \neq i$, to share a common distribution G_i , where G_i is a parametric distribution characterized by the player specific parameter δ_i . The difference in distributions of $\Delta_{i\cdot}$ and $\Delta_{\cdot k}$ gives a second dimension in contrasting players i and k .

Note that when we move away from the Bradley-Terry model, we are giving up both stochastic transitivity and the fundamental assumptions of linear paired comparison models. Hence, our models can also be viewed as an alternative general approach to model paired comparison data which encompasses the Bradley-Terry model as a special case and allows violation of stochastic transitivity. In the following sections, we introduce two variants of this hierarchical model structure.

1.3 MODEL VERSION I

The first model we propose assumes the t distribution with degrees of freedom ν as our choice of G_i , the distribution on Δ_{ij} . This Bayesian hierarchical model can be seen as a continuous bridge between the Bradley-Terry model and a saturated model. The complete model specification with prior distribution is displayed in Table 1.2.

Data model

$$\begin{aligned} Y_{ij} &\sim \text{Binom}(n_{ij}, p_{ij}) \\ \text{logit}(p_{ij}) &= \theta_i - \theta_j + \Delta_{ij} - \Delta_{ji} \\ &\quad i, j = 1, \dots, N; j > i. \end{aligned}$$

Player level parameters

$$\boldsymbol{\theta} \sim \mathcal{N}(0, \lambda^2 \mathbf{I}) \text{ s.t. } \sum_{i=1}^N \theta_i = 0$$

$$\sigma_i = k_i \lambda$$

$$k_i \sim \text{Exp}(k_0)$$

$$i = 1, \dots, N.$$

Player-pair parameters

$$\frac{\Delta_{ij}}{\sigma_i} \sim t_\nu \quad i, j = 1, \dots, N; j > i.$$

Prior on**global parameters**

$$k_0 \sim \Gamma(k_\alpha, k_\beta)$$

$$\lambda^2 \sim \text{Inv-}\Gamma(\lambda_\alpha, \lambda_\beta)$$

Table 1.2: Complete specification of Model 1.

We define $\Delta_{ii} = 0$ for convenience. We introduced player-level parameters k_i which can be interpreted as a measure of stability in performance of player i . We will demonstrate in Section 1.3.1.2 that the parameter k_i has the important role of controlling the prevalence of stochastic intransitivity in the model.

Similar to the Bradley-Terry model, we use θ_i to represent player i 's latent ability. Since the model likelihood is invariant to additive shifts in $\boldsymbol{\theta}$, we need to add a constraint. Instead of working with an arbitrary reference player by setting $\theta_{i_0} = 0$ for some arbitrary player i_0 , we choose to work with the identifying constraint $\sum_{i=1}^N \theta_i = 0$. This gives θ_i the interpretation of difference in player i 's average strength from the average player in the population. The constraint can be incorporated into the prior

distribution by re-writing it as

$$\theta_{1:(N-1)} \sim \mathcal{N} \left(0, \lambda^2 \left[\mathbf{I}_{N-1} - \frac{1}{N} \mathbf{1}_{N-1} \mathbf{1}_{N-1}^\top \right] \right),$$

and set $\theta_N = -\sum_{i=1}^N \theta_i$. Here, $\theta_{1:(N-1)}$ refers to the first $N - 1$ components of $\boldsymbol{\theta}$, $\mathbf{I}_{N-1}$ is the identity matrix of dimension $N - 1$ and $\mathbf{1}_{N-1}$ is the $(N - 1)$ -vector consisting of all 1s. The parameter λ controls the spread in player strength within the population. We will discuss in detail the influence of λ on the model in Section 1.3.1.1.

The log odds of player i defeating player j is characterized by the difference between θ_i and θ_j and the difference between Δ_{ij} and Δ_{ji} . The parameter Δ_{ij} can be interpreted as the average deviation of player i from player i 's average when facing opponent player j . Magnitudes of $\Delta_{i1}, \dots, \Delta_{iN}$ for player i are controlled by the variance σ_i^2 . A player with small σ_i has consistent performances when facing different opponents. A player with large σ_i can perform at vastly different strengths when facing different opponents. The standard deviation σ_i can be decomposed as product of λ and the stability parameter k_i . The distribution of the Δ_{ij} scaled by σ_i is assumed to be t -distributed with degrees of freedom ν . When ν is high, the distribution is approximately standard Normal. When ν is small, the distribution of $\frac{\Delta_{ij}}{k_i \lambda}$ has heavy tails. Under such model specifications, there is more room for Δ_{ij} to vary, and may risk over fitting to empirical probabilities. By default, we choose to work with the standard Normal distribution.

The Thurstone-Mosteller model can be described through a latent variable specification, with

$$U_i = \theta_i + \epsilon_i$$

where $\epsilon_i \sim \mathcal{N}(0, \tilde{\sigma}_i^2)$ for each player i , and $p_{ij} = \mathbb{P}(U_i > U_j)$. The standard Thurstone-Mosteller model assumes $\tilde{\sigma}_i^2 = 1, i = 1, \dots, N$. Mosteller (1951) briefly discussed using a different variance

$\tilde{\sigma}_i^2$ for each player. If we change the link function on p_{ij} in Table 1.2 to the inverse CDF of a standard Normal distribution, Model 1 will coincide with the Thurstone-Mosteller model with player-specific variances when ν is large. However, Mosteller did not discuss this in the context of violation of stochastic transitivity. Our model specification also comes with a purposeful decomposition of the variance σ_i^2 .

If we examine parameters k_i , there are two limiting cases. When $k_i = 0$ for all players, the model reduces to a Bradley-Terry model. If $k_i \rightarrow \infty$ for all players, the model approaches a saturated model with nonidentifiability among the Δ_{ij} and Δ_{ji} .

1.3.1 SETTING HYPERPARAMETERS

Two sets of hyperparameters, (k_α, k_β) and $(\lambda_\alpha, \lambda_\beta)$, need to be determined. To understand how to set the hyperparameter values, we need to understand the role of λ and k_i in the model.

1.3.1.1 RELATIONSHIP BETWEEN λ AND THE DISTRIBUTION OF p_{ij}

Under Model 1 displayed in Table 1.2, λ describes the variability in player strength θ_i . This is reflected in the distribution of p_{ij} for player i . To investigate the effect of λ on distribution of p_{ij} , we fix N , λ and k_0 , and simulate from the prior predictive distribution of Model 1. Figure 1.1 gives the distribution of p_{ij} for $N = 15$ and $k_0 = 0.5$ and 1. Over four different values of λ , we can see that as λ increases, we observe more extreme values in pairwise probabilities. Hence, we can use the understanding of empirical distribution of pairwise probabilities to help us describe likely values of λ .

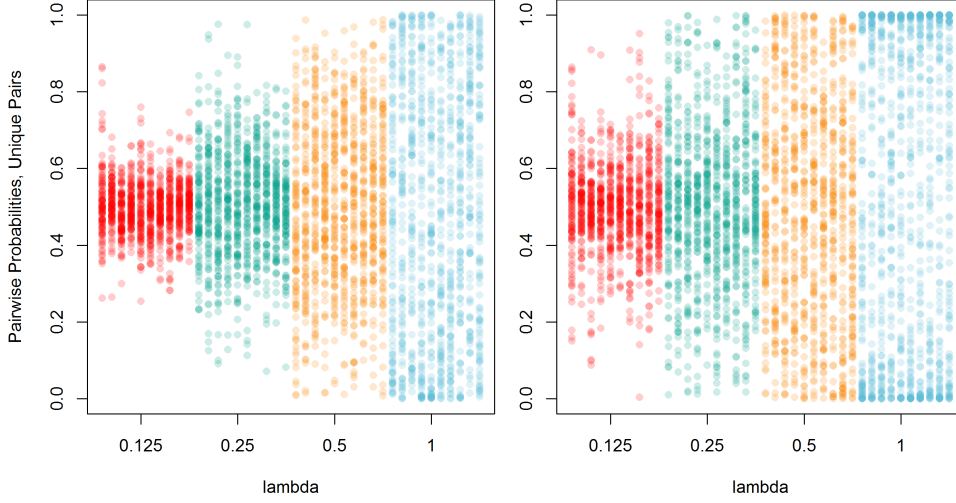


Figure 1.1: Prior predictive draws of $p_{ij}, j > i$ from Model 1, given different λ values. Left panel used $k_0 = 1$ and right panel used $k_0 = 0.5$. The plot gives result of 10 independent draws at each value of λ .

1.3.1.2 PARAMETER k_i AND STOCHASTIC INTRANSITIVITY

Consider the following two scenarios. In situation A, we have players i and j with average strength ϑ_i and ϑ_j , where $\vartheta_i - \vartheta_j > 0$ is large. In situation B, we have players i and j with average strength $\tilde{\vartheta}_i$ and $\tilde{\vartheta}_j$, where $\tilde{\vartheta}_i - \tilde{\vartheta}_j > 0$ is small. Measured by their average strength, player i is a stronger player under both situations. Consider the paired comparison between player i and player j under Model 1 with additional parameters Δ_{ij} and Δ_{ji} . Under situation A, the absolute value of the difference in Δ_{ij} and Δ_{ji} needs to be very high to make player j stronger than player i when comparing against each other. Under situation B, even small absolute differences in Δ_{ij} and Δ_{ji} can upset the relative strength of player i and player j . In order to disturb the ordinal relationship between players set by $\theta_1, \dots, \theta_N$ so that stochastic intransitivity can be induced, the magnitudes of the Δ_{ij} entries needs to be comparable to the sizes of gap between the θ_i . This is the motivation behind decomposing σ_i as the product of λ , a measure of variation in players' average ability, and k_i , a scaling parameter that marks the relative deviation in Δ compared to θ .

We again examine the prior predictive distribution of p_{ij} under Model 1 for different parameter values of k_0 . We are interested in understanding the effect of k_0 on inducing stochastic intransitivity. To quantify intransitivity, we measure the proportion of player triplets that violates stochastic transitivity. That is, the existence of distinct triplets (i, j, k) such that we have $p_{ij} > 0.5, p_{jk} > 0.5$ and $p_{ki} > 0.5$.

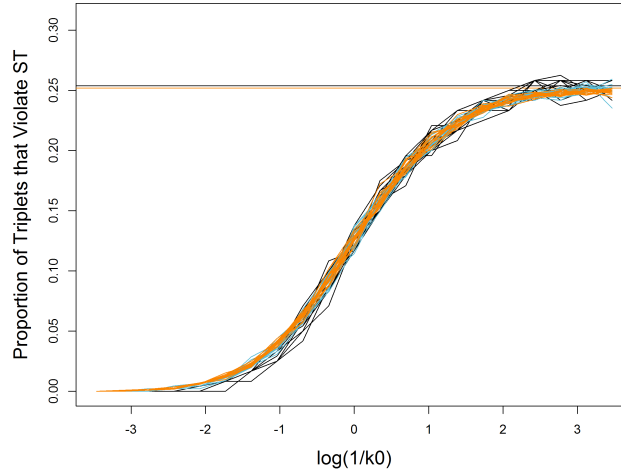


Figure 1.2: Median proportion of player triplets among prior predictive draws of p_{ij} from Model 1 that violate stochastic transitivity against k_0 . Each line reflects result based on 100 draws of p_{ij} for each k_0 value, given a λ parameter and N . The color reflects number of players N used in simulation, $N = 10, 15, 20$. Reference line gives the median proportion of triplets that violate stochastic transitivity based on 100 randomly generated probability matrix, where pairwise probabilities for unique pairs are sampled from $\text{Uniform}(0, 1)$ distribution.

We designed a sequence of 20 different k_0 , with $\log_2 k_0$ equally spaced between -5 and 5, and a sequence of 17 different λ , with $\log_2 \lambda$ equally spaced between -5 and 3, for 10, 15 and 20 players. Under each combination of parameters, we draw 100 sets of p_{ij} from its prior predictive distribution under Model 1. The median proportion of triplets that violates stochastic transitivity is reported in Figure 1.2. We observe that this proportion does not depend in any meaningful way on λ and N . The proportion of triplets that violate stochastic transitivity is controlled by parameter k_0 in the same way across all values of λ and N . This gives us a framework to interpret the prior specification of k_0 as

a specification on the extent of violation of stochastic transitivity among player triplets in the data a priori.

1.3.2 POSTERIOR INFERENCE FOR MODEL I

Inference from the model can be determined from draws simulated from the posterior distribution

$$f(\boldsymbol{\theta}, \boldsymbol{\Delta}, \mathbf{k}, \lambda^2, k_0 | \mathbf{y}) \propto f(\mathbf{y} | \boldsymbol{\theta}, \boldsymbol{\Delta}) f(\boldsymbol{\theta} | \lambda^2) f(\boldsymbol{\Delta} | \mathbf{k}, \lambda^2) f(\mathbf{k} | k_0) f(\lambda^2) f(k_0).$$

We choose to make inferences by drawing samples from the posterior distribution using Markov chain Monte Carlo (MCMC) methods. For our model, we can use the Gibbs sampler (Liu, 2008) to sample from the posterior by alternating draws among the conditional distributions:

$$\begin{aligned} f(\boldsymbol{\theta} | \boldsymbol{\Delta}, \mathbf{k}, \lambda^2, k_0, \mathbf{y}) &\propto f(\mathbf{y} | \boldsymbol{\theta}, \boldsymbol{\Delta}) f(\boldsymbol{\theta} | \lambda^2) \\ &\propto e^{-\frac{1}{2\lambda^2} \sum_{i=1}^N \theta_i^2} \prod_{j>i} e^{y_{ij}(\theta_i - \theta_j) - n_{ij} \log(1 + e^{\theta_i - \theta_j + \Delta_{ij} - \Delta_{ji}})} \end{aligned}$$

$$\begin{aligned} f(\boldsymbol{\Delta} | \boldsymbol{\theta}, \mathbf{k}, \lambda^2, k_0, \mathbf{y}) &\propto f(\mathbf{y} | \boldsymbol{\theta}, \boldsymbol{\Delta}) f(\boldsymbol{\Delta} | \mathbf{k}, \lambda^2) \\ &\propto \prod_{j>i} e^{y_{ij}(\Delta_{ij} - \Delta_{ji}) - n_{ij} \log(1 + e^{\theta_i - \theta_j + \Delta_{ij} - \Delta_{ji}})} \\ &\quad \times \prod_{j>i} e^{-\frac{1}{2\lambda^2 k_i^2} \Delta_{ij}^2 - \frac{1}{2\lambda^2 k_j^2} \Delta_{ji}^2} \end{aligned}$$

$$\begin{aligned} f(\mathbf{k} | \boldsymbol{\theta}, \boldsymbol{\Delta}, \lambda^2, k_0, \mathbf{y}) &\propto f(\boldsymbol{\Delta} | \mathbf{k}, \lambda^2) f(\mathbf{k} | k_0) \\ &\propto e^{-k_0 \sum_i k_i} \prod_{j>i} \frac{1}{k_i} e^{-\frac{1}{2\lambda^2 k_i^2} \Delta_{ij}^2} \frac{1}{k_j} e^{-\frac{1}{2\lambda^2 k_j^2} \Delta_{ji}^2} \end{aligned}$$

$$\begin{aligned} f(\lambda^2 | \boldsymbol{\theta}, \boldsymbol{\Delta}, \mathbf{k}, k_0, \mathbf{y}) &\propto f(\boldsymbol{\theta} | \lambda^2) f(\boldsymbol{\Delta} | \mathbf{k}, \lambda^2) f(\lambda^2) \\ &\propto (\lambda^2)^{-\lambda_\alpha - 1} e^{-\lambda_\beta / \lambda^2} \frac{N-1}{\lambda} e^{-\frac{1}{2\lambda^2} \sum_{i=1}^N \theta_i^2} \\ &\quad \times \prod_{j>i} \frac{1}{\lambda^2} e^{-\frac{1}{2\lambda^2 k_i^2} \Delta_{ij}^2 - \frac{1}{2\lambda^2 k_j^2} \Delta_{ji}^2} \end{aligned}$$

$$\propto (\lambda^2)^{-\lambda_\alpha - \frac{N-1}{2} - (\sum_{j>i} 1) - 1} e^{\left(-\frac{1}{\lambda^2} \left(\lambda_\beta + \sum_{i=1}^N \frac{\theta_i^2}{2} + \sum_{j>i} \left(\frac{\Delta_{ij}^2}{2k_i^2} + \frac{\Delta_{ji}^2}{2k_j^2} \right) \right) \right)}$$

$$f(k_0 | \boldsymbol{\theta}, \boldsymbol{\Delta}, \mathbf{k}, \lambda, \mathbf{y}) \propto f(\mathbf{k} | k_0) f(k_0)$$

$$\propto k_0^N e^{-k_0 \sum_i k_i} k_0^{k_\alpha - 1} e^{-k_\beta k_0}$$

$$\propto k_0^{N+k_\alpha-1} e^{-(k_\beta + \sum_i k_i) k_0}$$

Conditional distributions of λ and k_0 are inverse-Gamma and Gamma distributed, respectively. They can be sampled directly. Other conditional distributions can be handled using a random walk Metropolis-Hastings algorithm with a Normal distribution centered at 0 as the proposal distribution. The variance of the proposal distribution will be tuned at the sampling stage so that the acceptance rate of the algorithm is around 0.2 – 0.3 (Gelman et al., 1997). The following components of the MCMC sampler have particular issues worthy of comment.

SAMPLING OF $\boldsymbol{\theta}$: We modeled player abilities using a N -dimension vector $\boldsymbol{\theta}$ with a linear constraint $\sum_i^N \theta_i = 0$. To respect the constraint on $\boldsymbol{\theta}$, we can propose from a multivariate Normal distribution that satisfy the same constraint. Given any arbitrary N -dimensional covariance matrix $\tilde{\Sigma}$, and N -dimensional matrix A

$$\begin{pmatrix} 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ & & \ddots & & \\ 0 & 0 & \dots & 1 & 0 \\ 1 & 1 & \dots & 1 & 1 \end{pmatrix},$$

where the linear transformation $A\mathbf{x}$ gives the first $N - 1$ component of vector $\mathbf{x}$ and the sum of entries of $\mathbf{x}$ as the last component. We can set $\Sigma = A\tilde{\Sigma}A^\top$. Writting Σ as

$$\begin{pmatrix} \Sigma_{(1:N-1,1:N-1)} & \Sigma_{(N,1:N-1)} \\ \Sigma_{(1:N-1,N)} & \Sigma_{(N,N)} \end{pmatrix},$$

the proposal distribution

$$e_{1:(N-1)} \sim \mathcal{N}\left(0, h^2 \left[\Sigma_{(1:N-1,1:N-1)} - \Sigma_{(1:N-1,N)} \Sigma_{(N,N)}^{-1} \Sigma_{(N,1:N-1)} \right] \right)$$

and $e_N = -\sum_{i=1}^N e_i$ can be used for our random walk Metropolis-Hastings algorithm. This proposal is still symmetric and ensures that we are always proposing $\boldsymbol{\theta}' = \boldsymbol{\theta} + \mathbf{e}$ that respects the linear constraint we imposed. The parameter h is the step size which requires additional tuning. One possible choice for $\tilde{\Sigma}$ is $\lambda^2 \mathbf{I}$ which will correspond to the prior distribution of $\boldsymbol{\theta}$. Another possibility is to run the MCMC sampler first for an extended period of time, and then use empirical variance covariance matrix of posterior samples of $\boldsymbol{\theta}$.

SAMPLING OF $\mathbf{k}$: For the positive parameter $\mathbf{k}$, we can choose to apply the random walk Metropolis-Hastings on the log scale. This ensures that we are always proposing within the range of these parameters, and the space close to 0 are well explored.

1.4 MODEL VERSION 2

We developed Model 1 Using a t distribution on Δ_{ij} , which provides a continuous bridge between the Bradley-Terry model and a saturated model. In this section, we use a mixture distribution on Δ_{ij} as the basis for Model 2. With this mixture model assumption on the Δ_{ij} , the resulting posterior distribution also follows a mixture distribution, where one of the component corresponds to the

Bradley-Terry model. Model 2 embeds the Bradley-Terry model as a limiting case when $\omega_i = 0$ for all players. The full hierarchical model setup with prior specification is given in Table 1.3.

Data model	$Y_{ij} \sim \text{Binom}(n_{ij}, p_{ij})$ $\text{logit}(p_{ij}) = \theta_i - \theta_j + \Delta_{ij} - \Delta_{ji}$ $i, j = 1, \dots, N; j > i.$
Player level parameters	$\boldsymbol{\theta}_{1:n} \sim \mathcal{N}(0, \lambda^2 \mathbf{I}) \text{ s.t. } \sum_{i=1}^n \theta_i = 0$ $\omega_i \sim \text{Beta}(\alpha, \beta) \quad i = 1, \dots, N.$
Player-pair parameters	$\Delta_{ij} \sim (1 - \omega_i)\delta(0) + \omega_i\mathcal{N}(0, \sigma^2)$ $i, j = 1, \dots, N; j > i.$
Prior on global parameters	$\sigma = \kappa\lambda$ $\kappa^2 \sim \text{Inv-}\Gamma(\kappa_\alpha, \kappa_\beta)$ $\lambda^2 \sim \text{Inv-}\Gamma(\lambda_\alpha, \lambda_\beta)$ $\alpha \sim \text{Exp}(\alpha_0)$ $\beta \sim \text{Exp}(\beta_0)$

Table 1.3: Complete specification of Model 2.

Here, $\delta(0)$ is the Dirac delta mass at 0, and we set $\Delta_{ii} = 0$ for convenience. In this model, Δ_{ij} , the deviation of player i 's ability from his own average when facing player j , is either exactly 0, or comes from a Normal distribution with parameters shared by all players. We introduce player specific parameter ω_i , the probability of Δ_{ij} belonging to the zero-centered Normal distribution component. Players differ in θ_i , their average strength, and ω_i , a measure of player i 's performance stability. A

player with higher ω_i is more volatile, since this player is more likely to experience a non-zero Δ_{ij} when facing different opponents, that is, this player tends to perform at varied strengths when facing different opponents.

Interpretation of the parameter λ is similar to that of Model 1. When $\omega_i = 0$, the model is exactly the Bradley-Terry model. Hence, α and β , the hyperparameters of the Beta prior distribution on ω_i , will control the deviation from the Bradley-Terry model. It will also control the extent of violation of stochastic transitivity in this model. The parameter κ contributes to the variance Δ_{ij} for the non-zero entries. We can select a relatively flat prior distribution for κ .

1.4.1 POSTERIOR INFERENCE FOR MODEL 2

Again, we want to sample from posterior distribution to make inferences. To facilitate the sampling of the mixture component, we introduce a $N \times N$ matrix H where H_{ij} is a latent variable that keeps track of which mixture component Δ_{ij} is coming from, with

$$\Delta_{ij} \sim \begin{cases} \delta(0), & H_{ij} = 0 \\ \mathcal{N}(0, \sigma^2), & H_{ij} = 1. \end{cases}$$

The model specification from Table 1.3 gives us $\mathbb{P}(H_{ij} = 1) = \omega_i$. For convenience we can set $H_{ii} = 0$. Introducing the matrix of latent variable H into the model structure, the posterior distribution can be expressed as

$$\begin{aligned} & f(\boldsymbol{\theta}, \boldsymbol{\omega}, H, \boldsymbol{\Delta}, \kappa^2, \lambda^2, \alpha, \beta | \mathbf{y}) \\ & \propto f(\mathbf{y} | \boldsymbol{\theta}, \boldsymbol{\Delta}) f(\boldsymbol{\theta} | \lambda^2) f(\boldsymbol{\Delta} | H, \kappa^2, \lambda^2) f(H | \boldsymbol{\omega}) f(\boldsymbol{\omega} | \alpha, \beta) f(\lambda^2) f(\kappa^2) f(\alpha) f(\beta) \end{aligned}$$

As with inferences for Model 1, we can use a Gibbs sampler, alternating among the following set of conditional distributions:

$$f(\boldsymbol{\theta}|\boldsymbol{\omega}, H, \boldsymbol{\Delta}, \kappa^2, \lambda^2, \alpha, \beta, \mathbf{y}) \propto f(\mathbf{y}|\boldsymbol{\theta}, \boldsymbol{\Delta})f(\boldsymbol{\theta}|\lambda^2)$$

$$\propto e^{-\frac{1}{2\lambda^2} \sum_{i=1}^N \theta_i^2} \times \prod_{j>i} e^{y_{ij}(\theta_i - \theta_j) - n_{ij} \log(1 + e^{\theta_i - \theta_j + \Delta_{ij} - \Delta_{ji}})}$$

$$f(\boldsymbol{\omega}|\boldsymbol{\theta}, H, \boldsymbol{\Delta}, \kappa^2, \lambda^2, \alpha, \beta, \mathbf{y}) \propto f(H|\boldsymbol{\omega})f(\boldsymbol{\omega}|\alpha, \beta)$$

$$\propto \prod_i \omega_i^{\alpha-1} (1 - \omega_i)^{\beta-1} \prod_{j>i} \omega_i^{H_{ij}} (1 - \omega_i)^{1-H_{ij}} \omega_j^{H_{ji}} (1 - \omega_j)^{1-H_{ji}} \\ \propto \prod_i \omega_i^{\alpha + \sum_{j \neq i} H_{ij} - 1} (1 - \omega_i)^{\beta + \sum_{j \neq i} (1 - H_{ij}) - 1}$$

$$f(H, \boldsymbol{\Delta}|\boldsymbol{\theta}, \boldsymbol{\omega}, \kappa^2, \lambda^2, \alpha, \beta, \mathbf{y}) \propto f(\mathbf{y}|\boldsymbol{\theta}, \boldsymbol{\Delta})f(\boldsymbol{\Delta}|H, \kappa^2, \lambda^2)f(H|\boldsymbol{\omega})$$

$$f(\kappa^2|\boldsymbol{\theta}, \boldsymbol{\omega}, H, \boldsymbol{\Delta}, \lambda^2, \alpha, \beta, \mathbf{y}) \propto f(\boldsymbol{\Delta}|H, \kappa^2, \lambda^2)f(\kappa^2)$$

$$\propto (\kappa^2)^{-\kappa_\alpha - 1} e^{-\kappa_\beta / \kappa^2} \prod_{ij: H_{ij}=1} \frac{1}{\kappa} e^{-\frac{1}{\kappa^2} \frac{\Delta_{ij}^2}{2\lambda^2}} \\ \propto (\kappa^2)^{-\kappa_\alpha - \frac{\sum_{ij: H_{ij}=1} 1}{2} - 1} e^{-\frac{1}{\kappa^2} \left(\kappa_\beta + \sum_{ij: H_{ij}=1} \frac{\Delta_{ij}^2}{2\lambda^2} \right)}$$

$$f(\lambda^2|\boldsymbol{\theta}, \boldsymbol{\omega}, \boldsymbol{\Delta}, \kappa^2, \alpha, \beta, \mathbf{y}) \propto f(\boldsymbol{\theta}|\lambda^2)f(\boldsymbol{\Delta}|H, \kappa^2, \lambda^2)f(\lambda^2)$$

$$\propto (\lambda^2)^{-\lambda_\alpha - 1} e^{-\lambda_\beta / \lambda^2} \frac{N-1}{\lambda} e^{-\frac{1}{2\lambda^2} \sum_{i=1}^N \theta_i^2} \\ \times \prod_{ij: H_{ij}=1} \frac{1}{\lambda} e^{-\frac{1}{\lambda^2} \frac{\Delta_{ij}^2}{2\kappa^2}} \\ \propto (\lambda^2)^{-\lambda_\alpha - \frac{N-1 + \sum_{ij: H_{ij}=1} 1}{2} - 1} e^{-\frac{1}{\lambda^2} \left(\lambda_\beta + \sum_{i=1}^N \frac{\theta_i^2}{2} + \sum_{ij: H_{ij}=1} \frac{\Delta_{ij}^2}{2\kappa^2} \right)}$$

$$f(\alpha|\boldsymbol{\theta}, \boldsymbol{\omega}, H, \boldsymbol{\Delta}, \kappa^2, \lambda^2, \beta, \mathbf{y}) \propto f(\boldsymbol{\omega}|\alpha, \beta)f(\alpha)$$

$$\propto e^{-\alpha_0 \alpha} \prod_i \omega_i^{\alpha-1}$$

$$f(\beta|\boldsymbol{\theta}, \boldsymbol{\omega}, H, \boldsymbol{\Delta}, \kappa^2, \lambda^2, \alpha, \mathbf{y}) \propto f(\boldsymbol{\omega}|\alpha, \beta)f(\beta) \\ \propto e^{-\beta_0\beta} \prod_i (1 - \omega_i)^{\beta-1}$$

We will discuss the conditional distribution of H and $\boldsymbol{\Delta}$ in Section 1.4.1.1. Conditional on other parameters and the data, the parameters λ^2 and κ^2 follows an inverse-Gamma distribution, and each element of the parameter vector $\boldsymbol{\omega}$ follows a Beta distribution. These conditional distributions can be sampled directly. The parameters $\boldsymbol{\theta}$, α and β can be sampled using a random walk Metropolis-Hastings algorithm. The proposal distributions used will be a Normal distribution centered at 0. As in Model 1, the variance of the proposal distribution will be tuned based on acceptance rate of the algorithm. Other details on sampling of $\boldsymbol{\theta}$ and the positive parameters α and β are similar to what have been discussed for Model 1 in Section 1.3.2.

1.4.1.1 CONDITIONAL POSTERIOR DISTRIBUTION OF H AND $\boldsymbol{\Delta}$

As many as $N(N - 1)$ components of H and $\boldsymbol{\Delta}$ may need to be updated, depending on the player pairs that have competed. We can update each component using a Gibbs sampler. The conditional posterior distribution of the (i, j) th component of H and $\boldsymbol{\Delta}$, given all other components, is

$$f(H_{ij}, \Delta_{ij}|H_{-ij}, \boldsymbol{\Delta}_{-ij}, \boldsymbol{\omega}, \kappa^2, \lambda^2, \mathbf{y}) \\ \propto f(H_{ij}|\omega_i)f(\Delta_{ij}|H_{ij}, \kappa^2, \lambda^2)f(y_{ij}|\theta_i, \theta_j, \Delta_{ij}, \Delta_{ji}). \quad (1.2)$$

Let $c_0 = f(y_{ij}|s, \Delta_{ij} = 0)$ and $c_1 = \int f(y_{ij}|s, \Delta_{ij})\phi(\Delta_{ij}; 0, \kappa\lambda)d\Delta_{ij}$ where $s = \theta_i - \theta_j - \Delta_{ji}$, $f(y_{ij}|s, \Delta_{ij}) = \exp(y_{ij}(s + \Delta_{ij}) - n_{ij} \log(1 + e^{s+\Delta_{ij}}))$ and $\phi(\cdot; \mu, \sigma)$ is the density of the normal distribution mean μ and standard deviation σ . Integrating out Δ_{ij} from Equation 1.2,

we have

$$\begin{aligned}\mathbb{P}(H_{ij} = 1 | H_{-ij}, \mathbf{\Delta}_{-ij}, \boldsymbol{\omega}, \kappa^2, \lambda^2, \mathbf{y}) &\propto \omega_i c_1 \\ \mathbb{P}(H_{ij} = 0 | H_{-ij}, \mathbf{\Delta}_{-ij}, \boldsymbol{\omega}, \kappa^2, \lambda^2, \mathbf{y}) &\propto (1 - \omega_i) c_0.\end{aligned}$$

Conditioning on H_{ij} , $\Delta_{ij} = 0$ if $H_{ij} = 0$ and

$$f^*(\Delta_{ij}) = f(\Delta_{ij} | H_{ij} = 0, s, y_{ij}, \kappa^2, \lambda^2) \propto f(y_{ij} | s, \Delta_{ij}) \phi(\Delta_{ij}; 0, \kappa\lambda) \quad (1.3)$$

if $H_{ij} = 1$. Equation 1.3 can be simplified into

$$f^*(\Delta_{ij}) \propto \exp \left(y_{ij} \Delta_{ij} + n_{ij} \log(1 + e^{s + \Delta_{ij}}) - \frac{\Delta_{ij}^2}{2\kappa^2 \lambda^2} \right).$$

Using a Laplace approximation, we can approximate this distribution with

$$\mathcal{N} \left(\Delta_{ij,0}, \left[\frac{1}{\kappa^2 \lambda^2} + n p_0 (1 - p_0) \right]^{-1} \right),$$

where $\Delta_{ij,0} = \arg \max_{\Delta_{ij}} \log f^*(\Delta_{ij})$ and $p_0 = \frac{\exp(s + \Delta_{ij,0})}{1 + \exp(s + \Delta_{ij,0})}$.

Therefore, to sample from $f(H_{ij}, \Delta_{ij} | H_{-ij}, \mathbf{\Delta}_{-ij}, \boldsymbol{\omega}, \kappa^2, \lambda^2, \mathbf{y})$, we can use the following steps.

1. Sample the Bernoulli random variable H_{ij} with probability $\frac{\omega_i c_1}{\omega_i c_1 + (1 - \omega_i) c_0}$.
2. If $H_{ij} = 0$, update $\Delta_{ij} = 0$.
3. If $H_{ij} = 1$, sample from $f^*(\Delta_{ij})$ via the Metropolis-Hastings algorithm by using the normal approximation of $f^*(\Delta_{ij})$ through the Laplace approximation as the proposal distribution.

One round of sampling for H and $\mathbf{\Delta}$ ends when all (i, j) components have been updated.

The efficiency of this sampling procedure for H and Δ depends on having a good sampler for $f^*(\Delta_{ij})$. We calculated the effective sample size (ESS) (Liu, 1996) of the Laplace approximation, when it is used as the proposal distribution in an importance sampling that targets $f^*(\Delta_{ij})$. Table 1.4 reports the ESS for plausible values of constants s , $\kappa\lambda$, n_{ij} and y_{ij} . We can see from the table that most of the ESS values are close to 1, and the smallest values we observe is no less than 0.75. This suggests that the Laplace approximation will work well as a proposal distribution in the MCMC sampler.

The H matrix can also be helpful in posterior inference. The mean of H_{ij} for fixed i is the probability that Δ_{ij} is exactly 0. When all of the H_{ij} terms have an average close to 0, Model 2 is essentially a Bradley-Terry model.

n_{ij}		2			10			100		
y_{ij}/n_{ij}		0	0.5	1	0	0.50	1	0	0.5	1
s	$\kappa\lambda$									
-5	0.1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
-5	1	1.00	1.00	1.00	1.00	0.98	1.00	0.98	1.00	0.96
-5	2	0.99	0.91	0.95	0.95	0.98	0.95	0.82	1.00	0.77
-2	0.1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
-2	1	1.00	1.00	1.00	0.98	0.99	0.98	0.96	1.00	0.88
-2	2	0.95	0.78	0.96	0.87	1.00	0.91	0.86	1.00	0.86
0	0.1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
0	1	1.00	1.00	1.00	0.96	1.00	0.98	0.96	1.00	0.96
0	2	0.94	0.98	0.95	0.82	0.99	0.85	0.87	1.00	0.88
2	0.1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
2	1	1.00	1.00	1.00	0.99	1.00	0.98	0.97	1.00	0.90
2	2	0.94	0.97	0.94	0.92	0.99	0.84	0.90	1.00	0.79
-5	0.1	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
-5	1	1.00	1.00	1.00	1.00	0.98	1.00	0.98	1.00	0.96
-5	2	0.99	0.94	0.98	0.95	0.98	0.84	0.91	1.00	0.88

Table 1.4: ESS from importance sampling targeting $f^*(\Delta_{ij})$ with its Laplace approximation as proposal distribution with 1000 samples over plausible values of unknown constants. The smallest ESS values in table are colored in red.

1.5 A SIMULATION STUDY

In this section, we compare the performance of our proposed models against a Bradley-Terry model under several different simulation conditions. Knowing the ground truth of pairwise probabilities, we can measure model performance by comparing model inferences against true probabilities. The simulations are designed to compare the models under different settings. Simulation 1 examines how model performance relates to model assumptions. To do so, we simulated data from a Bradley-Terry model, and also from two set of pairwise probabilities that do not strictly follow the Bradley-Terry model assumptions. Simulation 2 examines issues arising from a case of parameter non-identifiability.

1.5.1 SIMULATION 1

As the basis of our simulation experiment, we refer to the data on citations among Statistics journals used in [Varin et al. \(2016\)](#). We selected journals which have exchanged more than 400 citations with other journals. This left us 10 journals, with a total of 45 paired comparisons. We kept the empirical number of exchanged citation between journal i and j , n_{ij} , as the count observed in the simulated data. Then, we designed 3 sets of data simulations using different designs for pairwise probabilities p_{ij} , the probability of journal i being cited by journal j . Outcomes between journals indexed i and j are sampled from $\text{Binom}(n_{ij}, p_{ij})$.

DATA SET 1: The pairwise probabilities used in first simulated data set, $\mathbf{p}_1 = (p_{12}, \dots, p_{ij}, \dots, p_{9,10})'$ were generated based on the Bradley-Terry model in the following manner. We first fit the Bradley-Terry model to the original citation data, and recorded the MLE of journals' academic influence from the fitted model. Setting the MLE as γ , entries of $\mathbf{p}_1$ were calculated as $\text{logit}(p_{ij}) = \gamma_i - \gamma_j$. This data set was used to compare our models against the Bradley-Terry model when the Bradley-Terry assumptions are true. We want to validate that our models still perform reasonably

when the Bradley-Terry assumptions are not violated.

DATA SET 2: The second data set was designed to introduce local disturbances in p_{ij} to create violations of the Bradley-Terry assumptions and, in particular, introduce violations of stochastic transitivity. Specifically, we first set $\mathbf{p}_2 = (\tilde{p}_{12}, \dots, \tilde{p}_{ij}, \dots, \tilde{p}_{9,10})' = \mathbf{p}_1$. Then, we randomly selected entries of $\mathbf{p}_2$ with 10% probability and set $\tilde{p}_{ij} = 1 - p_{ij}$. For the actual simulated data, the journal pairs that were affected were pairs indexed $(1, 2), (2, 9), (3, 6), (3, 9)$.

DATA SET 3: For the third data set, $\mathbf{p}_3$ is simply the empirical pairwise probability of journal i being cited by journal j observed in the data. This set of pairwise probabilities is not based on any stated model assumptions, and therefore do not strictly respect the Bradley-Terry assumptions.

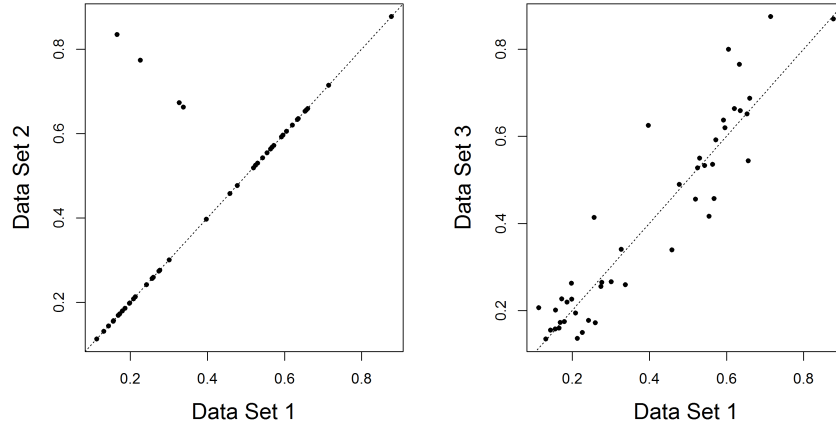


Figure 1.3: Pairwise probabilities used in simulation examples, for journal pairs (i, j) , $i < j$.

We display the pairwise probabilities used to generate Data Set 2 and 3 to the pairwise probabilities used for Data Set 1 in Figure 1.3. By design, most of the pairwise probabilities in Data Set 2 are identical to Data Set 1. Only 4 pairwise probabilities differ between the two data sets. On the other hand, the probabilities in Data Set 3 is just generally different from Data Set 1.

1.5.1.1 CHOICE OF PRIOR

For all analysis in Simulation 1, we set prior parameters for the model such that there is a preference towards Bradley-Terry model a priori.

PRIOR DISTRIBUTION USED IN MODEL 1: Because Model 1 embeds the Bradley-Terry model as a limiting case when $k_i = 0$ for all i , we set the hyperparameters k_α and k_β such that the prior predictive distribution of k_i is concentrated around values close to 0. The hyperparameters λ_α and λ_β are set to allow large range of values for λ^2 . The final parameters chosen are

$$k_0 \sim \text{Gamma}(10, 1)$$

$$\lambda^2 \sim \text{Inv-Gamma}(1, 1)$$

and the left panel in Figure 1.4 displays the prior predictive distributions of k_i under our selected prior.

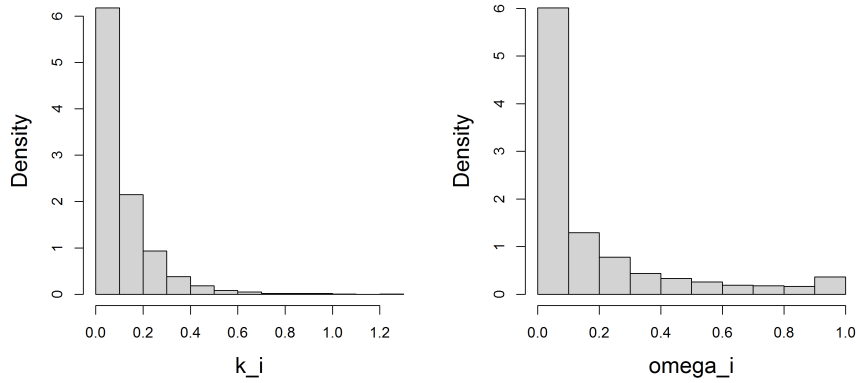


Figure 1.4: Left panel gives the prior predictive distribution of k_i given $k_0 \sim \text{Gamma}(10, 1)$ in Model 1. Right panel gives prior predictive distribution of ω_i given $\alpha \sim \text{Exp}(1)$ and $\beta \sim \text{Exp}(0.1)$ in Model 2.

PRIOR DISTRIBUTION USED IN MODEL 2: The interpretation of λ is similar across Model 1 and Model 2. Hence, we use the same prior distribution for λ^2 as in Model 1. The parameter κ^2 is part of the variance for the Normal mixture component on the distribution of Δ_{ij} . We can use the same prior distribution as λ^2 . To reflect our prior preference towards a Bradley-Terry model, we set the hyperparameters α_0 and β_0 such that ω_i is concentrated towards 0. The final parameter distributions chosen are

$$\alpha \sim \text{Exp}(1)$$

$$\beta \sim \text{Exp}(0.1)$$

$$\kappa \sim \text{Inv-Gamma}(1, 1)$$

$$\lambda \sim \text{Inv-Gamma}(1, 1)$$

and the prior predictive distribution of ω_i is displayed on the right panel in Figure 1.4.

1.5.1.2 INFERENCE

Along with using the Bayesian framework to make inferences on Model 1 and Model 2, we also fit a Bayesian version of the Bradley-Terry model introduced by [Leonard \(1977\)](#). Similar to what was used in Model 1 and Model 2, we set a multivariate Normal prior on θ and use the same prior distribution for λ^2 as specified in Section 1.5.1.1. To obtain posterior inferences, we ran MCMC samplers for 10000 iterations, with a burn-in period using the first 20% of the samples. To ensure convergence, we checked the behavior of trace plots and ensured that multiple chains initiated at different starting points converged to the same distribution. We also apply the Gelman-Rubin convergence diagnostic ([Gelman and Rubin, 1992](#); [Brooks and Gelman, 1998](#)) to chains initiated at different starting points, by calculating the potential scale reduction factor, R_c , for all parameters across different chains. We checked that the statistic R_c was below 1.2 for all the model parameters and R_c for hyperparameters

were below the more stringent cutoff of 1.1. The cutoff 1.2 is based on example from [Brooks and Gelman](#).

For each Data Set d , we are interested in how well the model estimated $\mathbf{p}_d$ on average. To compare the model performances, we examined the average error in estimating the pairwise log odds. Omitting the simulation setting index d , we examined the quantity

$$\frac{1}{S} \sum_{s=1}^S \left(\frac{1}{45} \sum_{i=1}^9 \sum_{j=i+1}^{10} (\widetilde{\text{logit}(p_{ij})}_s^{\mathcal{M}} - \text{logit}(p_{ij}))^2 \right)$$

where $\widetilde{\text{logit}(p_{ij})}_s^{\mathcal{M}}$ is a sample of $\text{logit}(p_{ij})$ from the posterior distribution under model $\mathcal{M}$. The average error in estimating pairwise log odds are reported in Table 1.5 and inferential results are presented in Figure 1.5.

	Bradley Terry	Model 1	Model 2
Data Set 1	0.0140	0.0236	0.0186
Data Set 2	0.3457	0.0894	0.0632
Data Set 3	0.2033	0.1477	0.1526

Table 1.5: Average error in estimating pairwise log-odds

RESULTS FROM DATA SET 1: For the first data set, all three models perform well, but the Bradley-Terry model performs the best. Visually inspecting the first panel in Figure 1.5, we see that the three models give similar point estimates, but Model 1 has wider credible intervals. Table 1.6 confirms that the credible intervals of Model 1 are the widest, while the credible intervals of Model 2 are close to those of the Bradley-Terry model.

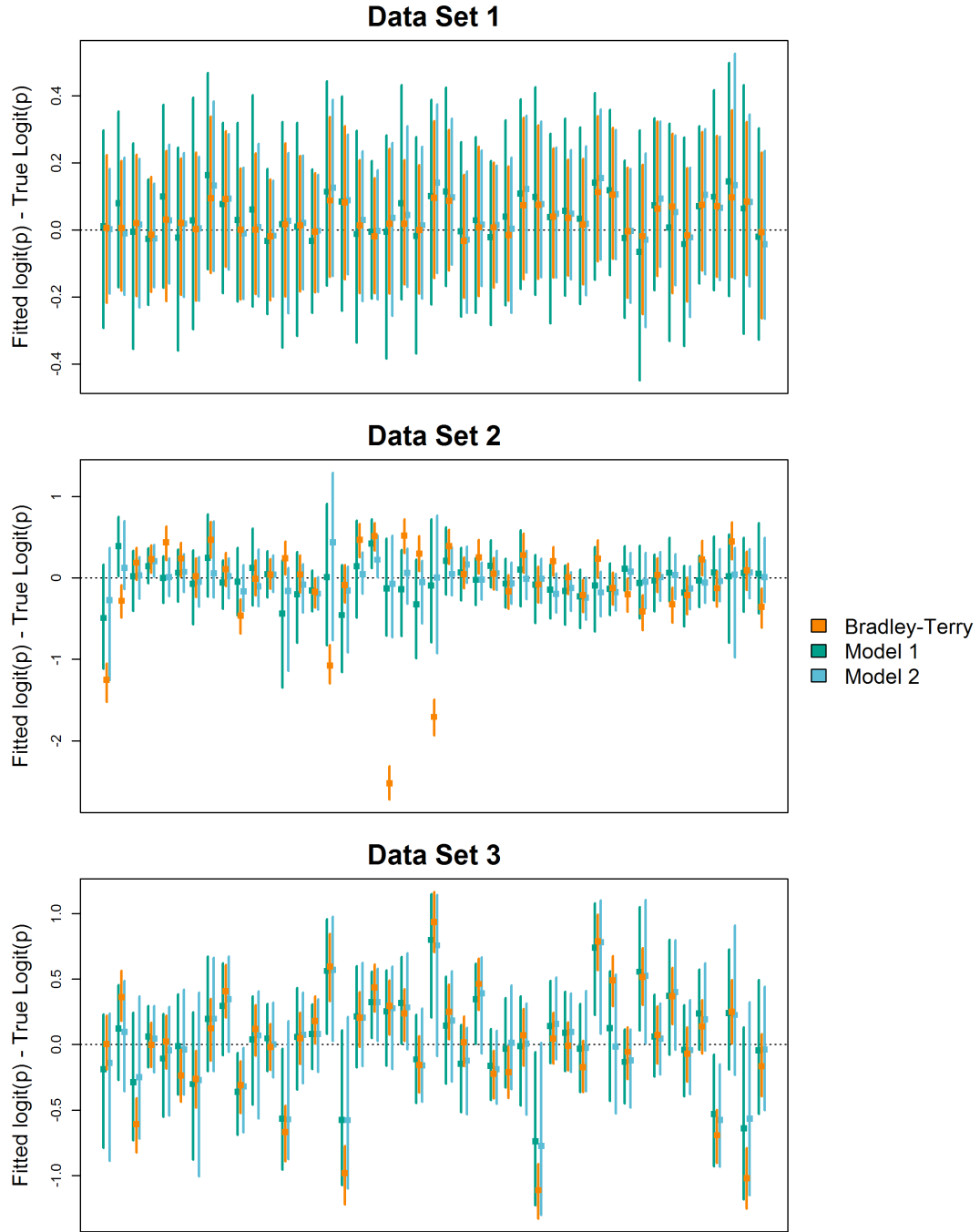


Figure 1.5: Difference between fitted log odds and true log odds. Posterior samples summarized by posterior median and 95% credible intervals using 0.025th and 0.975th quantile. Pairs are arranged in no particular order.

	Bradley Terry	Model 1	Model 2
Average CI Width	0.418	0.576	0.438

Table 1.6: Average width of 95% credible intervals on log odds under Data Set 1.

We reason that when the Bradley Terry assumptions are valid, Model 2 outperforms Model 1 because the posterior distribution of Model 2 is a mixture distribution, where one component of the mixture explicitly follows the Bradley-Terry model. Figure 1.6 displays the posterior distribution of stability parameters under Data Set 1. We can see that both sets of posterior parameters reflect that the data are coming from a model that is consistent with the Bradley-Terry assumptions. There is little variation across journals, and every journal's estimated stability parameter is low.

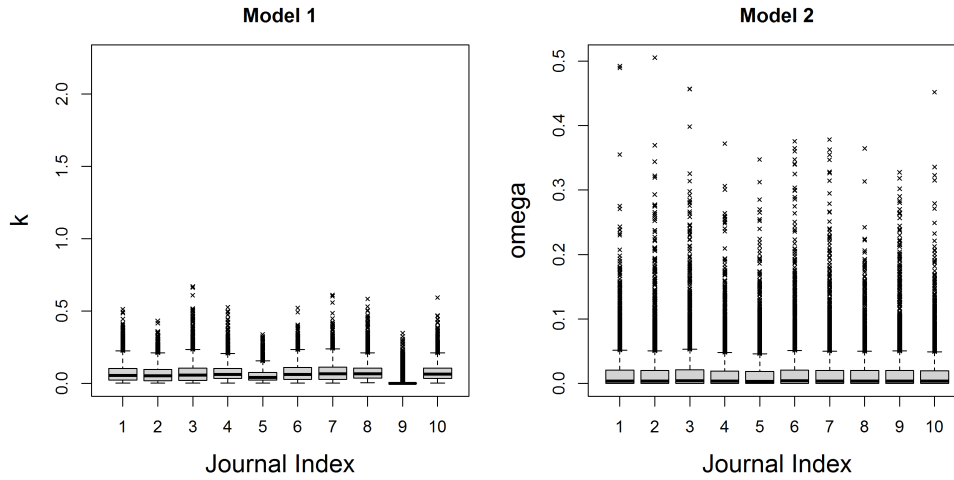


Figure 1.6: Posterior distribution of stability parameters for each journal under Model 1 and Model 2 for Data Set 1.

We also checked for evidence of violation of stochastic transitivity in the posterior samples. To do so, we examined all triplets of players in our data. For each set of parameters drawn from the posterior distribution, we checked for instances of violation of stochastic transitivity across all triplets. We then recorded the average proportion of violation of stochastic transitivity for each player triplets. For Data

Set 1, we can see from Figure 1.7 that almost all player triplets respect stochastic transitivity. There are singular cases under Model 1 where some player triplets violate stochastic transitivity for less than 20% of the time. This is very sensible under data generated from a Bradley-Terry model.

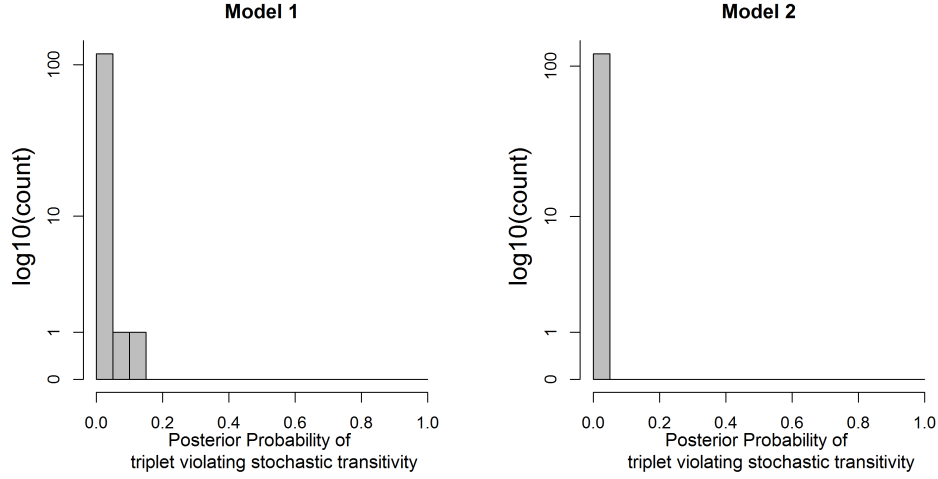


Figure 1.7: Histogram on player triplets by their average proportion of violating stochastic transitivity, for each set of parameters drawn from the posterior distribution, under Data Set 1. The y -axis in $\log_{10}$ scale.

RESULTS FROM DATA SET 2: For the second data set, based on the average error in estimating pairwise log odds given in Table 1.5, we see that the Bradley-Terry model performed the worst, while Model 2 moderately outperformed Model 1 in this instance. From the middle panel in Figure 1.5, we observe that the Bradley-Terry model cannot capture the disturbances we introduced into the pairwise probabilities. Some of the credible intervals under Bradley-Terry model are far away from the true log-odds. Model 1 and Model 2 give posterior interval estimates that cover the true log-odds, but with wider credible intervals.

The posterior distribution of stability parameters are given in Figure 1.8. We can see that the Model 1 identified Journals 2, 3 and 9 to be behaving differently, while Model 2 identified Journals 2, 3, 9 and to some extent, 6. These two models are consistent with each other. The detection of differential

behavior in Journals 2, 3, 6 and 9 based on the inferential results of Model 1 and 2 is consistent with the pairs (1,2) (2,9) (3,6) (3,9) being changed to simulate violations of stochastic transitivity.

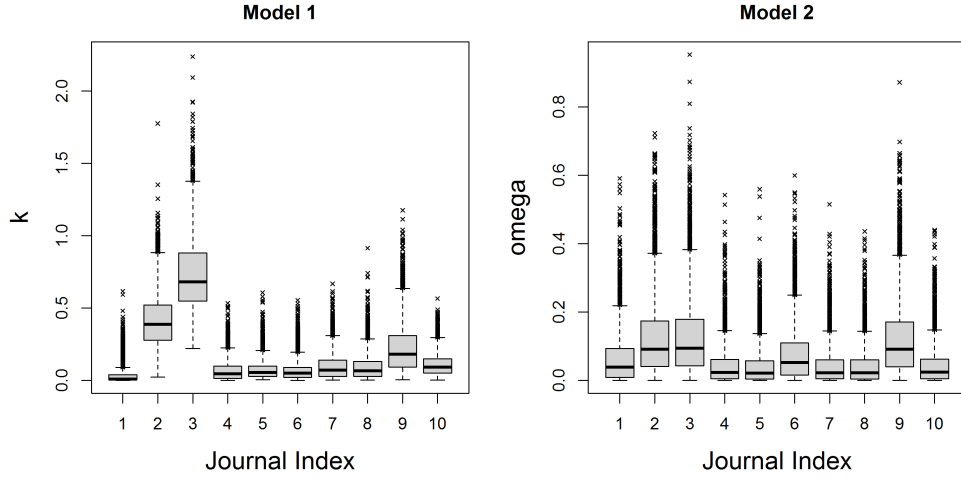


Figure 1.8: Posterior distribution of stability parameters for each journal under Model 1 and Model 2 for Data Set 2.

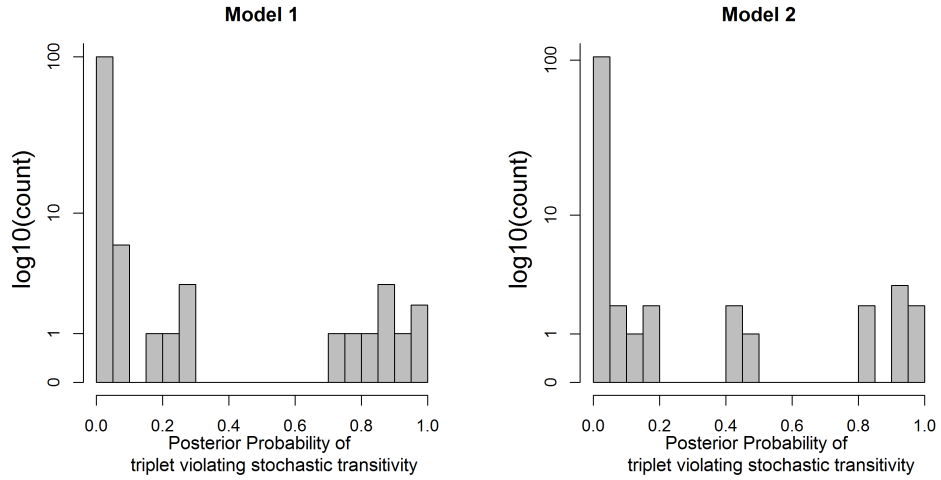


Figure 1.9: Histogram on player triplets by their average proportion of violating stochastic transitivity, for each set of parameters drawn from the posterior distribution, under Data Set 2. y -axis in $\log_{10}$ scale.

We also checked for violations of stochastic transitivity in the posterior samples. Figure 1.9 shows triplets of players that demonstrate violations of stochastic transitivity more than 80% of the time. Model 1 and Model 2 results are slightly different, but both models demonstrate the ability to capture violations of stochastic transitivity among selected triplets.

RESULTS FROM DATA SET 3: From the last panel in Figure 1.5, we do not observe clear evidence that one model outperforms the others. If we justify by average error in estimating true log odds, Table 1.5 indicates that Model 1 and Model 2 perform better than the Bradley-Terry model, with Model 1 being marginally better than Model 2. Data Set 3 was generated based on empirical pairwise probabilities that are not known to correspond to any of these models. The observation that Model 1 and Model 2 outperformed the Bradley-Terry model in estimating true log odds hints at Model 1 and Model 2 being useful alternative modeling choices over the Bradley-Terry model that can protect the inferences against the stringent assumptions of the Bradley-Terry model.

The posterior distribution of stability parameters are given in Figure 1.10. The posterior samples suggest a general deviation from basic Bradley-Terry models, but no single journal stands out being particularly different from the rest. Arguably, Journal 1 and 10 seem to be less stable than the rest.

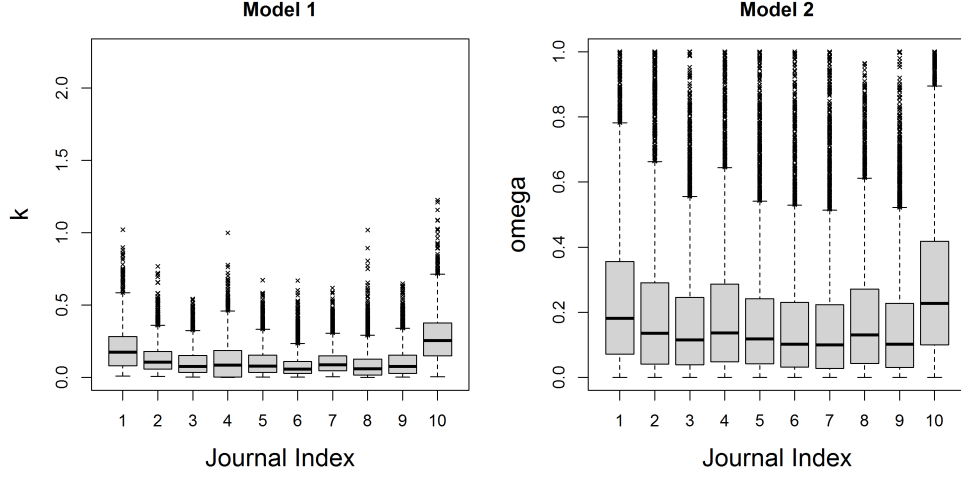


Figure 1.10: Posterior distribution of stability parameters for each journal under Model 1 and Model 2 for Data Set 3.

1.5.2 SIMULATION 2

Next, we use a simulation example to demonstrate what happens when the model may have identifiability concerns. Returning to how we designed pairwise probabilities for Data Set 2, consider the following setup. Suppose we have

$$\text{logit}(p_{ij}) = \theta_i - \theta_j + \theta_{ij}$$

where $\theta_{ij} = 0$ for all i, j except for a single pair (i', j') . Because the model assumes $\theta_{i'j'} = \Delta_{i'j'} + \Delta_{j'i'}$, we explore how the model would allocate $\theta_{i'j'}$ between Δ_{ij} and Δ_{ji} . To address this question, we designed a new set of pairwise probabilities $\mathbf{p}'_2$. We set the pairwise probabilities to be the same as $\mathbf{p}'_2 = \mathbf{p}_1 = (p_{12}, \dots, p_{ij}, \dots, p_{9,10})'$. Then, for a single pair of players (i', j') , we set up their pairwise probability as $1 - p_{i'j'}$. Then we simulated observed data y_{ij} given the set of pairwise probabilities $\mathbf{p}'_2$ from the Binomial distribution. For the actual simulated data, journal pair (4, 8) was affected.

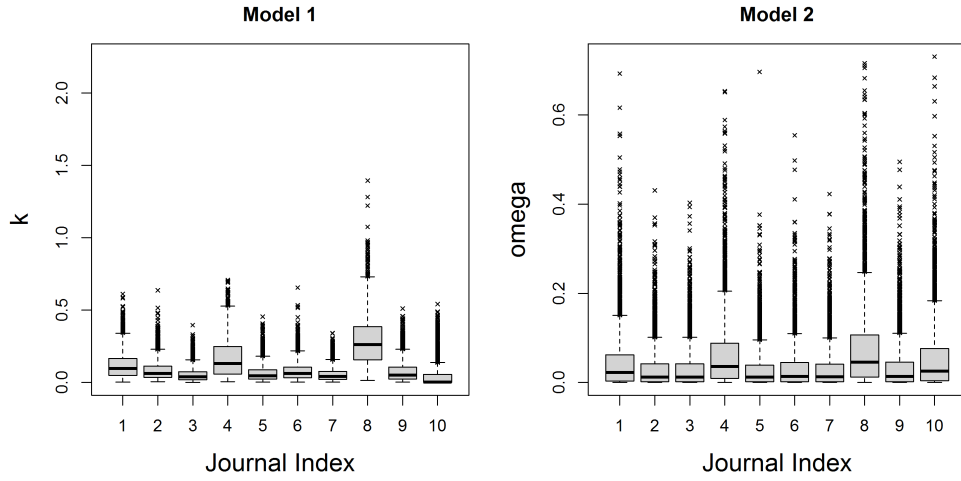


Figure 1.11: Posterior distribution of stability parameters for each journal under Model 1 and Model 2 for Simulation 2.

Posterior distribution of stability parameters are given in Figure 1.11. We observe that the estimated stability parameters for journals 4 and 8 are higher than other journals.

Figure 1.12 displays the posterior distribution of Δ_{ij} and Δ_{ji} for Model 1 and Model 2, for $i = 4$ and $j = 8$. The models appear to allocate deviations from the basic Bradley Terry model onto Δ_{ij} and Δ_{ji} equally. Either Δ_{ij} or Δ_{ji} deviates from 0. The models do not distinguish between the two journals. This contrasts what we observed in Figure 1.8. Under Data Set 2, although Journal 1 was affected by the switch in pairwise probabilities, the model did not detect any deviation for Journal 1. This is likely because there are other journals such as 2, 3 and 9, which are affected by the probability switches. Hence, the model was able to distinguish Journal 1 from the others, attributing model deviations to Journals 2, 3 and 9.

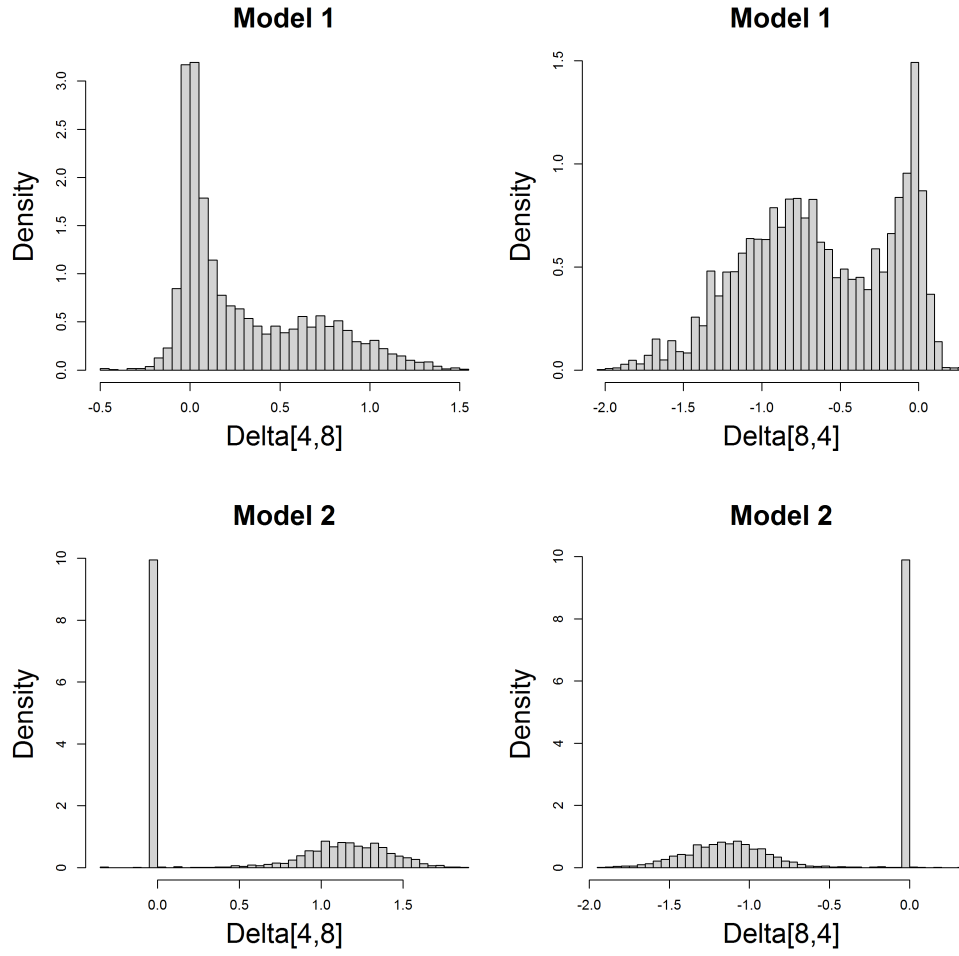


Figure 1.12: Posterior samples of Δ_{ij} and Δ_{ji} for $i = 4, j = 8$ under Model 1 and Model 2 for Simulation 2. Left panels are for $\Delta_{4,8}$, right panels are for $\Delta_{8,4}$.

Although the posterior distribution of the Δ_{ij} individually are multimodal, their posterior distributions are negatively correlated. The posterior distribution of log-odds, as given in Figure 1.13 is still an unimodal distribution that is approximately Normal.

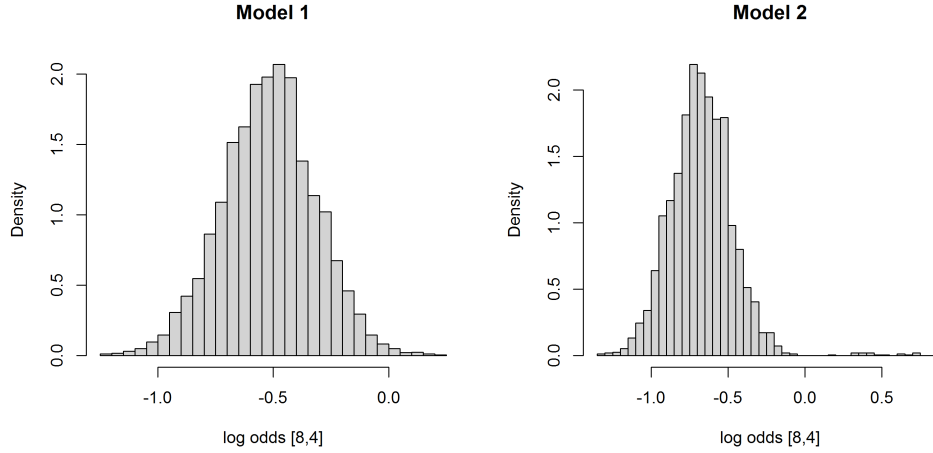


Figure 1.13: Posterior samples of log-odds of journal 8 citing journal 4 under Model 1 and Model 2 for Simulation 2.

Under this scenario, marginal posterior inferences of pairwise probabilities are unambiguous. However, if we are characterizing journals, one must be careful to make the conclusion that journal 4 and 8 demonstrate higher variability than rest of the journals. The model is suggesting that it could be either journal (or both) that is associated with larger variability.

1.6 APPLICATION TO EMPIRICAL DATA

In this section, we revisit the citation data analyzed by [Varin et al. \(2016\)](#). We selected statistics journals which have exchanged more than 200 citations with other statistics journals, which gives us 17 journals. We fit the Bayesian version of the Bradley-Terry model, Model 1 and Model 2 on this citation data. Details on choice of prior and MCMC convergence checks remain the same as Section 1.5.

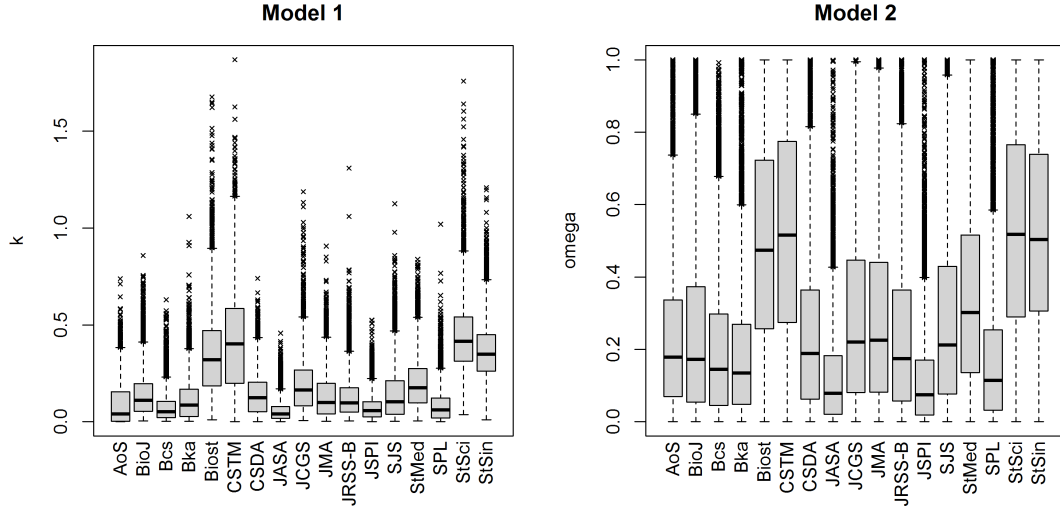


Figure 1.14: Posterior samples of stability parameter for each journal under Model 1 and Model 2.

As discussed in Section 1.2, the paired comparison analysis of citations can be viewed as modeling the influence of a journal on other journals. Figure 1.14 plots the posterior distribution of the stability parameters for each journal under Model 1 and Model 2. Observe that the estimated stability parameters differ across journals. Most journals in the sample have low stability parameters, implying that the average influence of the journal on other journals are relatively consistent. Some journals, such as *Statistical Science* (StSci) and *Statistical Sinica* (StSin) demonstrate instability. We examine *Statistical Science* as an example which is indexed at $i = 16$. We summarized the posterior distribution of $\Delta_{ij} - \Delta_{ji}$ for $i = 16, j = 1, \dots, 15, 17$. We find that compared to the benchmark set by the average academic influence of the journal *Statistical Science*, *Statistical Science* over-influences *Annals of Statistics* (AoS), *Communications in Statistics-Theory and Methods* (CSTM) and *Journal of the Royal Statistical Society, Series B* (JRSS-B) while under-influences *Biostatistics* (Biost), *Scandinavian Journal of Statistics* (SJS) and *Statistics in Medicine* (StMed).

Journal	Bradley-Terry		Model 1			Model 2		
	$\hat{\theta}$	95%CI, $\hat{\theta}$	$\hat{\theta}$	95%CI, $\hat{\theta}$	$\hat{k}$	$\hat{\theta}$	95%CI, $\hat{\theta}$	$\hat{\omega}$
JRSS-B	1.73	(1.6 - 1.9)	1.76	(1.52 - 2.02)	0.10	1.74	(1.52 - 1.95)	0.17
AoS	1.01	(0.89 - 1.12)	0.98	(0.81 - 1.18)	0.04	0.93	(0.78 - 1.11)	0.18
JASA	0.89	(0.8 - 0.99)	0.90	(0.76 - 1.05)	0.04	0.89	(0.76 - 1.03)	0.08
Bka	0.87	(0.75 - 1.01)	0.88	(0.68 - 1.07)	0.09	0.93	(0.77 - 1.14)	0.13
JCGS	0.40	(0.17 - 0.55)	0.42	(0.15 - 0.69)	0.16	0.42	(0.17 - 0.71)	0.22
Bcs	0.37	(0.23 - 0.49)	0.34	(0.17 - 0.52)	0.05	0.34	(0.18 - 0.51)	0.14
SJS	0.29	(0.07 - 0.49)	0.31	(0.07 - 0.56)	0.10	0.36	(0.11 - 0.66)	0.21
Biost	0.26	(0.1 - 0.41)	0.14	(-0.23 - 0.5)	0.32	0.13	(-0.24 - 0.42)	0.47
StSin	-0.11	(-0.25 - 0.04)	-0.02	(-0.3 - 0.26)	0.35	-0.11	(-0.45 - 0.27)	0.50
StSci	-0.35	(-0.53 - -0.15)	-0.25	(-0.64 - 0.12)	0.42	-0.28	(-0.6 - 0.01)	0.52
SPL	-0.42	(-0.57 - -0.28)	-0.40	(-0.65 - -0.19)	0.06	-0.38	(-0.57 - -0.21)	0.11
StMed	-0.44	(-0.57 - -0.31)	-0.43	(-0.66 - -0.21)	0.18	-0.42	(-0.65 - -0.17)	0.30
JSPI	-0.64	(-0.76 - -0.51)	-0.63	(-0.83 - -0.46)	0.06	-0.64	(-0.81 - -0.48)	0.07
JMA	-0.74	(-0.88 - -0.59)	-0.68	(-0.87 - -0.47)	0.10	-0.69	(-0.9 - -0.48)	0.23
BioJ	-0.89	(-1.09 - -0.68)	-0.83	(-1.13 - -0.59)	0.11	-0.83	(-1.07 - -0.55)	0.17
CSDA	-0.96	(-1.09 - -0.85)	-0.95	(-1.14 - -0.78)	0.13	-1.00	(-1.17 - -0.78)	0.19
CSTM	-1.25	(-1.43 - -1.07)	-1.49	(-1.96 - -1.1)	0.40	-1.41	(-1.85 - -1.1)	0.52

Table 1.7: Summary of estimated parameters on the citation data between Statistics journals based on draws from posterior distributions. Point estimate of parameters given by median of posterior samples, 95% credible interval given by 0.025 and 0.975 quantiles of posterior samples. Journals ranked by $\hat{\theta}$ under Bradley-Terry model.

Table 1.7 summarizes the estimated parameters across the three models. Comparing $\hat{\theta}$, we observe that there are differences across the models in the point estimates of average academic influence, but the rankings of journals' average academic influence are largely consistent across the three models. From Model 1 and Model 2, we recognize that a linear ranking of journals based on their average academic influence only captures part of the information. Several of the mid-ranked journals Biost, StSin and StSci have high estimated stability parameters.

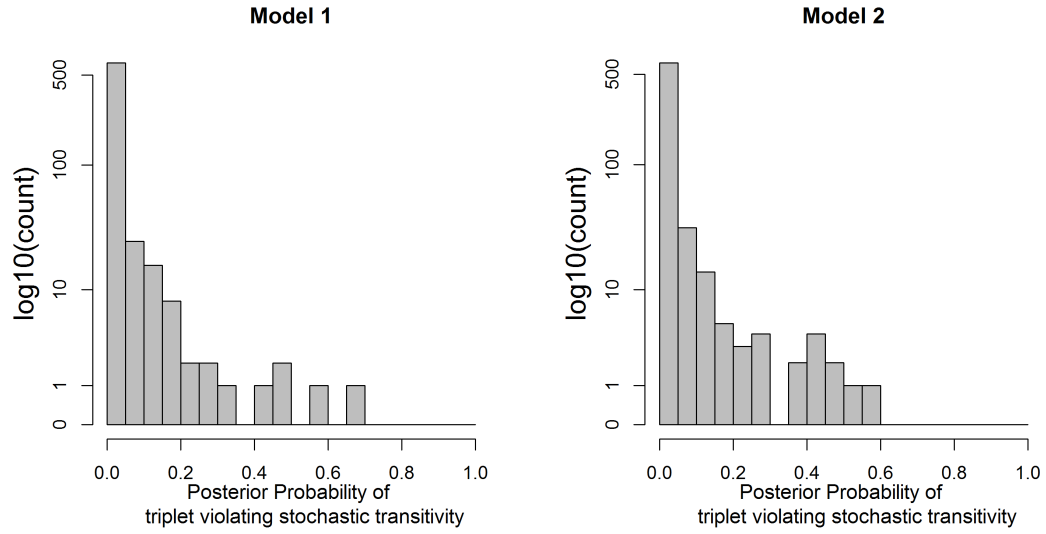


Figure 1.15: Histogram on player triplets by their average proportion of violating stochastic transitivity, for each set of parameters drawn from the posterior distribution. y -axis in $\log_{10}$ scale. For each set of posterior sample of parameters, we identify whether distinct triplets demonstrate violation of stochastic transitivity. Then, we tabulate the proportion of violation of stochastic transitivity for each triplet.

Figure 1.15 reports violation of stochastic transitivity among player triplets for Model 1 and Model 2 for estimated parameters. There exist multiple player triplets that violate stochastic transitivity for around 40% – 60% of the time.

1.6.1 MODEL COMPARISON VIA OUT-OF-SAMPLE PREDICTION

Below we compare the performance of our hierarchical models against the Bradley-Terry model through out-of-sample predictive accuracy.

There are several standard metrics on evaluating prediction accuracy of binary outcomes based on comparing correspondence between predicted outcomes and observed outcomes, where prediction of a positive outcome is accomplished using a threshold on fitted probabilities. Examples include the receiver operating curve (ROC) and the F1 score. However, we are comparing fitted probabilities

at the journal-pair level, and there are very limited observations between each journal pair. Metrics that rely on comparing predicted outcomes and observed outcomes are not sensitive to demonstrated differences in fitted probabilities under different models. Hence, we consider the following measure of model fit based on out of sample prediction.

We split the data into $B = 10$ subsets. Each subset $b = 1, \dots, B$ contains observed outcomes (n_{ij}^b, y_{ij}^b) for unique pairs $j > i$, with $\sum_b n_{ij}^b = n_{ij}$ and $\sum_b y_{ij}^b = y_{ij}$. For each index b , we define training data b as $(n_{ij} - n_{ij}^b, y_{ij} - y_{ij}^b)$ and test data b as (n_{ij}^b, y_{ij}^b) , for $j > i$. We can perform out-of-sample evaluation by fitting the model on training data b and evaluating the fitted model on test data (n_{ij}^b, y_{ij}^b) .

For each model $\mathcal{M}$ and testing data b , we define a measurement of how well the fitted model using training data b fits the observed outcomes (n_{ij}^b, y_{ij}^b) as

$$\mathcal{D}_{ij,b}^{\mathcal{M}} = \frac{1}{S} \sum_{s=1}^S \left(\binom{n_{ij}^b}{y_{ij}^b} (\tilde{p}_{ij,bs}^{\mathcal{M}})^{y_{ij}^b} (1 - \tilde{p}_{ij,bs}^{\mathcal{M}})^{n_{ij}^b - y_{ij}^b} \right)$$

for some large constant S . The quantity $\tilde{p}_{ij,bs}^{\mathcal{M}}$ is a draw from the posterior distribution of p_{ij} under Model $\mathcal{M}$ fitted using training data b . The index s denotes the s th sample. The quantity

$$\binom{n_{ij}^b}{y_{ij}^b} (\tilde{p}_{ij,bs}^{\mathcal{M}})^{y_{ij}^b} (1 - \tilde{p}_{ij,bs}^{\mathcal{M}})^{n_{ij}^b - y_{ij}^b}$$

is the probability of observing (n_{ij}^b, y_{ij}^b) given pairwise probability $\tilde{p}_{ij,bs}^{\mathcal{M}}$. Hence, $\mathcal{D}_{ij,b}^{\mathcal{M}}$ is approximating the posterior expectation of the probability of observing (n_{ij}^b, y_{ij}^b) under Model $\mathcal{M}$. Between two models, the model with a higher $\mathcal{D}_{ij,b}^{\mathcal{M}}$ does better at predicting out-of-sample observations (n_{ij}^b, y_{ij}^b) . This is akin to using the “log loss” for binary outcomes, except we adjust it by the binomial coefficient.

We can aggregate this measure over all testing data sets by defining

$$\mathcal{D}_{ij}^{\mathcal{M}} = \frac{1}{B} \sum_{b=1}^B \mathcal{D}_{ij,b}^{\mathcal{M}}$$

for each pair $j > i$.

Figure 1.16 compares the out-of-sample performance of Model 1 and Model 2 against the Bradley-Terry model. We observe that for most pairs of journals, the out of sample prediction of the three models are comparable. However, we observe quite a few points that lie above and away from the $y = x$ line. For these journal pairs, Model 1 and Model 2 did a better job at out-of-sample prediction than the Bradley-Terry model.

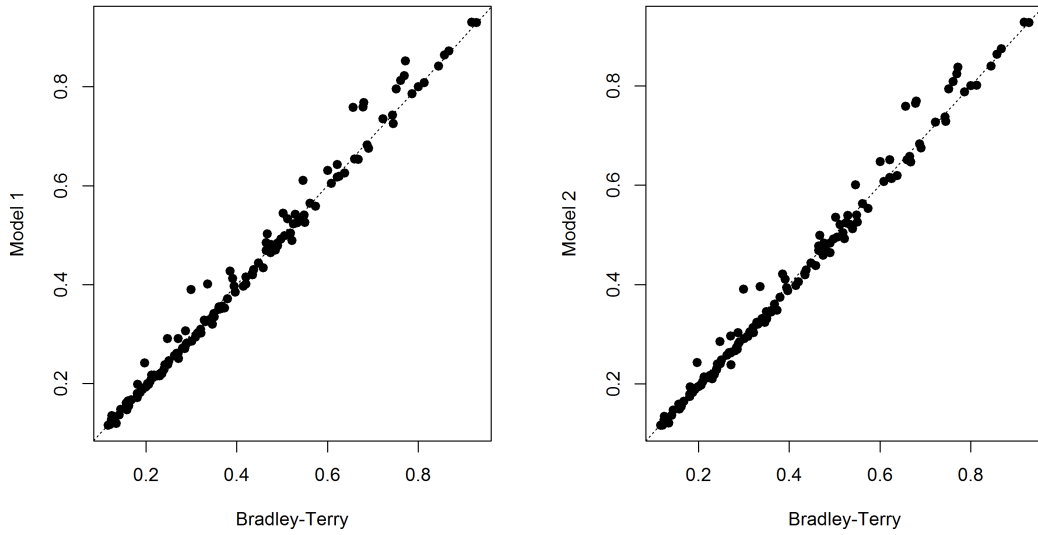


Figure 1.16: Comparison of $\mathcal{D}_{ij}^{\mathcal{M}}$ for $B = 10$ on the citation data. Left panel compares Model 1 against the Bradley-Terry model. Right panel compares Model 2 against the Bradley-Terry model. Dotted line is the $y = x$ line. Higher value of $\mathcal{D}_{ij}^{\mathcal{M}}$ indicates better out-of-sample prediction.

1.7 CONCLUDING REMARKS

When modeling paired comparison data, the Bradley-Terry model characterizes players with a single scalar strength parameter. To accommodate violation of stochastic transitivity, we proposed to add a stability parameter which provides extra flexibility. We proposed two variants of a hierarchical model that embeds the Bradley-Terry model as a limiting case. We demonstrated that the models are no worse than the Bradley-Terry model in general and can outperform the Bradley-Terry model when the data exhibit violations of transitivity.

Our approach also offers an additional perspective in comparing players. For example, when coaches are selecting individual players to represent institutions in sports event, they are faced with the decision of selecting individuals based on their merits. Our modeling approach gives a quantification of both average strength and stability of each player. The additional quantification of stability can be used to distinguish players with similar average strength.

Violation of stochastic transitivity in paired comparisons can be a natural phenomenon. Hence, there exist a need to accommodate such behavior when modeling paired comparisons. The basic linear paired comparison models do not account for existence of violation of stochastic transitivity, and can perform poorly when violations of stochastic transitivity are observed. Our proposed model have the flexibility to accommodate violation of stochastic transitivity. Our approach of modeling players using strength and stability can serve as the basis for further work on modeling violation of stochastic transitivity.

2

A Dynamic Paired Comparison Model with Strength and Stability

2.1 INTRODUCTION

Motivated by modeling violations of stochastic transitivity, we introduced two hierarchical models for paired comparison data in Chapter 1. The hierarchical models represent the ability of player i by an average strength parameter θ_i , following the approach of linear paired comparison models. The key extension we introduced is a stability parameter for player i that describes the variation in players' strength when facing different opponents. One key assumption in Chapter 1 is that players' characteristics are constant. This is not a realistic assumption when we observe players over extended periods of time. Hence, in this chapter, we describe a dynamic model for paired comparisons where players' strengths and stabilities may change over time. Based on a dynamic paired comparison model, we propose a non-iterative updating algorithm so that the model can be adapted into a rating system, where players' characteristics are updated over time as we collect more observations. The methodology used in developing the updating algorithm is similar to what was used by Glickman (1999, 2001) to develop the Glicko and Glicko-2 rating systems.

In sports, quantification of athletes' abilities at a point in time are referred to as ratings. Rating systems are typically updated in an online fashion, taking in newly observed game outcomes and updating the quantification of athletes' abilities based on observed game outcomes. Stefani (2011) surveyed 159 international sports and found that 99 of these sports adopt some kind of rating system. Rating systems are useful in designing tournaments. For example, a typical tournament in sports such as Tennis and Badminton pair up athletes, and the winner proceeds to the next round till there is a single winner. With a reliable rating system, the strongest players in the context of knockout tournaments are placed far apart so they will only face each other towards the latter stages of the tournament. The player ratings ensure that the latter stages of the tournament are most exciting. Ratings are also useful for shortlisting athletes. For example, coaches of a national team may be interested in drafting athletes to compete in international competitions. Having a quantification of athletes' skills gives a direct

way of isolating top players. One of the most well-known rating system is the Elo rating system (Elo, 1978), which is adopted by sports such as tennis and League of Legends. The Elo rating system was also widely adopted in chess (Glickman and Jones, 1999). However, many national chess federations and online chess platforms have moved on to adopt the Glicko rating system since its development. Many rating systems are built based on probabilistic models. A typical analysis in sports involves large number of athletes and games. Hence, the rating systems are much more computationally tractable than complete likelihood-based analysis using models. Furthermore, ratings systems have the advantage of making inference on player strengths in an online fashion. When new games are observed, inferences on player strengths are updated without any need to refit the entire model.

This chapter is outlined as follows. Section 2.2 details the dynamic model with strength and stability parameters we propose. Section 2.3 describes an approximate updating algorithm that adapts the dynamic model into a rating system, and Section 2.4 illustrates this rating system using data from professional Go. In Section 2.5, we discuss an approximation strategy to reduce computation efforts at each step of the rating system, and examines the validity of this approximation strategy on the professional Go data. This chapter concludes in Section 2.6.

2.2 DYNAMIC PAIRED COMPARISON MODELS

We first propose a dynamic paired comparison model. Starting at time t_1 , we assume that there are equally spaced rating periods marked by time points $t_r, r = 1, \dots, R$. Let $(Y_{ij}^{(r)}, n_{ij}^{(r)})$ be the game outcomes collected during rating period r between players i and j for unique player pairs $j > i$ among N players, where $n_{ij}^{(r)}$ games were observed between players i and j and $Y_{ij}^{(r)}$ of them won by player i . During each rating period, we assume that the player characteristics are constant. Hence, the

outcomes $(Y_{ij}^{(r)}, n_{ij}^{(r)})$ are modeled as

$$Y_{ij}^{(r)} \sim \text{Binom} \left(n_{ij}^{(r)}, p_{ij}^{(r)} \right)$$

where $p_{ij}^{(r)}$, the probability player i defeats j , is a function of player characteristics in the rating period r as specified by the model. We define $Y_{ij}^{(r)} = n_{ij}^{(r)} - Y_{ji}^{(r)}$ for $i > j$.

2.2.1 DYNAMIC PAIRED COMPARISON MODELS BASED ON BRADLEY-TERRY MODELS

Let $\theta_i^{(r)}$ represent the strength parameter of player i in the rating period r . Using the Bradley-Terry model as the basis of the dynamic model gives us

$$\text{logit} \left(p_{ij}^{(r)} \right) = \theta_i^{(r)} - \theta_j^{(r)}.$$

Both rating systems Glicko (Glickman, 1999) and Glicko-2 (Glickman, 2001) are based on a dynamic model with the Bradley-Terry model as the building block.

The Glicko rating system is built upon a dynamic version of the Bradley-Terry model which assumes that the time-varying strength parameter $\theta_i^{(r)}$ evolves over time based on the Markovian model

$$\theta_i^{(r)} | \theta_i^{(r-1)}, \nu^2 \sim \mathcal{N} \left(\theta_i^{(r-1)}, \nu^2 \right)$$

where ν^2 represents the increase in the uncertainty about a player's strength between rating periods. This dynamic specification asserts that over time, without newly observed games for player i , we become increasingly uncertain about the knowledge of player i 's strength. The underlying dynamic model for the Glicko-2 rating system extends that of the Glicko by introducing a time-varying param-

eter on the uncertainty of $\theta_i^{(r)}$. The underlying dynamic model of Glicko-2 assumes

$$\begin{aligned}\theta_i^{(r)} | \theta_i^{(r-1)}, \sigma_i^{2(r)} &\sim \mathcal{N} \left(\theta_i^{(r-1)}, \sigma_i^{2(r)} \right), \\ \log \left(\sigma_i^{2(r)} \right) | \sigma_i^{2(r-1)}, \tau^2 &\sim \mathcal{N} \left(\log \left(\sigma_i^{2(r-1)} \right), \tau^2 \right)\end{aligned}\tag{2.1}$$

where τ^2 represents increases in uncertainty of $\log(\sigma_i^2)$ per rating period. Compared to Glicko, Glicko-2 supports time periods where drastic changes in $\theta_i^{(r)}$ may happen. Since both Glicko and Glicko-2 are built upon Bradley-Terry models, pairwise probabilities based on Glicko and Glicko-2 will always respect stochastic transitivity.

2.2.2 A DYNAMIC PAIRED COMPARISON MODEL WITH PLAYER STRENGTH AND STABILITY

In the spirit of hierarchical models introduced in Chapter 1, we propose a dynamic paired comparison model with each player being characterized by their strength and stability. We will refer to this model as Model DyS² in this chapter. Table 2.1 presents the complete specifications of this model.

Data model

$$y_{ij}^{(r)} | n_{ij}^{(r)}, p_{ij}^{(r)} \sim \text{Binom} \left(n_{ij}^{(r)}, p_{ij}^{(r)} \right)$$
$$\eta_{ij}^{(r)} = \text{logit}(p_{ij}^{(r)}) | \theta_i^{(r)}, k_i^{(r)}, \theta_j^{(r)}, k_j^{(r)} \sim \mathcal{N} \left(\theta_i^{(r)} - \theta_j^{(r)}, k_i^{2(r)} + k_j^{2(r)} \right)$$
$$i, j = 1, \dots, N; j > i.$$

**Dynamic
model
components**

$$\theta_i^{(r)} | \theta_i^{(r-1)}, \nu^2 \sim \mathcal{N} \left(\theta_i^{(r-1)}, \nu^2 \right)$$
$$\log \left(k_i^{(r)2} \right) | k_i^{(r-1)2}, h^2 \sim \mathcal{N} \left(\log \left(k_i^{(r-1)2} \right), h^2 \right)$$
$$i = 1, \dots, N.$$

Prior at t_1

$$\theta_i^{(1)} \sim \mathcal{N}(\theta_0, \lambda^2)$$
$$\log \left(k_i^{2(1)} \right) \sim \mathcal{N}(k_0, \sigma_k^2) \quad i = 1, \dots, N.$$
$$\lambda \sim \text{Inv-Gamma}(\lambda_\alpha, \lambda_\beta)$$
$$\nu^2 \sim \text{Inv-Gamma}(\nu_\alpha, \nu_\beta)$$
$$h^2 \sim \text{Inv-Gamma}(h_\alpha, h_\beta)$$

Table 2.1: Complete specification of a dynamic paired comparison model with strength and stability as player characteristics.

The parameter $\theta_i^{(r)}$ represents the average strength of player i in the rating period r . The stability parameter $k_i^{(r)}$ is an indication of the extent to which player i 's strength changes when facing different opponents. When $k_i^{(r)} \rightarrow 0$, player i always performs at their average strength $\theta_i^{(r)}$ during the rating period r regardless of which opponent they are facing. In the limiting case when $k_i^{(r)} = 0$ for all players i and all rating periods r , paired comparisons in our model at each rating period can be described by a Bradley-Terry model, and our model resembles the underlying model of the Glicko rating system. Parameters ν^2 and h^2 are global parameters that reflect the incremental variances in parameters $\theta_i^{(r)}$ and $\log \left(k_i^{2(r)} \right)$ per rating period respectively. Similar to [Glickman \(1999, 2001\)](#), this setup reflects that, without any new information, our uncertainty about player characteristics increase over time.

The constant θ_0 is a fixed starting point for a previously unobserved player. We set $\theta_0 = 0$. With

this anchor, the parameter $\theta_i^{(r)}$ can be interpreted as measuring player i 's strength relative to an average player in the population at time t_r . The parameter λ^2 measures the variability in a newly observed player's average strength within the population. The prior distribution for λ^2 , as well as ν^2 and h^2 , can be assumed to be non-informative inverse-Gamma distributions. The constant k_0 describes prior information on player stability. We set $k_0 = \log(10^{-3})$ and let $\sigma_k^2 = 1000$. This is essentially an approximately uniform prior on $\log(k_i^{2(1)})$.

2.2.2.1 THE FULL BAYESIAN POSTERIOR

Omitting the time index, let $\mathbf{y}$ and $\mathbf{n}$ represent the vector of paired comparison outcomes y_{ij} and n_{ij} for unique pairs $j > i$. Let $\boldsymbol{\theta}$ and $\mathbf{k}$ represent vector of player characteristics θ_i and k_i for $i = 1, \dots, N$ respectively. The posterior density of Model DyS² given observed outcomes $\mathbf{y}^{(1:R)}$ and $\mathbf{n}^{(1:R)}$ over rating periods 1 to R is given by

$$\begin{aligned} & f(\boldsymbol{\theta}^{(1:R)}, \log(\mathbf{k}^{2(1:R)}), \nu^2, h^2, \lambda^2 | \mathbf{y}^{(1:R)}, \mathbf{n}^{(1:R)}) \\ & \propto f(\nu^2) f(h^2) f(\lambda^2) \left(\prod_{i=1}^N \phi(\theta_i^{(1)}; \theta_0, \lambda^2) \phi(\log(k_i^{2(1)}; k_0, \sigma_k^2)) \right) \\ & \quad \times \prod_{r=1}^R \prod_{j>i} p(y_{ij}^{(r)} | n_{ij}^{(r)}, \theta_i^{(r)}, \theta_j^{(r)}, k_i^{2(r)}, k_j^{2(r)}) \\ & \quad \times \prod_{r=1}^R \prod_{i=1}^N \phi(\theta_i^{(r)}; \theta_i^{(r-1)}, \nu^2) \phi(\log(k_i^{2(r)}); \log(k_i^{2(r-1)}), h^2) \end{aligned}$$

where $\phi(\cdot; \mu, \sigma^2)$ is the density of the normal distribution with mean μ and variance σ^2 and prior densities $f(\nu^2)$, $f(h^2)$, and $f(\lambda^2)$ are inverse-Gamma densities following the specifications in Table 2.1.

The probability mass function evaluated at $y_{ij}^{(r)}$ is

$$\begin{aligned}
& p\left(y_{ij}^{(r)} | \theta_i^{(r)}, \theta_j^{(r)}, k_i^{2(r)}, k_j^{2(r)}, n_{ij}^{(r)}\right) \\
&= \int p\left(y_{ij}^{(r)} | \eta_{ij}, n_{ij}^{(r)}\right) \phi\left(\eta_{ij}; \theta_i^{(r)} - \theta_j^{(r)}, k_i^{2(r)} + k_j^{2(r)}\right) d\eta_{ij} \\
&\propto \int \left(2\pi \left(k_i^{2(r)} + k_j^{2(r)}\right)\right)^{-\frac{1}{2}} \\
&\quad \times e^{\frac{y_{ij}^{(r)} \eta_{ij} - n_{ij}^{(r)} \log(1+e^{\eta_{ij}}) - \frac{1}{2(k_i^{2(r)} + k_j^{2(r)})} (\eta_{ij} - (\theta_i^{(r)} - \theta_j^{(r)}))^2}{}} d\eta_{ij}.
\end{aligned} \tag{2.2}$$

As in Chapter 1, sampling from the posterior distribution can be performed using the Gibbs sampler. Conditioning on $\boldsymbol{\theta}^{(1:R)}$ and $\mathbf{k}^{(1:R)}$, the parameters λ^2 , ν^2 and h^2 follows the inverse-Gamma distribution by conjugacy. They can be sampled directly. Computation of conditional distributions of $\boldsymbol{\theta}^{(1:R)}$ and $\mathbf{k}^{(1:R)}$ involves calculation of the integral in Equation (2.2). This integral can be numerically approximated using methods such as the Simpson's rule or Gaussian quadrature (Davis et al., 2014) over a finite interval such as $\theta_i^{(r)} - \theta_j^{(r)} \pm 5\sqrt{k_i^{2(r)} + k_j^{2(r)}}$. The parameters $\boldsymbol{\theta}^{(1:R)}$ and $\mathbf{k}^{(1:R)}$ can be updated in blocks, recognizing that $\boldsymbol{\theta}^{(r)}$ and $\mathbf{k}^{(r)}$ only depend on $\boldsymbol{\theta}^{(r-1, r+1)}$ and $\mathbf{k}^{(r-1, r+1)}$. However, the mixing of this Gibbs sampler may be very slow.

Model DyS² is an example of a non-linear state-space model. Parameter inference for such models can also be implemented using particle methods. Kantas et al. (2015) presented a detailed review on particle methods for parameter estimation in state-space models. Examples of algorithms that can be used include the particle MCMC algorithm (Andrieu et al., 2010) and the SMC² algorithm (Chopin et al., 2013).

2.3 AN APPROXIMATE BAYESIAN UPDATING ALGORITHM

Full Bayesian inference for Model DyS² based on sampling from the exact posterior distribution requires carefully designed sampling algorithms. As the number of players N and length of observations

R increase, computation cost increases. Following [Glickman \(1999, 2001\)](#), we develop an approximate Bayesian updating scheme through forward filtering. We trade loss of efficiency in estimation for a simpler algorithm that can be updated easily as more games are observed over new rating periods. This simplified algorithm can function as a rating system that characterizes players on strength and stability. We will refer to this rating system as DyS².

Consider the problem of updating player characteristics at rating period r . For a previously observed player, we assume that the information on player characteristics at time t_{r-1} (after data during period $r - 1$ have been recorded) can be summarized approximately using

$$\begin{aligned}\theta_i^{(r-1)} &\sim \mathcal{N}(\mu_i, \sigma_i^2), \\ \log(k_i^{(r-1)2}) &= c_i.\end{aligned}$$

We retain information on the uncertainty in $\theta_i^{(r-1)}$ but only retain a point estimate for $\log(k_i^{(r-1)2})$.

Using the dynamics of player characteristics over time given in Table 2.1, we have

$$\begin{aligned}\theta_i^{(r)} &\sim \mathcal{N}(\mu_i, \sigma_i^2 + \nu^2), \\ \log(k_i^{2(r)}) &\sim \mathcal{N}(c_i, h^2).\end{aligned}$$

For a completely new player, we apply the prior

$$\begin{aligned}\theta_i^{(r)} &\sim \mathcal{N}(\theta_0, \lambda^2) \\ \log(k_i^{(r)2}) &\sim \mathcal{N}(k_0, \sigma_k^2).\end{aligned}$$

For simplicity, we will not distinguish the completely new players in our derivations later. They are simply special cases for parameter values c_i , h^2 and σ_i^2 .

This system requires knowledge of global parameters ν^2 , λ^2 and h^2 . One option is to fit a full Bayesian model using a computationally tractable data set and obtain posterior summaries for these parameters. Alternatively, in Appendix A.1, we propose a simple procedure to compute point estimates for these parameters. The procedure in Appendix A.1 requires observations from two rating periods.

Omitting index (r) on all parameters and observations and always conditioning on prior parameters $\mu_{1:N}$, $\sigma_{1:N}$, $c_{1:N}$, ν^2 , h^2 and λ^2 , we can write the posterior update for using observations from rating period r as

$$f(\boldsymbol{\theta}, \log(\mathbf{k}^2) | \mathbf{y}, \mathbf{n}) \propto \left[\prod_{i=1}^N \phi(\theta_i; \mu_i, \sigma_i^2 + \nu^2) \phi(\log(k_i^2); c_i, h^2) \right] \left[\prod_{j>i} p(y_{ij} | n_{ij}, \theta_i, \theta_j, k_i^2, k_j^2) \right].$$

The marginal posterior for parameters associated with player i is

$$\begin{aligned} f(\theta_i, \log(k_i^2) | \mathbf{y}, \mathbf{n}) &\propto \int \dots \int f(\boldsymbol{\theta}, \log(\mathbf{k}^2) | \mathbf{y}, \mathbf{n}) d\boldsymbol{\theta}_{-i} d\mathbf{k}_{-i} \\ &= \phi(\theta_i; \mu_i, \sigma_i^2 + \nu^2) \phi(\log(k_i^2); c_i, h^2) \\ &\quad \times \int \dots \int \left[\prod_{j=1, j \neq i}^N \phi(\theta_j; \mu_j, \sigma_j^2 + \nu^2) \phi(\log(k_j^2); c_j, h^2) \right] \\ &\quad \times \left[\prod_{j \neq i} p(y_{ij} | n_{ij}, \theta_i, \theta_j, k_i^2, k_j^2) \right] \left[\prod_{m>l, m \neq i} p(y_{lm} | n_{lm}, \theta_l, \theta_m, k_l^2, k_m^2) \right] d\boldsymbol{\theta}_{-i} d\mathbf{k}_{-i} \end{aligned}$$

where we use $d\mathbf{k}_{-i}$ and $d\boldsymbol{\theta}_{-i}$ as a shorthand for indicating the multiple integration is over k_j and θ_j for $j \neq i$ respectively. Following the approach employed by Glickman (1999) and Glickman (2001), we make two simplifications to the marginal posterior associated with player i .

1. Updating on θ_i and k_i^2 can be accomplished by integrating θ_j (for opponent j) over its prior distribution rather than integrating over the entire posterior. That is, when updating characteristics of player i , we discard information from opponents' games against each other during this rating period.
2. When updating θ_i and k_i^2 , we assume that $\log(k_j^2)$ is a constant at its expectation c_j .

Following these simplifications, the approximate posterior is given by

$$\begin{aligned}
& f(\theta_i, \log(k_i^2) | \mathbf{y}, \mathbf{n}) \\
& \stackrel{\text{approx}}{\propto} \phi(\theta_i; \mu_i, \sigma_i^2 + \nu^2) \phi(\log(k_i^2); c_i, h^2) \\
& \quad \times \int \dots \int \left[\prod_{j=1, j \neq i}^N \phi(\theta_j; \mu_j, \sigma_j^2 + \nu^2) \right] \left[\prod_{j \neq i} p(y_{ij} | n_{ij}, \theta_i, \theta_j, k_i^2, k_j^2) \right] d\boldsymbol{\theta}_{-i} \\
& = \phi(\theta_i; \mu_i, \sigma_i^2 + \nu^2) \phi(\log(k_i^2); c_i, h^2) \\
& \quad \times \prod_{j=1, j \neq i}^N \int p(y_{ij} | n_{ij}, \theta_i, \theta_j, k_i^2, k_j^2) \phi(\theta_j; \mu_j, \sigma_j^2 + \nu^2) d\theta_j.
\end{aligned}$$

Exchanging the order of integration we have

$$\begin{aligned}
& \int p(y_{ij} | n_{ij}, \theta_i, \theta_j, k_i^2, k_j^2) \phi(\theta_j; \mu_j, \sigma_j^2 + \nu^2) d\theta_j \\
& = \int \int p(y_{ij} | n_{ij}, \eta_{ij}) \phi(\eta_{ij} | \theta_i - \theta_j, k_i^2 + k_j^2) d\eta_{ij} \phi(\theta_j; \mu_j, \sigma_j^2 + \nu^2) d\theta_j \quad (2.3) \\
& = \int p(y_{ij} | n_{ij}, \eta_{ij}) \int \phi(\eta_{ij} | \theta_i - \theta_j, k_i^2 + k_j^2) \phi(\theta_j; \mu_j, \sigma_j^2 + \nu^2) d\theta_j d\eta_{ij}
\end{aligned}$$

Recognizing that with

$$\theta_j \sim \mathcal{N}(\mu_j, \sigma_j^2 + \nu^2) \text{ and } \eta_{ij} | \theta_i, \theta_j, k_i^2, k_j^2 \sim \mathcal{N}(\theta_i - \theta_j, k_i^2 + k_j^2),$$

we have

$$\eta_{ij}|\theta_i, k_i^2, k_j^2 \sim \mathcal{N}(\theta_i - \mu_j, \sigma_j^2 + \nu^2 + k_i^2 + k_j^2).$$

Letting $m_j = \theta_i - \mu_j$ and $\omega_j^2 = \sigma_j^2 + \nu^2 + k_i^2 + k_j^2$, Equation (2.3) can be written as

$$\begin{aligned} & \int p(y_{ij}|n_{ij}, \eta_{ij}) \int \phi(\eta_{ij}|\theta_i - \theta_j, k_i^2 + k_j^2) \phi(\theta_j; \mu_j, \sigma_j^2 + \nu^2) d\theta_j d\eta_{ij} \\ &= \int p(y_{ij}|n_{ij}, \eta_{ij}) \frac{1}{\sqrt{2\pi\omega_j^2}} \exp\left(-\frac{1}{2\omega_j^2}(\eta_{ij} - m_j)^2\right) d\eta_{ij}. \end{aligned}$$

Given $\mu_{1:N}, \sigma_{1:N}, c_{1:N}, \nu^2, h^2$ and λ^2 , we can update parameters θ_i and $\log(k_i^2)$ by numerically solving for the mode of the approximate log-posterior

$$\begin{aligned} & \log f(\theta_i, \log(k_i^2)|\mathbf{y}, \mathbf{n}) \\ & \approx -\frac{1}{2(\nu^2 + \sigma_i^2)} (\theta_i - \mu_i)^2 - \frac{1}{2h^2} (\log(k_i^2) - c_i)^2 \\ & \quad + \sum_{j \neq i} \log \int p(y_{ij}|n_{ij}, \eta_{ij}) \frac{1}{\sqrt{2\pi\omega_j^2}} \exp\left(-\frac{1}{2\omega_j^2}(\eta_{ij} - m_j)^2\right) d\eta_{ij}. \end{aligned} \tag{2.4}$$

This is a 2-dimension optimization problem that can be carried out in parallel for all players. Given the mode $\log(\hat{k}_i^2)$ and $\hat{\theta}_i$, we numerically solve for the second derivative of Equation (2.4) with respect to θ_i at $\log(\hat{k}_i^2)$ and $\hat{\theta}_i$ and can approximate the variance of $\hat{\theta}_i$ by the inverse of the second derivative.

We expect the target in Equation (2.4) to be well behaved in θ_i . Using approximations following Glickman (1999, 2001), given $\log(\hat{k}_i^2)$, Equation (2.4) can be approximated with log-density of a Normal distribution with one local maximum that is also a global maximum. This approximation and related approaches are discussed in detail in Section 2.5. However, such nice behavior may not be expected for $\log(k_i^2)$. Each of the likelihood term in Equation (2.3) is well behaved in $\omega_j^2 = \sigma_j^2 +$

$\nu^2 + k_j^2 + k_i^2$ where σ_j^2 , ν^2 and k_j^2 are all fixed. Hence, individual likelihood terms may push k_i^2 towards 0 when σ_j^2 , ν^2 and k_j^2 are large enough across all j . To overcome this, we constrain the range of $\log(k_i^2)$ over $[\log(0.0005), \log(5)]$. The upper bound is set at a high value such that when $\log(k_i^2)$ hits the upper bound, it reflects high level of instability for player i .

2.4 AN APPLICATION TO PROFESSIONAL GO DATA

In this section, we apply the DyS² rating system to game outcomes among professional Go players. In Section 2.5.2, we will compare the results in this section against results from approximations of the DyS² rating system.

Go, also referred to as “Weiqi” in Chinese and “Baduk” in Korean, is a two-player board game. Players use stones to capture territory on a square grid. The player who captures greater territory on the board wins. Go is a game with essentially a binary outcome. In games such as Chess, ties are common, whereas in Go, ties are rare in and can only result from certain idiosyncratic rules. According to the Japan Go Association, only about 0.01% of official games at the Japanese Go Association end in draws.

We retrieved game results from professional Go players online.¹ We examined data between calendar years 2000 and 2020, focusing on professional players who have played more than 200 games between 2000 and 2020. Using year as the rating period, our data include 32976 games from $N = 204$ players over $R = 21$ rating periods. Hyperparameters for the inverse-Gamma distributions on λ^2 , ν^2 and h^2 have all been set to 0.1, and we obtained point estimates for λ^2 , ν^2 and h^2 using the approach described in Appendix A.1.

¹<https://www.goratings.org>

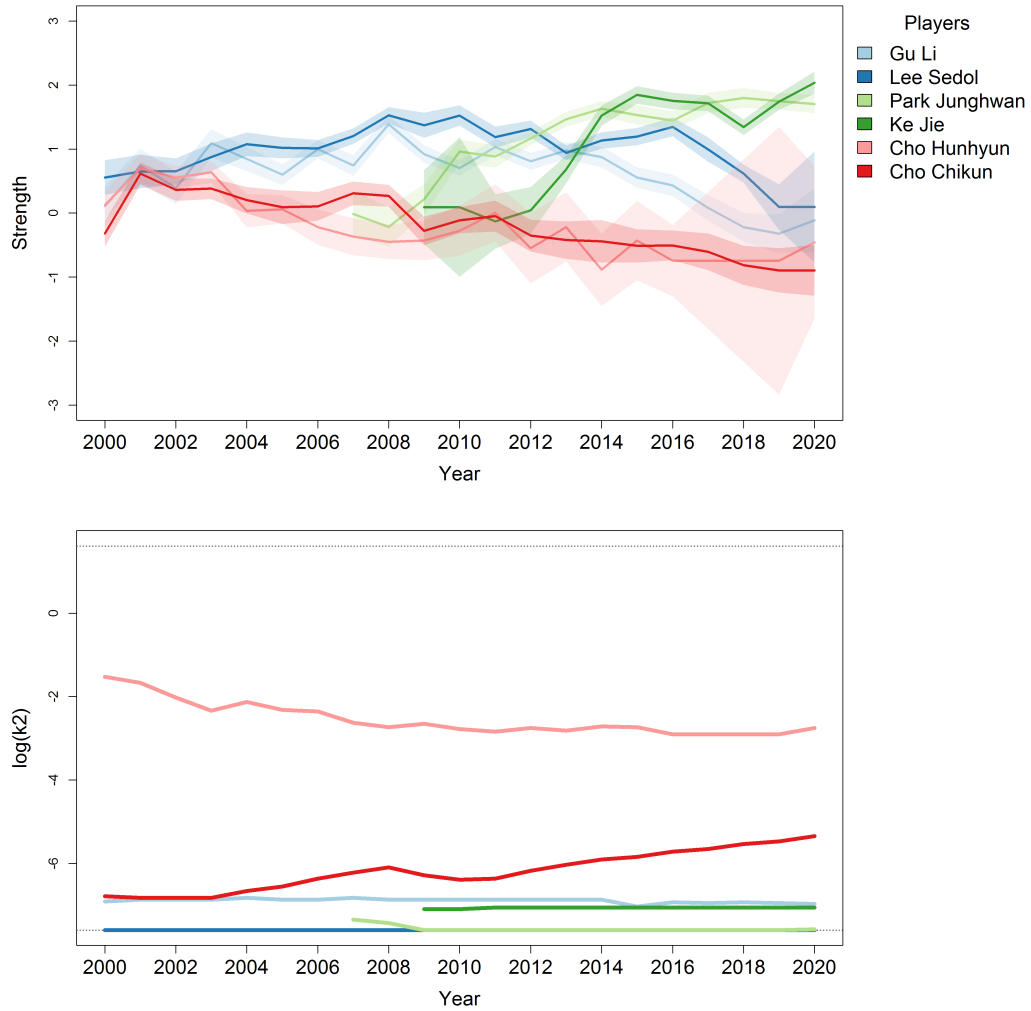


Figure 2.1: Estimated parameters of 3 pairs of contemporary professional Go players. The top panel plots the posterior mean of strength parameters of 3 pairs of contemporary professional Go players with approximate pointwise 95% posterior interval. The lower panel plots the posterior mean of stability parameters. Dotted lines are plotted at $\log(0.0005)$ and $\log(5)$, which are lower and upper bound placed on $\log(k_i^2)$ during optimization.

In Figure 2.1 we compare the estimated parameters from three pairs of contemporary professional Go players. Cho Hunhyun and Cho Chikun, born in 1953 and 1956, respectively, belong to an older generation of players. Lee Sedol and Gu Li were born in 1983. They are a pair of contemporaries who competed against each other frequently as finalists in top competitions in late 2000s. Park Junghwan

and Ke Jie, born in 1993 and 1997, respectively, are currently top players in the professional Go scene. From the top panel in Figure 2.1, we observe that strengths of Cho Hunhyun and Cho Chikun, who are past their peak, steadily declined over the period of observation. Strengths of Gu Li and Lee Sedol followed similar trajectories during the decade 2001 – 2010. Gu Li’s strength started to steadily decline around 2013, but the continual decline in Lee Sedol’s strength started later around 2016. Incidentally, the revolutionary match between AlphaGo and Lee Sedol occurred in 2016. We only start to observe younger players Park Junghwan and Ke Jie towards the late 2000s. We can observe their climb towards the top over the decade 2010 – 2020. Wide posterior intervals are partially reflective of low number of games played during the rating period. For example, Lee Sedol retired in November 2019. He did not participate in any professional games in 2020, hence, the posterior intervals widened substantially.

We examine the estimated stability parameters in the lower panel of Figure 2.1. Contemporaries Cho Hunhyun and Cho Chikun are on similar trajectories in terms of their strength parameters. However, their stability parameters differ in level and trend. Players Gu Li, Lee Sedol, Park Junghwan and Ke Jie all have estimated stability parameters over time that are close to 0. The estimated stability parameter of Gu Li and Ke Jie are marginally higher than that of Lee Sedol and Park Junghwan. We can see that stability parameters can be used to differentiate players on top of their strength parameters.

Figure 2.2 plots the distribution of player characteristics at the last rating period. The distribution of the strength parameter across all players is bell shaped. The distribution of stability parameter looks like a two-component mixture. A majority of the observed players have estimated stability parameters close to 0, and a small fraction of players have estimated $\log(k_i^2)$ above -4 . This is in fact true throughout the rating periods. This suggests that most professional players we observe perform at a consistent strength. In Figure 2.3, we illustrate examples of different trajectories we observe for the stability parameter of a selection of top Go players. We observe that the evolution trajectories of the stability parameter may adopt many different shapes.

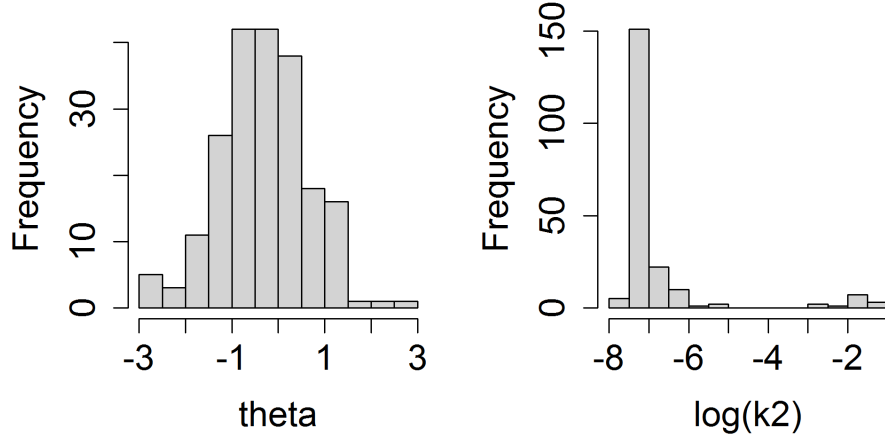


Figure 2.2: Histogram of estimated parameters of all players at the end of year 2020. Left panel is for strength parameter θ_i and right panel is stability parameter $\log(k_i^2)$.

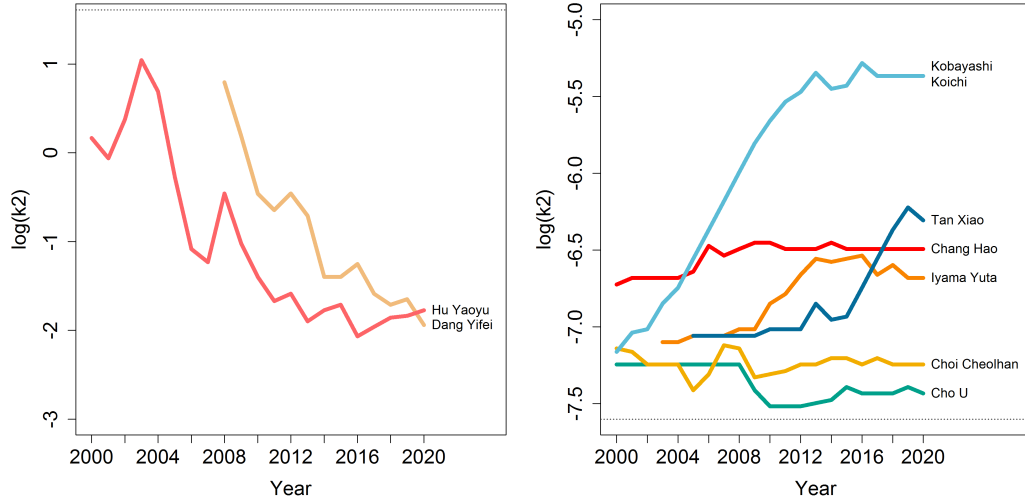


Figure 2.3: Estimated stability parameters of selected players. Two panels are plotted over different y -axis values. Dotted lines are plotted at $\log(5)$ on the right and $\log(0.0005)$ on the left, indicating upper and lower bound placed on $\log(k_i^2)$ during optimization respectively.

Inclusion of the stability parameter allows predicted pairwise probabilities to violate stochastic transitivity. For example, at the end of the 2003, player Hu Yaoyu is marginally better than player Cho Chikun, with predicted probability of winning at 0.51. Player Cho Chikun is stronger than

player Ma Xiaochun, with predicted probability of winning at 0.74. However, the predicted probability of winning for player Hu Yaoyu against player Ma Xiaochun is only 0.67. This set of predicted probabilities violates strong stochastic transitivity. Such a phenomenon would not have been possible under rating systems such as Elo, which is based on Bradley-Terry models.

2.5 APPROXIMATING THE BINOMIAL-NORMAL INTEGRAL

The DyS² rating system requires optimization of the target in Equation (2.4) for each player at every rating period. Although this is only a 2-dimension optimization problem, it requires computing integrals for every opponent player i faced during the rating period. Numerically approximating the integral demands computation effort. In the Glicko and Glicko-2 rating systems, [Glickman \(1999, 2001\)](#) bypassed computation of integrals by proposing an approximation for the integral term. In a similar spirit, we attempted to approximate the integrals in Equation (2.4) as well. Our analysis finds that we cannot completely replace the integral using a single approximation, but the approximation can help us to speed up part of the updating.

Ideally, we want to approximate the integral

$$\begin{aligned} & \int p(y_{ij}|n_{ij}, \eta_{ij}) \phi(\eta_{ij}|m_j, \omega_j^2) d\eta_{ij} \\ & \propto \int \frac{(\exp(\eta_{ij}))^{y_{ij}}}{(1 + \exp(\eta_{ij}))^{n_{ij}}} \frac{1}{\sqrt{2\pi\omega_j^2}} \exp\left(-\frac{1}{2\omega_j^2}(\eta_{ij} - m_j)^2\right) d\eta_{ij} \end{aligned} \quad (2.5)$$

using a function of θ_i and k_i^2 .

[Glickman \(1999, 2001\)](#) estimated an integral of this type using the following approximation

$$\int \frac{(\exp(\eta_{ij}))^{y_{ij}}}{(1 + \exp(\eta_{ij}))^{n_{ij}}} \frac{1}{\sqrt{2\pi\omega_j^2}} \exp\left(-\frac{1}{2\omega_j^2}(\eta_{ij} - m_j)^2\right) d\eta_{ij}$$

$$\begin{aligned}
&= \int \prod_{s=1}^{n_{ij}} \frac{(\exp(\eta_{ij}))^{y_{ijs}}}{1 + \exp(\eta_{ij})} \frac{1}{\sqrt{2\pi\omega_j^2}} \exp\left(-\frac{1}{2\omega_j^2}(\eta_{ij} - m_j)^2\right) d\eta_{ij} \\
&\approx \prod_{s=1}^{n_{ij}} \int \frac{(\exp(\eta_{ij}))^{y_{ijs}}}{1 + \exp(\eta_{ij})} \frac{1}{\sqrt{2\pi\omega_j^2}} \exp\left(-\frac{1}{2\omega_j^2}(\eta_{ij} - m_j)^2\right) d\eta_{ij}
\end{aligned}$$

where y_{ijs} denotes the outcome of the s -th game between player i and player j with $\sum_{s=1}^{n_{ij}} y_{ijs} = y_{ij}$.

The approximation step can be described as

$$\mathbb{E} \left[\prod_{s=1}^{n_{ij}} \frac{(\exp(X))^{y_{ijs}}}{1 + \exp(X)} \right] \approx \prod_{s=1}^{n_{ij}} \mathbb{E} \left[\frac{(\exp(X))^{y_{ijs}}}{1 + \exp(X)} \right]$$

with $X \sim \mathcal{N}(m_j, \omega_j^2)$. The motivation behind the approximation is imagining a process where instead of competing at a constant mean strength difference η_{ij} throughout this period, each game between players i and j is played at a strength which is a fresh draw from distribution of η_{ij} .

[Crooks \(2009\)](#) gives an approximation for the logistic normal integral

$$\begin{aligned}
b(m, \omega) &= \int_{-\infty}^{\infty} \frac{1}{1 + e^{-y}} \frac{1}{\sqrt{2\pi\omega^2}} \exp\left(-\frac{1}{2\omega^2}(y - m)^2\right) dy \\
&\approx \frac{1}{1 + e^{-\frac{m}{g(\omega)}}}, \quad g(\omega) = \sqrt{1 + \frac{\pi\omega^2}{8}}.
\end{aligned}$$

Using this approximation, the Glicko approach approximates Equation (2.5) as

$$\begin{aligned}
&\prod_{s=1}^{n_{ij}} \int \frac{(\exp(\eta_{ij}))^{y_{ijs}}}{1 + \exp(\eta_{ij})} \frac{1}{\sqrt{2\pi\omega_j^2}} \exp\left(-\frac{1}{2\omega_j^2}(\eta_{ij} - m_j)^2\right) d\eta_{ij} \\
&\approx \prod_{s=1}^{n_{ij}} \frac{(e^{m_j/g(\omega_j)})^{y_{ijs}}}{1 + e^{m_j/g(\omega_j)}} = \frac{(e^{m_j/g(\omega_j)})^{y_{ij}}}{(1 + e^{m_j/g(\omega_j)})^{n_{ij}}}
\end{aligned}$$

with $g(\omega_j) = \sqrt{1 + \frac{\pi\omega_j^2}{8}}$.

We also investigate an alternative approximation for the integral in Equation (2.5) using a Taylor approximation. Re-writing Equation (2.5) we have

$$\begin{aligned} & \int p(y_{ij}|n_{ij}, \eta_{ij}) \phi(\eta_{ij}|m_j, \omega_j^2) d\eta_{ij} \\ & \propto \int \frac{1}{\sqrt{2\pi\omega_j^2}} \exp\left(-\frac{1}{2\omega_j^2}(\eta_{ij} - m_j)^2 + y_{ij}\eta_{ij} - n_{ij} \log(1 + e^{\eta_{ij}})\right) d\eta_{ij} \end{aligned} \quad (2.6)$$

Define the function

$$s(\eta) = -\frac{1}{2\omega^2}(\eta - m)^2 + y\eta - n \log(1 + e^\eta) \quad (2.7)$$

with $s'(\eta) = -\frac{1}{\omega^2}(\eta - m) + y - n \frac{e^\eta}{1+e^\eta}$ and $s''(\eta) = -\left(\frac{1}{\omega^2} + n \frac{1}{1+e^\eta} \frac{e^\eta}{1+e^\eta}\right)$. We consider a second-order Taylor expansion around $\eta = m$ given by

$$s(\eta) \approx s(m) + s'(m)(\eta - m) + \frac{1}{2}s''(m)(\eta - m)^2.$$

Completing the square, we have

$$\begin{aligned} s(\eta) & \approx s(m) + s'(m)(\eta - m) + \frac{1}{2}s''(m)(\eta - m)^2 \\ & = -\frac{1}{2}|s''(m)| \left(\left(\eta - m - \frac{s'(m)}{|s''(m)|} \right)^2 \right) + \frac{1}{2}|s''(m)| \left(\frac{s'(m)}{|s''(m)|} \right)^2 + s(m). \end{aligned}$$

Using the approximation based on this Taylor expansion round m_j , the integral in Equation (2.5) can be approximated by

$$\int \frac{1}{\sqrt{2\pi\omega_j^2}} e^{-\frac{1}{2\omega_j^2}(\eta_{ij}-m_j)^2 + y_{ij}\eta_{ij} - n_{ij} \log(1+e^{\eta_{ij}})} d\eta_{ij}$$

$$\begin{aligned}
&\approx \int \frac{1}{\sqrt{2\pi\omega_j^2}} e^{-\frac{1}{2}|s''(m_j)| \left(\left(\eta - m_j - \frac{s'(m_j)}{|s''(m_j)|} \right)^2 \right) + \frac{1}{2} \frac{(s'(m_j))^2}{|s''(m_j)|} + s(m_j)} d\eta_{ij} \\
&= (\omega_j^2 |s''(m_j)|)^{-0.5} \exp \left(\frac{1}{2} \frac{(s'(m_j))^2}{|s''(m_j)|} + s(m_j) \right)
\end{aligned}$$

with $s(m_j) = y_{ij}m_j - n_{ij} \log(1+e^{m_j})$, $s'(m_j) = y_{ij} - n_{ij} \frac{e^{m_j}}{1+e^{m_j}}$ and $|s''(m_j)| = \left(\frac{1}{\omega_j^2} + n_{ij} \frac{1}{1+e^{m_j}} \frac{e^{m_j}}{1+e^{m_j}} \right)$.

We first numerically compare the two approaches in estimating the integral given by Equation (2.5).

Define

$$\begin{aligned}
\mathcal{I}(n, y, m, \omega) &= \int \frac{(e^\eta)^y}{(1+e^\eta)^n} \frac{1}{\sqrt{2\pi\omega^2}} \exp \left(-\frac{1}{2\omega^2} (\eta - m)^2 \right) d\eta, \\
\tilde{\mathcal{I}}_{\text{Glicko}}(n, y, m, \omega) &= \frac{(e^{m/g(\omega)})^y}{(1+e^{m/g(\omega)})^n}, \\
\tilde{\mathcal{I}}_{\text{Taylor,m}}(n, y, m, \omega) &= (\omega^2 |s''(m)|)^{-0.5} \exp \left(\frac{1}{2} \frac{(s'(m))^2}{|s''(m)|} + s(m) \right),
\end{aligned}$$

with $g(\omega) = \sqrt{1 + \frac{\pi\omega^2}{8}}$ and $s(m)$ given by Equation (2.7). We compare the approximation of $\mathcal{I}(n, y, m, \omega)$ over a grid of parameter values given by

- $\log_2 \omega$ equally spaced between -5 and 1, with an increment of 0.5.
- m equally spaced between -3 and 3, with an increment of 0.5.
- n equally spaced between 1 and 10, with an increment of 1.
- Given n, y takes all integer values between 0 and n .

The target integral is approximated through numerical integration. We compute the error quantities

$$\begin{aligned}
\text{Absolute Error}(n, y, m, \omega) &= |\tilde{\mathcal{I}}(n, y, m, \omega) - \mathcal{I}(n, y, m, \omega)|, \\
\text{Absolute Relative Error}(n, y, m, \omega) &= \frac{|\tilde{\mathcal{I}}(n, y, m, \omega) - \mathcal{I}(n, y, m, \omega)|}{\mathcal{I}(n, y, m, \omega)}
\end{aligned}$$

for each of the approximation methods. The error between two methods are plotted in Figure 2.4 on the log scale.

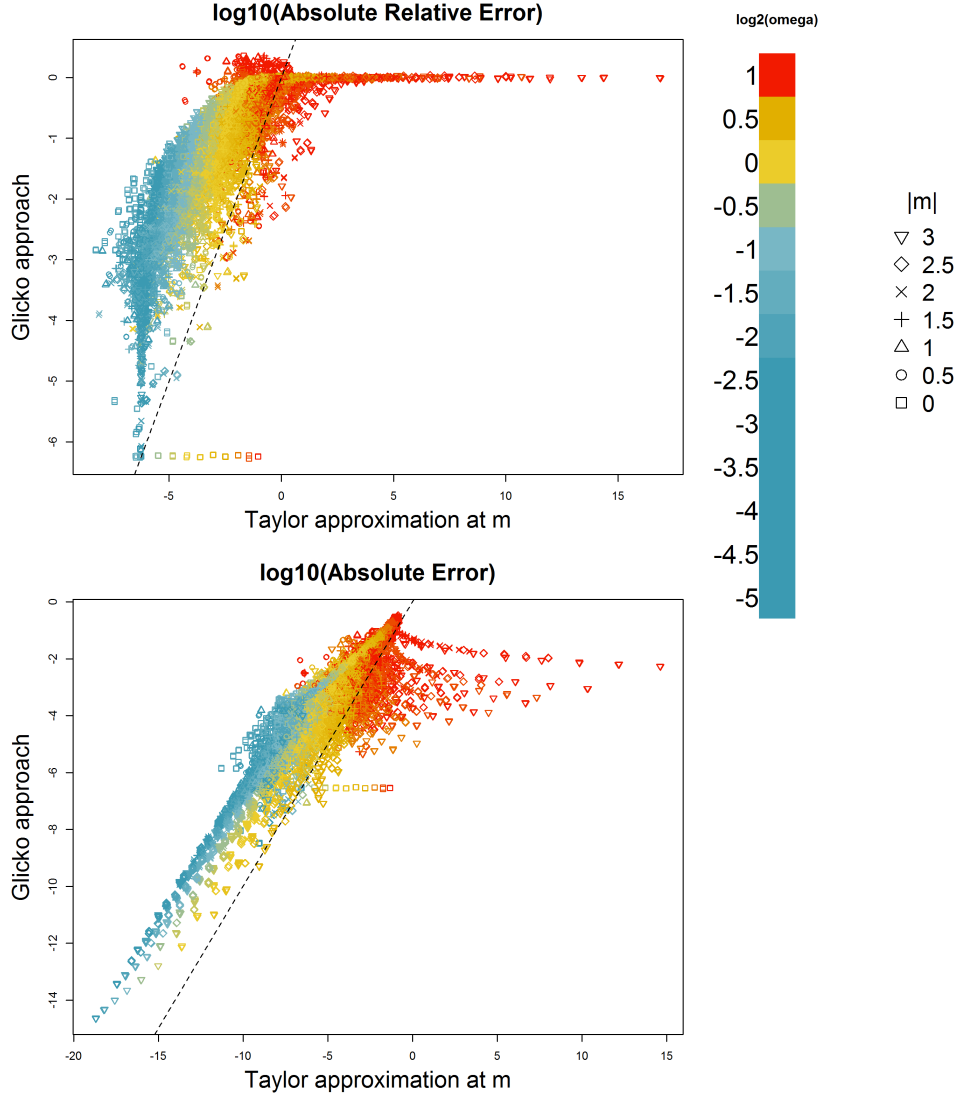


Figure 2.4: Absolute error and absolute relative error for approximating the integral $\mathcal{I}(n, y, m, \omega)$ using the Glicko approach and a Taylor expansion around m in $\log_{10}$ scale. Small jitters have been applied to both axis so that overlapping points can be distinguished. The dotted line is the reference line $y = x$.

Between the two approximations, we find that for most of the parameter values used in the nu-

merical analysis, the Taylor approximation outperforms the Glicko approximation in both absolute error and relative absolute error. However, when ω is high, magnitude of error in Taylor approximation approach can be extremely high. We can also check that $\mathcal{I}(n, y, m, \omega)$ and $\tilde{\mathcal{I}}_{\text{Glicko}}(n, y, m, \omega)$ are both bounded between 0 and 1 but $\tilde{\mathcal{I}}_{\text{Taylor},m}(n, y, m, \omega) \geq 0$. More importantly, we note that the error in approximation depends on ω . This raises a red flag since we are actually interested in optimizing the integral over k_i^2 , which contributes to ω .

Next, we want to know how well $\tilde{\mathcal{I}}(n, y, m, \omega)$ approximates $\mathcal{I}(n, y, m, \omega)$ as a function of ω and m . Given n^*, y^* and m^* , the ideal situation is that the functions $\tilde{\mathcal{I}}(\omega) = \tilde{\mathcal{I}}(n^*, y^*, m^*, \omega)$ and $\mathcal{I}(\omega) = \mathcal{I}(n^*, y^*, m^*, \omega)$ are maximized at the same ω value, and the same can be said for m when we fix n, y and ω . As a first step, we check whether $\tilde{\mathcal{I}}(\omega)$ preserves the ordering of $\mathcal{I}(\omega)$ over ω , and repeat this for m . We use Spearman's correlation coefficient to measure how well $\tilde{\mathcal{I}}(\omega)$ can be represented by a monotonic transformation of $\mathcal{I}(\omega)$.

The lower panels on Figure 2.5 examine the extent to which $\tilde{\mathcal{I}}(m) = \tilde{\mathcal{I}}(n^*, y^*, m, \omega^*)$ is monotonically related to $\mathcal{I}(m) = \mathcal{I}(n^*, y^*, m, \omega^*)$. The lower left panel shows that given data and ω , the Glicko approximation of $\mathcal{I}(m)$ is strongly correlated with $\mathcal{I}(m)$ over different values of m . Over the entire range of parameter values we considered, the Spearman correlation is observed not to fall below 0.85, with the correlation being exactly 1 over more than 70% of the sets of parameter values we considered. This is evidence that $\tilde{\mathcal{I}}_{\text{Glicko}}(m)$ has a monotonic relationship with $\mathcal{I}(m)$. This is consistent with the observation that this approximation approach has worked well for Glicko and Glicko-2 rating systems. For both systems, the methodology has been used to approximate a similar integral as a function of m . On the other hand, the lower right panel shows that the correlation between $\tilde{\mathcal{I}}_{\text{Taylor},m}(m)$ and $\mathcal{I}(m)$ can fall anywhere over the range of -1 to 1 , depending on the data and parameter ω . Hence, even though Figure 2.4 reports that the Taylor approximation can sometimes outperform the Glicko approach in terms of approximation error, the Taylor approximation is not appropriate for approximating the integral as a function of m .

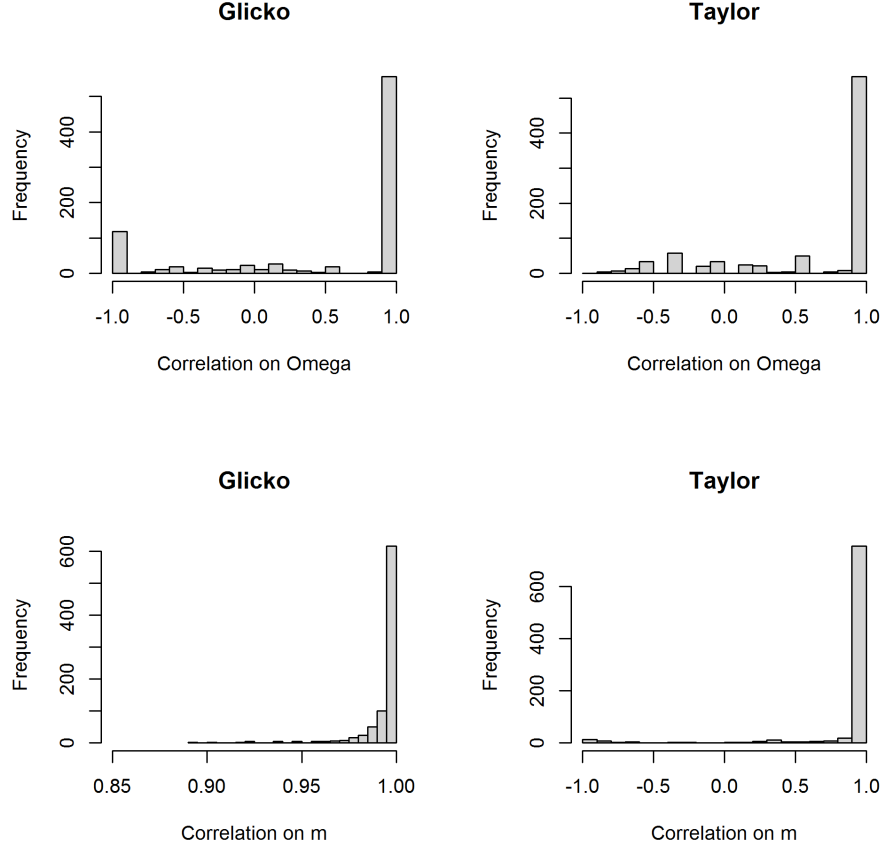


Figure 2.5: Upper panels display Spearman's correlation between $\mathcal{I}(n^*, y^*, m^*, \omega)$ and $\tilde{\mathcal{I}}(n^*, y^*, m^*, \omega)$ over ω . Lower panels display Spearman's correlation between $\mathcal{I}(n^*, y^*, m, \omega^*)$ and $\tilde{\mathcal{I}}(n^*, y^*, m, \omega^*)$ over m . Left panels are for Glicko approach to approximating the integral, right panels are for Taylor approximation around m . Note that the range of x -axis is different for the lower left panel so that details of the histogram can be viewed.

The upper panels on Figure 2.5 examine whether $\tilde{\mathcal{I}}(\omega)$ is monotonically related to $\mathcal{I}(\omega)$. The results show that the direction of relationship can deviate from a correlation of 1. In fact, the Spearman correlation is below 0 for 25% and 20% of parameter values we considered for the Glicko and Taylor approximation approaches respectively. This implies that $\tilde{\mathcal{I}}(\omega)$ and $\mathcal{I}(\omega)$ may achieve a maximum at different values of ω . Hence, we find that although DyS² are developed following very similar methods

as Glicko and Glicko-2, the approximation strategy employed by Glickman (1999, 2001) cannot be adapted to completely replace the integral in Equation (2.4) using some function of θ_i and k_i^2 . Instead, we use the observation that $\tilde{\mathcal{I}}_{\text{Glicko}}(m)$ approximates $\mathcal{I}(m)$ well as a function of m to speed up part of the updating process, as we discuss below.

2.5.1 INTRODUCING APPROXIMATION INTO THE RATING SYSTEM

We can re-frame the optimization of Equation (2.4) as a 1-dimension optimization problem on the function

$$\begin{aligned} & \log f(\log(k_i^2) | \mathbf{y}, \mathbf{n}) \\ & \approx -\frac{1}{2(\nu^2 + \sigma_i^2)} \left(\hat{\theta}_i(k_i^2) - \mu_i \right)^2 - \frac{1}{2h^2} (\log(k_i^2) - c_i)^2 \\ & + \sum_{j \neq i} \log \int p(y_{ij} | n_{ij}, \eta_{ij}) \frac{1}{\sqrt{2\pi\omega_j^2}} \exp \left(-\frac{1}{2\omega_j^2} (\eta_{ij} - (\hat{\theta}_i(k_i^2) - \mu_j))^2 \right) d\eta_{ij}. \end{aligned} \quad (2.8)$$

with

$$\hat{\theta}_i(k_i^2) = \operatorname{argmax}_{\theta_i} \log f(\theta_i | \log(k_i^2), \mathbf{y}, \mathbf{n}). \quad (2.9)$$

The conditional log posterior $\log f(\theta_i | \log(k_i^2), \mathbf{y}, \mathbf{n})$ is given by

$$\begin{aligned} & \log f(\theta_i | \log(k_i^2), \mathbf{y}, \mathbf{n}) \\ & = -\frac{1}{2(\nu^2 + \sigma_i^2)} (\theta_i - \mu_i)^2 + \sum_{j \neq i} \log \int p(y_{ij} | n_{ij}, \eta_{ij}) \frac{1}{\sqrt{2\pi\omega_j^2}} e^{\left(-\frac{1}{2\omega_j^2} (\eta_{ij} - m_j)^2 \right)} d\eta_{ij} \end{aligned} \quad (2.10)$$

up to some additive constant. Conditioning on $\log(k_i^2)$, we can approximate the integral

$$\int p(y_{ij}|n_{ij}, \eta_{ij}) \frac{1}{\sqrt{2\pi\omega_j^2}} e^{\left(-\frac{1}{2\omega_j^2}(\eta_{ij}-m_j)^2\right)} d\eta_{ij}$$

as a function of θ_i by adopting the methodology used by Glickman (1999, 2001). This gives us an approximation of Equation (2.10) given by

$$\begin{aligned} & \log \tilde{f}(\theta_i | \log(k_i^2), \mathbf{y}, \mathbf{n}) \\ &= -\frac{1}{2(\nu^2 + \sigma_i^2)} (\theta_i - \mu_i)^2 + \sum_{j \neq i} \sum_{j \neq i} y_{ij} \frac{\theta_i - \mu_j}{g(\omega_j)} - n_{ij} \log \left(1 + e^{\frac{\theta_i - \mu_j}{g(\omega_j)}} \right). \end{aligned} \quad (2.11)$$

We can approximate $\hat{\theta}_i(k_i^2)$ defined in Equation (2.9) with

$$\tilde{\theta}_i(k_i^2) = \underset{\theta_i}{\operatorname{argmax}} \log \tilde{f}(\theta_i | \log(k_i^2), \mathbf{y}, \mathbf{n}). \quad (2.12)$$

This approximation removes the need to perform numerical integration when searching for $\hat{\theta}_i(k_i^2)$.

In Appendix A.2 we show that

$$\tilde{\theta}_i(k_i^2) \approx \mu_i + \frac{a'(\mu_i)}{1/(\sigma_i^2 + \nu^2) - a''(\mu_i)} \quad (2.13)$$

$$\text{with } a'(\mu_i) = \sum_{j \neq i} \frac{y_{ij}}{g(\omega_j)} - \frac{n_{ij}}{g(\omega_j)} \frac{e^{\frac{\mu_i - \mu_j}{g(\omega_j)}}}{1 + e^{\frac{\mu_i - \mu_j}{g(\omega_j)}}} \text{ and } a''(\mu_i) = - \sum_{j \neq i} \frac{n_{ij}}{g(\omega_j)^2} \frac{1}{1 + e^{\frac{\mu_i - \mu_j}{g(\omega_j)}}} \frac{e^{\frac{\mu_i - \mu_j}{g(\omega_j)}}}{1 + e^{\frac{\mu_i - \mu_j}{g(\omega_j)}}}.$$

2.5.2 ERROR IN APPROXIMATION ON GO DATA

We embed the approximation steps into the updating step of the rating system and apply the different versions to the Go data used in Section 2.4. We compare three versions of the rating system:

1. The “exact algorithm” refers to optimizing Equation (2.8) with $\hat{\theta}_i(k_i^2)$ as defined by Equation (2.9).
2. The GlickoApprox algorithm refers to optimizing Equation (2.8) with $\hat{\theta}_i(k_i^2)$ approximated by the quantity defined in Equation (2.13).
3. The GlickoApproxWithNR algorithm refers to optimizing Equation (2.8) with $\hat{\theta}_i(k_i^2)$ approximated by $\tilde{\theta}_i(k_i^2)$ as defined in Equation (2.12). The quantity $\tilde{\theta}_i(k_i^2)$ is obtained using a Newton-Raphson algorithm initialized at the quantity defined in Equation (2.13).

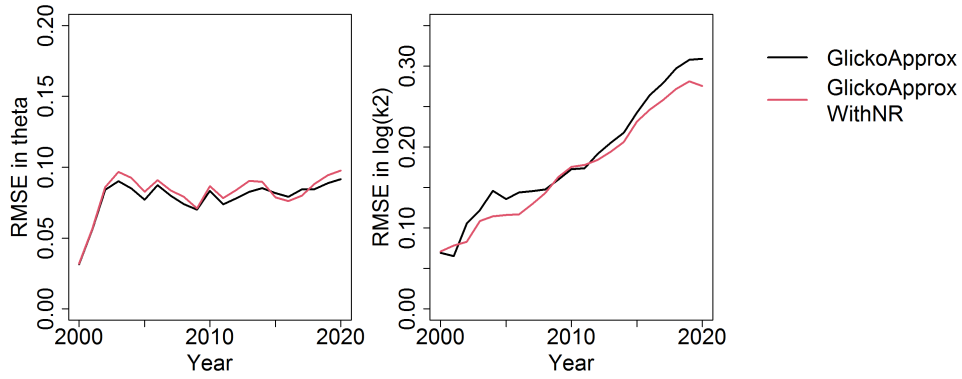


Figure 2.6: Root mean squared error in estimated parameters between exact algorithm and algorithms with approximation at each rating period. Left panel is for θ_i , and right panel for $\log(k_i^2)$.

Figure 2.6 reports the root mean squared error of the estimated parameters between the exact algorithm and the two versions of the algorithm with approximations embedded. Figure 2.7 reports the Spearman’s correlation of estimated parameters between the exact algorithm and the approximate algorithms. From Figure 2.6, we observe that the error in estimation of θ_i stabilized around 0.1, while the error in estimation of $\log(k_i^2)$ is increasing over time. From Figure 2.7, we observe that the correlations in estimated θ_i between the exact and approximate algorithms are consistently close to 1. The correlations in k_i^2 decrease over time but are still above 0.8 at the end of periods of observation. In-

roducing a Newton-Raphson step for the approximation marginally improves the approximation by giving smaller error and better correlations in estimated $\log(k_i^2)$.

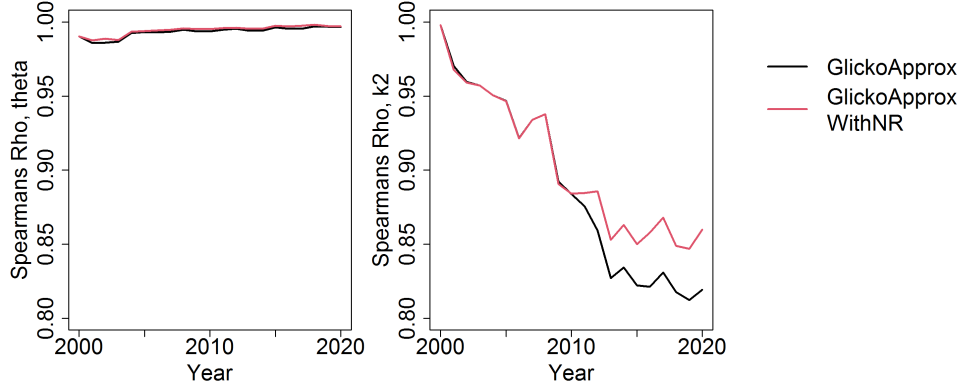


Figure 2.7: Spearman's correlation for estimated parameters between exact algorithm and algorithms with approximation at each rating period. Left panel is for θ_i and right panel for $\log(k_i^2)$.

Figure 2.8 plots the estimated $\log(k_i^{2(r)})$ between exact and approximate algorithms at the beginning, middle and end of competition period. We find that although there are errors in estimation of $\log(k_i^{2(r)})$, players with $\log(k_i^{2(r)}) > -4$ are always identified correctly. Hence, we find that the results from using the approximate algorithms are acceptable.

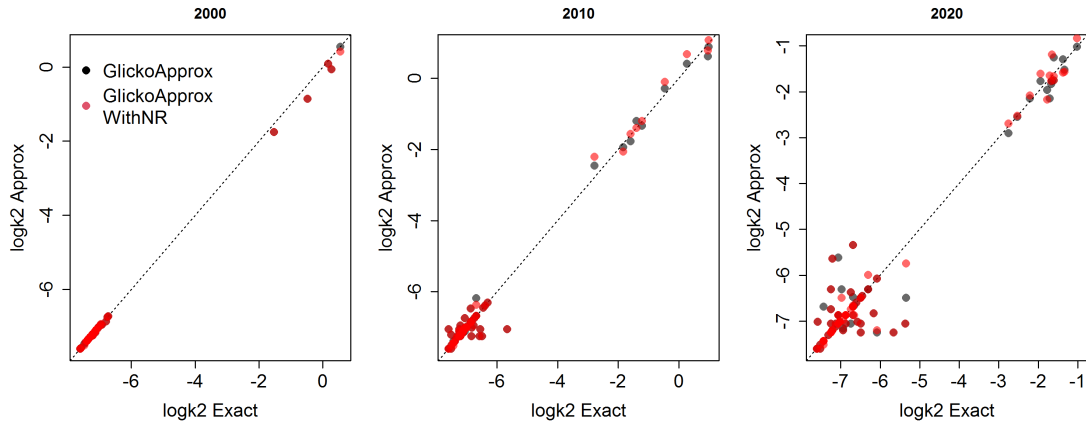


Figure 2.8: Scatter plot for estimated $\log(k_i^{2(r)})$ between exact algorithm and algorithms with approximation at $t_r = 2000, 2010$ and 2020 from left to right. Reference line is $y = x$.

2.6 CONCLUSION

The DyS² rating system can be seen as an extension of the Glicko rating system with an added stability parameter for each player. Both the Glicko-2 and DyS² rating systems have a time changing player-specific parameter associated with variability. In Glicko-2, it is the quantification of uncertainty associated with estimation of player strength. In DyS², it is the quantification of players' stability when facing different opponents. The DyS² rating system is built on a model that explicitly accounts for variability between the player-pair-specific log-odds, $\text{logit}(p_{ij})$, and the average strength differential between player i and player j , $\theta_i - \theta_j$.

Suppose there is a strong inconsistency between the observed outcomes between player i and j and $\theta_i - \theta_j$. This inconsistency will be interpreted by the Glicko-2 system as “unreliable” paired comparisons and the data will be reflected through the model as an increase in uncertainty of θ_i . On the other hand, DyS² captures this inconsistency as a characteristic of player i rather than through the uncertainty in inferences. The two rating systems offer different interpretations of the same observed information because of differences in the model specifications.

More importantly, the Glicko-2 system is built upon the Bradley-Terry framework. The fundamental assumption is that player i displays the same average strength θ_i when facing different opponents. Drawing from the predictive distribution of Glicko-2 will never result in a set of pairwise probabilities that violates stochastic transitivity. On the other hand, Model DyS² assumes that player i can display different strengths against different opponents. Hence, drawing from the predictive distribution of DyS² can result in player triplets that violate stochastic transitivity.

3

The Analysis of Rank Order Data using Paired Comparison Methods

3.1 INTRODUCTION

Given a collection of objects of interest, rank order data arise when we observe a preference ordering. For example, in multi-competitor sports such as swimming and marathons, the outcome of the competition can be reduced to an ordering of competing athletes from best to worst based on their performance. In a study of preferences, consumers may be asked to rank different items within a single broad category. In an election within an association, members can be asked to rank all candidates in order of preference. Given a data set of rank orderings, a typical question is how to assess the probabilities of new rank orders, or the probability of one object being preferred to another.

A common model used for analyzing rank order data is the Plackett-Luce model (Plackett, 1975; Luce, 1959). This model has wide applications in sports analytics. However, we notice that not all practitioners in the field of sports analytic are familiar with the Plackett-Luce model. Many of those who work in this field are much more familiar and comfortable with paired comparison models such as the Bradley-Terry model (Bradley and Terry, 1952). We observe instances where these practitioners will choose to convert the rank order data into a set of paired comparisons and fall back to analyzing the data using the Bradley-Terry model. Intuitively, this does not seem completely wrong but we are not aware of any formal discussion of this approach. This discrepancy has provided the motivation to explore this knowing wrong approach and see if we can help practitioners better appreciate what they are doing and how well they are doing using a “wrong” model.

Rank order data can contain complete rankings, subset rankings or top c rankings. Complete ranking gives an ordering of every item of interest. Each observation under subset rankings gives an ordering on only a subset of the items of interest. Observations from top c rankings give explicit ranking on c of the items, but it is assumed that these c items are preferred to the non-ranked items. For this chapter, we will focus only on rank order data from complete ranking. We will assume ties do not occur in rank orderings in our discussion.

3.1.1 OVERVIEW OF MODELS FOR RANK ORDER DATA

There are several broad classes of models for rank order data. Detailed overviews of these models can be found in works such as [Critchlow et al. \(1991\)](#) and [Yu et al. \(2019\)](#). Consider the complete ranking of $n + 1$ items indexed $0, \dots, n$, and let A represent a permutation of the $n + 1$ items, with $A = (a(1), \dots, a(n + 1))'$ where $a(k)$ gives the index of the item in k th position. One class of models is distance-based models characterized by a modal ranking and dispersion parameter. This is often referred to as the Mallows model ([Mallows, 1957](#); [Diaconis, 1988](#)). Under this model, the distribution of A is given by

$$\mathbb{P}(A) \propto \exp(-\xi d(A, A_0)) \quad (3.1)$$

where $\xi \geq 0$ is the dispersion parameter of the model. The ranking A_0 is known as the modal ranking of the model and $d(\cdot, \cdot)$ is a metric on the set of permutation of $n + 1$ items with properties such as positivity, symmetry and right-invariance ([Critchlow et al., 1991](#)). Given two rankings of the same $n + 1$ items, we refer to a pair of items as concordant if both rankings give the same ordinal relationship between the items, and discordant if the ordinal relationship between the items differ across the two rankings. The Mallows model with Kendall's τ metric has $d(A, A_0)$ to be the number of discordant pairs between rankings A and A_0 .

Another class of models based on ordered statistics can be formulated using a latent variable approach, following [Thurstone \(1927\)](#). Let W_i be the latent variable associated with item i , where $W_0, W_1, \dots, W_n$ follow some specified distribution with location parameters $\theta_0, \theta_1, \dots, \theta_n$ respectively. Under this class of models, the distribution of A is given by

$$\mathbb{P}(A) = \mathbb{P}(W_{a(1)} > W_{a(2)} > \dots > W_{a(n+1)})$$

The model with W_i following a Normal distribution is referred to as the Thurstone-Mosteller-Daniels model (Daniels, 1950). Stern (1990) defined a class of gamma ranking models with $W_i = -\log V_i$ with $V_i \sim \text{Gamma}(r, e^{\theta_i})$ where r and e^{θ_i} are the shape and scale parameter of the gamma distribution respectively. When $r = 1$, W_i follows a Gumbel distribution with location parameter θ_i . This corresponds to the classic Plackett-Luce model (Luce, 1959; Plackett, 1975) which we discuss in more detail below.

3.1.2 THE PLACKETT-LUCE MODEL

The Plackett-Luce model is widely used in analysis of rank order data. For example, van Etten et al. (2019) used this model to analyze farmers' preference for crop variety. They used the results to demonstrate how crowdsourcing to citizen scientists can be used to generate recommendations for adapting crop variety to climate conditions. Non-parametric extensions of the Plackett-Luce model have been used to analyze data on New York Times bestsellers (Caron and Teh, 2012) and preferences for college degree programmes amongst Irish middle school graduates (Caron et al., 2014). Variants of this model have been used to analyze results of multi-competitor sports such as golf (Baker and McHale, 2015) and Formula One (Henderson et al., 2018). Glickman and Hennessy (2015) developed a rating system for multi-competitor sports based on this model. More recently, Plackett-Luce models have gained popularity in the field of Learning-to-Rank algorithms. Singh and Joachims (2019) and Diaz et al. (2020) demonstrated that Plackett-Luce models are useful in creating fair ranking policies, where fairness refers to the exposure of an item should be proportional to its merit.

We interchangeably use θ_i and $\gamma_i = \exp(\theta_i)$ in our notation. We can interpret θ_i and γ_i as the worth of item i in the context of consumer preference, and interpret them as the strength of player i in the context of multi-competitor sports. Using the latent variable formulation, the likelihood of

observing a particular ranking A is given by

$$\begin{aligned} L_{PL}(\boldsymbol{\theta}) &= \mathbb{P}(W_{a(1)} > W_{a(2)} > \cdots > W_{a(n+1)}) \\ &= \frac{\gamma_{a(1)}}{\sum_{j=1}^{n+1} \gamma_{a(j)}} \frac{\gamma_{a(2)}}{\sum_{j=2}^{n+1} \gamma_{a(j)}} \cdots \frac{\gamma_{a(n)}}{\sum_{j=n}^{n+1} \gamma_{a(j)}} = \prod_{k=1}^n \frac{\gamma_{a(k)}}{\sum_{j=k}^{n+1} \gamma_{a(j)}}. \end{aligned} \quad (3.2)$$

Equivalently, the log-likelihood is

$$\ell_{PL}(\boldsymbol{\theta}) = \sum_{k=1}^n \log \left(\frac{\gamma_{a(k)}}{\sum_{j=k}^{n+1} \gamma_{a(j)}} \right)$$

The parameter $\boldsymbol{\theta}$ is not uniquely identified. If we shift $\boldsymbol{\theta}$ by a constant C , the likelihood is unaffected. Hence, some constraint on $\boldsymbol{\theta}$ is required for estimation. A convenient choice is $\theta_0 = 0$, using an arbitrary item 0 as the reference group. As pointed out by [Hunter \(2004\)](#), there is a one-to-one correspondence between the constraint $\theta_0 = 0$ and $\sum_{i=0}^n \gamma_i = 1$.

Frequentist inference using the Plackett-Luce model can be done using maximum likelihood estimation. We use

$$\hat{\boldsymbol{\theta}}_{PL} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \ell_{PL}(\boldsymbol{\theta})$$

to denote the maximum likelihood estimator (MLE) of $\boldsymbol{\theta}$ under the Plackett-Luce model. By choice of constraint, $\hat{\theta}_0 = 0$. The estimand of $\hat{\theta}_i$ is in fact $\theta_i - \theta_0$. We set $\theta_0 = 0$ as the default so there is direct correspondence between $\hat{\theta}_i$ and θ_i . If in every possible partition of the items into two nonempty subsets, some item in the second set is ranked higher than some item in the first set at least once, then there exists a unique MLE $\hat{\boldsymbol{\theta}}_{PL}$ ([Hunter, 2004](#); [Turner et al., 2020](#)). This estimate $\hat{\boldsymbol{\theta}}_{PL}$ is determined by optimizing the log-likelihood function. Other than generic optimization techniques such as the Newton-Raphson algorithm or the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm, [Hunter \(2004\)](#) developed a minorization-maximization (MM) algorithm for Plackett-Luce models. More recently, [Maystre and Grossglauser \(2015\)](#) shown that the MLE of the Plackett-Luce model can

be expressed as the stationary distribution of a Markov chain and developed relevant algorithms for estimating the MLE. Several packages are also available in R to provide parameter estimation on rank order data using the Plackett-Luce model (Turner et al., 2020).

The first part of this chapter focuses on providing insights into the analysis of rank order data as paired comparisons. In particular, we assume the setup where the practitioner has access to software that fits a standard Bradley-Terry model (or software that fits a standard logistic regression). In Section 3.2, we formalize the model setup and establish the validity of the paired comparison approach using existing statistical frameworks. We derive the exact correction to the covariance matrix of MLE produced by a standard Bradley-Terry model. In Section 3.3, we describe the loss in efficiency from using the misspecified model to analyze rank order data from numerical results. The second half of this chapter builds upon the formulation of rank order data as paired comparisons. In Section 3.4, we demonstrate that this formulation gives a way of directly computing moments of rank order data under the Plackett-Luce model. Relying on these moment computations, we propose a new tool for model diagnostic for the Plackett-Luce model. We demonstrate the behavior of our diagnostic through a simulation study. This chapter concludes in Section 3.5.

3.2 FRAMING RANK ORDER DATA AS PAIRED COMPARISON

As explained in Chapter 1, paired comparison data refers to outcomes of direct comparison between two items. The Bradley-Terry model (Bradley and Terry, 1952) is a popular model used in analyzing such data. Let Y_{ij} be the binary outcome of a paired comparison between item i and item j ; $Y_{ij} = 1$ if i is preferred, and 0 otherwise. The Bradley-Terry model assumes that all paired comparisons are independent conditional on the model parameters. With parameter ϑ_i representing the worth of item

i , the Bradley-Terry model states that the probability p_{ij} item i is preferred to j is given by

$$p_{ij} = \frac{e^{\vartheta_i}}{e^{\vartheta_i} + e^{\vartheta_j}}. \quad (3.3)$$

By convention and without loss of generality, we will assume $i < j$. Comparing against Equation (3.2), we can see that the Bradley-Terry model is just a special case of the Plackett-Luce model when exactly two items are ranked each time.

Assuming that all pairs are compared exactly T times, the log-likelihood of T independent sets of paired comparisons for every unique pair given paired comparison outcomes $Y_{ijt}, t = 1, \dots, T; i, j = 0, \dots, n, j > i$ is given by

$$\ell_{BT}(\vartheta) = \sum_{t=1}^T \sum_{i=0}^{n-1} \sum_{j=i+1}^n \left(Y_{ijt} \log p_{ij} + (1 - Y_{ijt}) \log(1 - p_{ij}) \right).$$

3.2.1 CONVERT A RANKING INTO A SET OF PAIRED COMPARISONS

Given an observed ranking $A = (a(1), \dots, a(n+1))'$, define $r(i)$ where $a(k) = i \Leftrightarrow r(i) = k$, so that $r(i)$ gives us the rank of item indexed i . Based on observed ranking A of $n+1$ items, we can create a set of $\binom{n+1}{2}$ paired comparisons by defining the outcome between each pair as

$$Y_{ij} = \begin{cases} 1 & \text{if } r(i) < r(j), \\ 0 & \text{otherwise} \end{cases} \quad (3.4)$$

for $i = 0, \dots, n-1, j = i+1, \dots, n$. As before, we let $Y_{ij} = 1 - Y_{ji}$ for $j < i$.

3.2.1.1 PROPERTIES OF PAIRED COMPARISON DATA DERIVED FROM RANK ORDER DATA UNDER THE PLACKETT-LUCE MODEL

Assuming that the ranking A is generated from a Plackett-Luce model with parameters $(\theta_0, \dots, \theta_n)'$, we have

$$p_{ij} = \mathbb{P}_{PL}(Y_{ij} = 1) = \frac{e^{\theta_i}}{e^{\theta_i} + e^{\theta_j}}. \quad (3.5)$$

In addition, for distinct triplet i, j and k , we have

$$\text{Cov}_{PL}(Y_{ij}, Y_{jk}) = -\frac{\gamma_j}{\gamma_i + \gamma_j + \gamma_k} \frac{\gamma_i}{\gamma_i + \gamma_j} \frac{\gamma_k}{\gamma_k + \gamma_j} \quad (3.6)$$

and

$$\text{Cov}_{PL}(Y_{ij}, Y_{kj}) = \text{Cov}_{PL}(Y_{ji}, Y_{jk}) = -\text{Cov}_{PL}(Y_{ji}, Y_{kj}) = -\text{Cov}_{PL}(Y_{ij}, Y_{jk}). \quad (3.7)$$

For distinct items i, j, k, l ,

$$\text{Cov}_{PL}(Y_{ij}, Y_{kl}) = 0. \quad (3.8)$$

These second moments can be derived by considering the latent variable framework with Gumbel random variables. The correlation between paired comparison outcomes with item pairs that share a common item j arise from the fact that the same realization of the latent Gumbel random variable W_j is used in both paired comparisons.

When paired comparisons are derived from rank order data under the Plackett-Luce model, we observe that the model specification of the Bradley-Terry model given by Equation (3.3) is correct marginally but the independence assumption of the Bradley-Terry model is violated. When we ob-

serve T independent rankings $A_t, t = 1, \dots, T$, the rank order data convert to T sets of paired comparisons $Y_{ijt}, t = 1, \dots, T$, where there is independence across t . Within each index t , the correlation structure is given by Equations (3.5) - (3.8).

Suppose we have paired comparison data Y_{ijt} converted from rank order data generated from a Plackett-Luce model under parameters $\boldsymbol{\theta} = (\theta_0, \dots, \theta_n)'$ with $\theta_0 = 0$. We can fit a standard Bradley-Terry model to the data. We use

$$\hat{\boldsymbol{\theta}}_{BT} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}} \ell_{BT}(\boldsymbol{\theta})$$

to denote the MLE of $\boldsymbol{\theta}$ under the Bradley-Terry model.

3.2.2 BEHAVIOR OF $\hat{\boldsymbol{\theta}}_{BT}$ GIVEN RANK ORDER DATA FROM A PLACKETT-LUCE MODEL

3.2.2.1 EXISTENCE OF UNIQUE MLE

It is well established that a unique MLE can be found for Bradley-Terry models if, in every possible partition of items into two nonempty groups, there exists some item in the second group that defeats some item in the first group at least once (Ford Jr, 1957; Hunter, 2004). Framing this in the context of paired comparison data generated from rank order data, the condition can be restated as for every possible partition of items into two nonempty groups, there exists some item in the second group that is ranked higher than some item in the first group at least once. This is, in fact, also the sufficient condition for uniqueness of MLE under the Plackett-Luce model (Hunter, 2004; Turner et al., 2020). So uniqueness of the MLE depends on the same sufficient condition whether we are using the Bradley-Terry or the Plackett-Luce model. When the data does not meet the sufficient condition, Hunter (2004) removed items that are ranked below (or above) all other items while Turner et al. (2020) added pseudo counts so that the sufficient conditions are satisfied.

When the sufficient conditions for a unique MLE are met, we can express the MLE of the Bradley-

Terry model as solutions to the following first order conditions:

$$\sum_{t=1}^T \sum_{j=0, j \neq i}^n \left(Y_{ijt} - \frac{e^{\hat{\theta}_i}}{e^{\hat{\theta}_i} + e^{\hat{\theta}_j}} \right) = 0, \quad i = 1, \dots, n. \quad (3.9)$$

The MLE can be solved by combining the first order conditions with the linear constraint $\theta_0 = 0$. As pointed out by [Simons and Yao \(1999\)](#), the MLE of Bradley-Terry model is also a method of moments estimator, since

$$\mathbb{E} \left[\sum_{j=0, j \neq i}^n Y_{ij} \right] = \sum_{j=0, j \neq i}^n \frac{e^{\theta_i}}{e^{\theta_i} + e^{\theta_j}} = \sum_{j=0, j \neq i}^n p_{ij}. \quad (3.10)$$

3.2.2.2 CONSISTENCY OF $\hat{\theta}_{BT}$

One of the most important questions to ask is whether $\hat{\theta}_{BT}$ is a consistent estimator for θ in the Plackett-Luce model. For finite n , we offer two perspectives to see why $\hat{\theta}_{BT}$ is consistent for θ as $T \rightarrow \infty$ even though the Bradley-Terry model assumptions are not completely appropriate.

The first perspective builds upon thinking about the Bradley-Terry MLE as a method of moments estimator. Since the sample moment

$$\frac{1}{T} \sum_{t=1}^T \sum_{j=0, j \neq i}^n Y_{ij,t}, \quad i = 1, \dots, n$$

will converge to its expectation as $T \rightarrow \infty$, $\sum_{j=0, j \neq i}^n \frac{e^{\hat{\theta}_i}}{e^{\hat{\theta}_i} + e^{\hat{\theta}_j}}$ converges to $\sum_{j=0, j \neq i}^n \frac{e^{\theta_i}}{e^{\theta_i} + e^{\theta_j}}$ for $i = 1, \dots, n$. Hence, $\hat{\theta}$ is consistent for θ under consistency of method of moments estimators.

The second perspective is drawn from analysis of repeated independent observations of correlated data, for example, longitudinal data analysis and spatial data analysis. Since the marginal model in

Equation (3.3) is correctly specified, Equation (3.9) can also be seen as solving a system of unbiased estimating functions formed by assuming independence across all observations. The estimator $\hat{\theta}_{BT}$ produced is not optimal but is still a valid estimator that is consistent (Zeger and Liang, 1986).

The discussion of the behavior of the estimator when n goes to infinity is non-trivial. For the original Bradley-Terry model, Simons and Yao (1999) showed that when $M_n = \max_{i,j} \frac{\gamma_i}{\gamma_j} = o\left(\sqrt{\frac{n}{\log n}}\right)$ as $n \rightarrow \infty$, the MLE of the model is consistent. However, this result does not extend to our case since the observations Y_{ijt} are correlated within each index t . At the same time, no similar results have been established for the Plackett-Luce model when $n \rightarrow \infty$.

This estimator based on pairwise comparisons was studied by Jeon and Kim (2013), who referred to the estimator as the maximum pairwise comparison likelihood estimator under the general class of order statistics based ranking models. Jeon and Kim argued that the likelihood based on pairwise comparisons can be seen as a pseudo likelihood with marginal probabilities, and focused exclusively on proving the consistency of this estimator. However, here we argue that under the Plackett-Luce model, the consistency of the estimator can be guaranteed by well-known existing results through viewing the MLE as a method of moments estimator or as a solution to estimating equations. The more interesting aspect of understanding this estimator lies in appreciating the quantification of uncertainty of the estimator.

3.2.2.3 ACCOUNTING FOR CORRELATION IN ESTIMATION

Using the second perspective based on the analysis of longitudinal data analysis where we have independent data points (rankings) with multiple observations per data point (paired comparisons within each ranking), we can offer a short digression on alternative methods for analysis of the paired comparison data converted from rankings. There are many general methods where the correlation within each set of paired comparisons can be incorporated into the estimation framework.

For example, we have already noted that $\hat{\theta}_{BT}$ is an estimator derived from using an unbiased

estimating function assuming independence. Crowder (1987) derived that the optimal estimating function can be achieved by weighting the estimating function by the inverse of the correlation of the data. Zeger and Liang (1986) proposed the famous generalized estimating equations (GEE) methodology for analyzing such correlated data, where a working correlation matrix is used. They showed that even without knowing the exact correlation structure, the resulting estimators are still consistent and an improvement over naive estimators based on independence assumptions. An alternative framework of analyzing these correlated data is to use models with random effects. Correlation within each independent ranking is incorporated into the model by assigning a random effect for each ranking. The integration of Bradley-Terry models with random effects is not new. For example, Lancaster and Quade (1983) analyzed sets of paired comparison data from individuals by assigning a random effect to each unique individual involved in the paired comparisons.

Despite the rich selection of methodologies available to account for correlation in paired comparison data derived from rank order data, we emphasize that such approaches are still based on inaccurately post-processing the rank order data into paired comparisons. We do not advocate such an approach. As framed in our introduction, we will focus on the estimator $\hat{\theta}_{BT}$ from a standard Bradley-Terry model, obtained by practitioners that have access to such a software. We have already justified that the estimator is consistent. Next, we will examine the covariance matrix of the estimator.

3.2.3 CORRECTING $\text{VAR}(\hat{\theta}_{BT})$

Let $\ell_{BT}(\theta)$ be the log-likelihood of θ under the Bradley-Terry model with exactly 1 paired comparison from each item pair, $Y_{ij}, i = 0, 1, \dots, n-1; j = i+1, \dots, n$. This log-likelihood is

$$\ell_{BT}(\theta) = \sum_{i=0}^{n-1} \sum_{j=i+1}^n \left(Y_{ij}\theta_i + (1 - Y_{ij})\theta_j - \log \left(e^{\theta_i} + e^{\theta_j} \right) \right).$$

Define

$$v_{ij} = \frac{e^{\theta_i} e^{\theta_j}}{(e^{\theta_i} + e^{\theta_j})^2}. \quad (3.11)$$

This is the variance of Y_{ij} , the outcome of the paired comparison between item i and j . The first and second derivatives of the log-likelihood w.r.t. θ_i is given by

$$\begin{aligned} \frac{\partial \ell_{BT}(\boldsymbol{\theta})}{\partial \theta_i} &= \sum_{j=0, j \neq i}^n \left(Y_{ij} - \frac{e^{\theta_i}}{e^{\theta_i} + e^{\theta_j}} \right), & i = 1, \dots, n \\ \frac{\partial^2 \ell_{BT}(\boldsymbol{\theta})}{\partial \theta_i^2} &= - \sum_{j=0, j \neq i}^n v_{ij}, & i = 1, \dots, n \\ \frac{\partial^2 \ell_{BT}(\boldsymbol{\theta})}{\partial \theta_i \partial \theta_j} &= -v_{ij}, & i, j = 1, \dots, n; i \neq j \end{aligned}$$

where we define $Y_{ij} = 1 - Y_{ji}$ for $j < i$. Also let $\nabla \ell_{BT}(\boldsymbol{\theta}) = (\frac{\partial \ell_{BT}(\boldsymbol{\theta})}{\partial \theta_1}, \dots, \frac{\partial \ell_{BT}(\boldsymbol{\theta})}{\partial \theta_n})'$, with $\frac{\partial \ell_{BT}(\boldsymbol{\theta})}{\partial \theta_0}$ being omitted, since item 0 is set to be the reference group. Let the $n \times n$ matrix $\mathcal{J} = -\nabla^2 \ell_{BT}(\boldsymbol{\theta})$, with

$$\mathcal{J}_{ii} = \sum_{k=0, k \neq i}^n v_{ik} \quad i = 1, \dots, n \quad (3.12)$$

$$\mathcal{J}_{ij} = -v_{ij} \quad i, j = 1, \dots, n; i \neq j. \quad (3.13)$$

When the data truly come from a Bradley-Terry model, it is a well known result that

$$\text{Var}_{BT}(\hat{\boldsymbol{\theta}}_{BT}) = \mathbb{E}_{BT} [-\nabla^2 \ell_{BT}(\boldsymbol{\theta})]^{-1} = \mathcal{J}^{-1}$$

where the expectation is taken with respect to the true data generating process of the Bradley-Terry model. We use the subscript BT to emphasize that data are coming from a Bradley-Terry model. Since $\nabla^2 \ell_{BT}(\boldsymbol{\theta})$ is a constant, the expectation operator returns the constant. An estimator of the variance can be formed by evaluating $\mathcal{J}^{-1}$ at $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}_{BT}$. Packages that perform standard Bradley-Terry analysis will return a covariance matrix for $\hat{\boldsymbol{\theta}}_{BT}$ by computing an estimator of the quantity $\mathcal{J}^{-1}$.

3.2.3.1 $\text{Var}(\hat{\boldsymbol{\theta}}_{BT})$ UNDER DATA FROM A PLACKETT-LUCE MODEL

When data are not generated from a Bradley-Terry model, $\mathcal{J}^{-1}$ is no longer a valid representation of the variance of $\hat{\boldsymbol{\theta}}_{BT}$. White (1982) showed that the variance of the MLE, when the model is misspecified, is given by the “sandwich” form

$$\text{Var}_{PL}(\hat{\boldsymbol{\theta}}_{BT}) = \mathbb{E}_{PL}[-\nabla^2 \ell_{BT}(\boldsymbol{\theta})]^{-1} \text{Var}_{PL}[\nabla \ell_{BT}(\boldsymbol{\theta})] \mathbb{E}_{PL}[-\nabla^2 \ell_{BT}(\boldsymbol{\theta})]^{-1}.$$

where the expectation and variance are taken over the true data generating process. In our case, the expectation and variance are taken with respect to the distribution of data generated from a Plackett-Luce model. We add the subscript PL to the expectation and variance operators to emphasize this. We have already derived that the expression $\nabla^2 \ell_{BT}(\boldsymbol{\theta})$ does not depend on any random variable. Hence, $\mathbb{E}_{PL}[-\nabla^2 \ell_{BT}(\boldsymbol{\theta})]^{-1} = \mathcal{J}^{-1}$.

Let $n \times n$ matrix $\mathcal{Q} = \text{Var}_{PL}[\nabla \ell_{BT}(\boldsymbol{\theta})]$, noting that the only random part of the i th term of $\nabla \ell_{BT}(\boldsymbol{\theta})$ is $\sum_{j=0, j \neq i}^n Y_{ij}$. We can explicitly derive each component of the covariance matrix $\mathcal{Q}$ as follows.

$$\begin{aligned}
\mathcal{Q}_{ii} &= \text{Var}_{PL} \left(\sum_{j=0, j \neq i}^n Y_{ij} \right) \\
&= \sum_{\substack{j=0 \\ j \neq i}}^n \left[v_{ij} + \sum_{k=0, k \neq i, j}^n \text{Cov}_{PL}(Y_{ij}, Y_{ik}) \right] \\
&= \sum_{j=0, j \neq i}^n \left[v_{ij} + \sum_{k=0, k \neq i, j}^n \frac{(\gamma_i + \gamma_j)(\gamma_i + \gamma_k)}{(\gamma_i + \gamma_j + \gamma_k)\gamma_i} v_{ij} v_{ik} \right], \quad i = 1, \dots, n. \\
\mathcal{Q}_{ij} &= \text{Cov}_{PL} \left(\sum_{k=0, k \neq i}^n Y_{ik}, \sum_{l=0, l \neq j}^n Y_{jl} \right) \\
&= \sum_{k=0, k \neq i}^n \sum_{l=0, l \neq j}^n \text{Cov}_{PL}(Y_{ik}, Y_{jl}) \\
&= -v_{ij} + \sum_{k=0, k \neq i, j}^n [\text{Cov}_{PL}(Y_{ik}, Y_{jk}) - \text{Cov}_{PL}(Y_{ij}, Y_{kj}) - \text{Cov}_{PL}(Y_{ij}, Y_{ik})] \\
&= -v_{ij} + \sum_{k=0, k \neq i, j}^n \left[\frac{-2\gamma_k^2 \gamma_i \gamma_j}{(\gamma_i + \gamma_j + \gamma_k)(\gamma_i + \gamma_j)(\gamma_i + \gamma_k)(\gamma_k + \gamma_j)} \right], \quad i, j = 1, \dots, n; i \neq j.
\end{aligned} \tag{3.14}$$

To appreciate the difference between $\text{Var}_{BT}(\hat{\theta}_{BT})$ and $\text{Var}_{PL}(\hat{\theta}_{BT})$, note that

$$\begin{aligned}
\text{Var}_{BT}(\hat{\theta}_{BT}) &= \frac{1}{n} \left(\frac{1}{n} \mathcal{J} \right)^{-1}, \\
\text{Var}_{PL}(\hat{\theta}_{BT}) &= \mathcal{J}^{-1} \mathcal{Q} \mathcal{J}^{-1} = \left(\frac{1}{n} \mathcal{J} \right)^{-1} \left(\frac{1}{n^2} \mathcal{Q} \right) \left(\frac{1}{n} \mathcal{J} \right)^{-1}.
\end{aligned}$$

The diagonal terms of the matrix $\mathcal{J}$ is the sum of n components of v_{ij} , while the diagonal terms of the matrix $\mathcal{Q}$ is on the order of the square of sum of n components of v_{ij} . Suppose we can control the magnitude of $\frac{1}{n} \sum_{k=0, k \neq i}^n v_{ik}$ for $i = 1, \dots, n$ as n changes. It follows that $\text{Var}_{BT}(\hat{\theta}_{BT})$ is inversely

proportional to n but there is no directly relationship between $\text{Var}_{PL}(\hat{\theta}_{BT})$ and n . This reflects the fact that under the Bradley-Terry model, as n increases, more independent observations are being accumulated. On the other hand, when data are converted from ranking data, even with more items, we are still evaluating information from a single ranking.

An estimator for $\text{Var}_{PL}(\hat{\theta}_{BT})$ is given by $\hat{\mathcal{J}}^{-1}\hat{\mathcal{Q}}\hat{\mathcal{J}}^{-1}$. For the practitioner using a standard package to fit the Bradley-Terry model, $\hat{\mathcal{J}}^{-1}$ should have been returned by the software. An estimator $\hat{\mathcal{Q}}$ can be obtained by evaluating the $\mathcal{Q}$ matrix at $\hat{\theta}_{BT}$. Observe that $\mathcal{Q}$ is in fact the covariance matrix of a linear combination of the Y_{ij} . Given any vector θ , the $\mathcal{Q}$ matrix can be evaluated by first defining a covariance matrix on Y_{ij} for all unique pairs $j > i$, a square matrix of dimension $\binom{n+1}{2}$; and then defining the matrix of linear combination of dimension $n \times \binom{n+1}{2}$ using the fact that the i -th component of $\nabla \ell(\theta)$ is given by $\sum_{j=0, j \neq i}^n Y_{ij} = \sum_{j=0}^{i-1} (1 - Y_{ji}) + \sum_{j=i+1}^n Y_{ij}$. Easier book-keeping may be achieved by defining a larger covariance matrix over the vector of outcomes Y_{ij} for $i, j = 0, \dots, n$. The trade-off is the computation involving a higher dimensional matrix.

3.3 LOSS OF EFFICIENCY IN $\hat{\theta}_{BT}$

In Section 3.2, given rank order data from a Plackett-Luce model, we detailed how to convert them into sets of paired comparisons and described the analysis of the paired comparison data using a Bradley-Terry model. We argued that $\hat{\theta}_{BT}$ is a reasonable estimator of θ because it is consistent, and it demands the same sufficient condition for uniqueness as $\hat{\theta}_{PL}$. We have shown that $\text{Var}(\hat{\theta}_{BT}) = \mathcal{J}^{-1}\mathcal{Q}\mathcal{J}^{-1}$, where we omit the subscript PL in this Section because we assume to be always working with rank order data from Plackett-Luce models. We described how to correctly calculate $\widehat{\text{Var}}(\hat{\theta}_{BT})$ from the outputs of a standard package that fits the Bradley-Terry model to account for the correlation among each set of paired comparison data. In this section, we focus on comparing estimators $\hat{\theta}_{BT}$ and $\hat{\theta}_{PL}$ on their covariance matrix. In particular, we expect a loss of efficiency in using $\hat{\theta}_{BT}$ since it is an

estimator built upon an incorrect model, but we would like to understand whether we can further describe this loss of efficiency, and understand this loss as $n + 1$ (the number of items being compared) changes.

Following classical results of MLE, we know that $\text{Var}(\hat{\theta}_{PL})$ is given by the inverse of the Fisher information matrix of θ under the Plackett-Luce model. Let π_{ij}^k denote a permutation of a subset of k items out of $n + 1$ items indexed $0, \dots, n$, excluding items i and j . Let $\pi_{ij}^k(1), \dots, \pi_{ij}^k(k)$ denote the index of items at position $1, \dots, k$ respectively. Define the quantity for $i, j = 0, 1, \dots, n, i \neq j$,

$$\begin{aligned} \Delta_{ij} = & \gamma_i \gamma_j + \sum_{\pi_{ij}^1} \gamma_{\pi_{ij}^1(1)} \frac{\gamma_i}{1 - \gamma_{\pi_{ij}^1(1)}} \frac{\gamma_j}{1 - \gamma_{\pi_{ij}^1(1)}} \\ & + \sum_{\pi_{ij}^2} \gamma_{\pi_{ij}^2(1)} \frac{\gamma_{\pi_{ij}^2(2)}}{1 - \gamma_{\pi_{ij}^2(1)}} \frac{\gamma_i}{1 - \gamma_{\pi_{ij}^2(1)} - \gamma_{\pi_{ij}^2(2)}} \frac{\gamma_j}{1 - \gamma_{\pi_{ij}^2(1)} - \gamma_{\pi_{ij}^2(2)}} \\ & \vdots \\ & + \sum_{\pi_{ij}^{n-1}} \left(\gamma_{\pi_{ij}^{n-1}(1)} \prod_{l=2}^{n-1} \frac{\gamma_{\pi_{ij}^{n-1}(l+1)}}{1 - \sum_{m=1}^l \gamma_{\pi_{ij}^{n-1}(m)}} \right) \frac{\gamma_i}{1 - \sum_{m=1}^{n-1} \gamma_{\pi_{ij}^{n-1}(m)}} \frac{\gamma_j}{1 - \sum_{m=1}^{n-1} \gamma_{\pi_{ij}^{n-1}(m)}} \end{aligned}$$

where each summation is over all sets of permutations. Let $\mathcal{M}$ denote the $n \times n$ Fisher information matrix of θ under the Plackett-Luce model with the constraint $\theta_0 = 0$. The (i, j) th element of matrix $\mathcal{M}$ is given by

$$\mathcal{M}_{ii} = \sum_{j=0, j \neq i}^n \Delta_{ij}, \quad i = 1, \dots, n, \quad (3.16)$$

$$\mathcal{M}_{ij} = -\Delta_{ij}, \quad i, j = 1, \dots, n. \quad (3.17)$$

Details of the derivation is given in Appendix B.1. Based on this matrix, we have

$$\text{Var}(\hat{\theta}_{PL}) = \mathcal{M}^{-1}.$$

We recognize that calculation of $\mathcal{M}$ involves bookkeeping of $(n - 1)!$ permutations. Thus the computation of an estimate for $\text{Var}(\hat{\theta}_{PL})$ is computationally intensive as n increases.

Given the expressions for $\text{Var}(\hat{\theta}_{BT})$ and $\text{Var}(\hat{\theta}_{PL})$, we would like to compare these two covariance matrices. One approach is to compare the two matrices entry-wise. In particular, we compare entry-wise ratios of the diagonal entries of the covariance matrices. However, the exact entries of the covariance matrix is sensitive to the choice of the reference group when selecting a linear constraint for estimation (Firth, 2003; Firth and De Menezes, 2004). Since the choice of the reference item is arbitrary, we do not want our results to depend on the identification constraint. We choose instead to compare the determinant of the covariance matrices, since this value does not depend on choice of reference item or linear constraint ensuring identifiability. Wilks (1932) referred to this quantity as the generalized variance of a multivariate random variable. Since a determinant scales with dimension of the matrix, we use the quantity

$$\mathcal{R}_n = \left(\frac{\det(\mathcal{J}^{-1} \mathcal{Q} \mathcal{J}^{-1})}{\det(\mathcal{M}^{-1})} \right)^{\frac{1}{n}} \quad (3.18)$$

for our evaluation.

3.3.1 $\text{VAR}(\hat{\theta}_{BT})$ AND $\text{VAR}(\hat{\theta}_{PL})$ UNDER A MODEL WITH UNIFORMLY DISTRIBUTED RANKINGS

Consider the special case where $\theta_0 = \dots = \theta_n = 0$, or equivalently $\gamma_0 = \dots = \gamma_n = \frac{1}{n+1}$. Under this setting, all items are equally preferred. The Plackett-Luce model with such a set of parameters is the same as an uniform distribution over all possible rankings. Under this special setup, we can explicitly derive the matrices $\mathcal{J}$, $\mathcal{Q}$ and $\mathcal{M}$ as

$$\mathcal{M} = \frac{1}{n(n+1)} \left(\sum_{k=1}^n \frac{n-k+1}{n-k+2} \right) ((n+1)I_n - \mathbf{1}_n \mathbf{1}_n^\top),$$

$$\mathcal{Q} = \frac{n+2}{12} ((n+1)I_n - \mathbf{1}_n \mathbf{1}_n^\top),$$

$$\mathcal{J} = \frac{1}{4} ((n+1)I_n - \mathbf{1}_n \mathbf{1}_n^\top),$$

where $\mathbf{1}_n$ is the n -vector consisting of all 1s. Hence, under the model with uniformly distributed rankings, we have

$$\mathcal{R}_n^{\text{Unif}} = \frac{4}{3} \frac{n+2}{n(n+1)} \sum_{k=1}^n \frac{n-k+1}{n-k+2} = \frac{4}{3} \frac{n+2}{n(n+1)} \left(n - \sum_{k=1}^n \frac{1}{n-k+2} \right).$$

As $n \rightarrow \infty$, $\mathcal{R}_n^{\text{Unif}} \rightarrow \frac{4}{3}$. In other words, when all the items are equally preferred, the loss of efficiency in using $\hat{\theta}_{BT}$ instead of $\hat{\theta}_{PL}$ is bounded from above by 33.3% as number of items being ranked increases.

3.3.2 NUMERICAL RESULTS

In general we do not have an analytical form for $\mathcal{R}_n$. Hence, we examine the behavior of $\mathcal{R}_n$ using some numerical examples.

3.3.2.1 NUMERICAL RESULTS BASED ON RANDOMLY DRAWN γ_i

We start with $n = 2, \dots, 7$, where we cap the number of items at 8 so that calculation of $\mathcal{M}$ is manageable. We note that each element of matrices $\mathcal{Q}$ and $\mathcal{J}$ is constructed using v_{ij} 's. By definition of v_{ij} from Equation (3.11), v_{ij} is a function of the gap between the worth of items i and j . When $\theta_i = \theta_j$, $v_{ij} = 0.25$. When the gap between the two items widens, v_{ij} decreases. This quantity is also symmetric in labels i and j with $v_{ij} = v_{ji}$. Hence, we compare $\mathcal{R}_n$ against $v_m = \min_{i,j} v_{ij}$, $i, j = 0, \dots, n$. We have

$$v_m \leq v_{ij} \leq \frac{1}{4}, \quad (3.19)$$

for $i, j = 0, \dots, n$. Letting $r_M = \frac{\max_i \gamma_i}{\min_i \gamma_i} = \max_{i,j} \frac{\gamma_i}{\gamma_j}$, $i, j = 0, \dots, n$, we have the relationship

$$v_m = \frac{r_M}{(1 + r_M)^2}. \quad (3.20)$$

The quantity v_m is also an indication of the gap between the items with highest and lowest worth. To illustrate realistic values of v_m in data, we examined examples of ranking data used by [Hunter \(2004\)](#), [Turner et al. \(2020\)](#) and [Croissant \(2020\)](#). For each data set, we fit a Plackett-Luce model. Table 3.1 records the estimates of v_m for each data from the fitted worth of the items.

Data	Netflix	Sushi	T-shirt	Game	NASCAR
Number of Items ($n + 1$)	4	10	11	6	83
$\log(v_m)$	-1.96	-2.23	-2.68	-1.84	-4.92

Table 3.1: Maximum difference between estimated worth of items for sample rank order data. Netflix, Sushi and T-shirt data taken from <https://www.preflib.org/>, Game data taken from package `mlogit` and NASCAR data taken from package `PlackettLuce`. The NASCAR data set involves rankings from subsets of $n + 1$ competitors; all other data sets involve rank orders of all $n + 1$ items/competitors.

We used 3 sampling schemes to simulate γ^n , where the superscript n marks the number of items included in the ranking. The first is a uniform sampling of γ^n using a Dirichlet distribution with parameters $(\frac{1}{n+1}, \dots, \frac{1}{n+1})$. As an anchor point, we always include the vector $\gamma^n = (\frac{1}{n+1}, \dots, \frac{1}{n+1})'$ as the first sample. The second is based on sampling θ^n from $\mathcal{N}(0, \sigma^2)$, with $\sigma \sim \text{Unif}(1, 5)$. The vector γ^n is obtained by normalizing $\exp(\theta^n)$. For the first two sampling schemes, we draw $500(n + 1)^2$ samples under each scheme. The third sampling scheme involves conditionally sampling γ^n given v_m in the following manner. Let $\gamma_k^n = (\gamma_{k0}^n, \dots, \gamma_{kn}^n)'$ be a vector of parameters for the k th simulation draw under $n + 1$ items. Given a sequence of $v_m^{(c)} = \frac{r_M^{(c)}}{(1 + r_M^{(c)})^2}$, we draw $K = 100(n + 1)$ samples for each $v_m^{(c)}$ using the following steps.

1. Sample $\tilde{\gamma}_{k0}^n \sim \text{Exp}(1)$, set $\tilde{\gamma}_{kn}^n = r_M^{(c)} \gamma_{k0}^n$.
2. Sample $U_{k1}, \dots, U_{k(n-1)} \sim \text{Beta}(\alpha_k, \beta_k)$ with $\alpha_k \sim \text{Exp}(1)$, $\beta_k \sim \text{Exp}(1)$.

3. Set $\tilde{\gamma}_{ki}^n = \tilde{\gamma}_{k0}^n + U_{ki}(\tilde{\gamma}_{kn}^n - \tilde{\gamma}_{k0}^n)$, $i = 1, \dots, n-1$.
4. Normalize $\tilde{\gamma}_{k0}^n, \dots, \tilde{\gamma}_{kn}^n$ to get γ_k^n .

We calculate $\mathcal{R}_n$ for each sample of γ^n drawn. Since the primary goal of this simulation study is to give a general understanding of extreme values of $\mathcal{R}_n$ as v_m changes, we combine results from all three sampling schemes and plot the results in Figure 3.1.

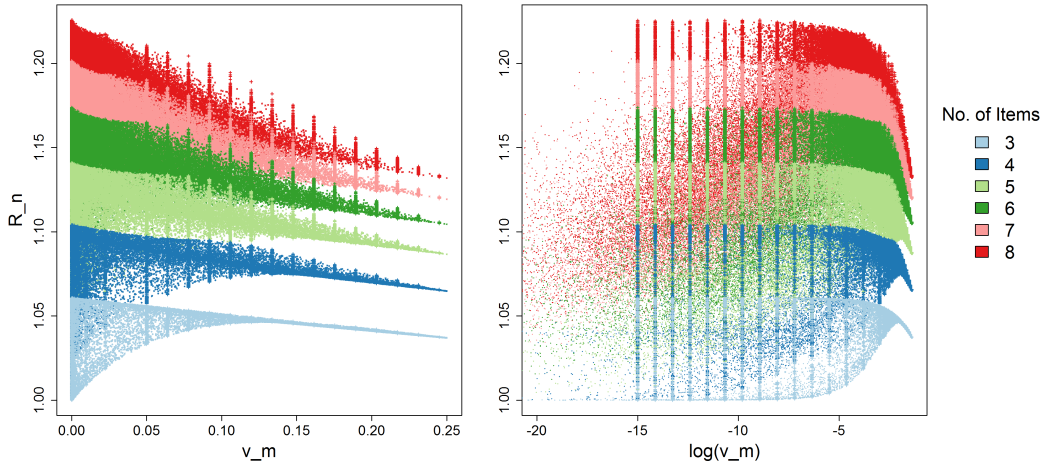


Figure 3.1: Distribution of $\mathcal{R}_n$ against v_m on linear and log scale of v_m , $n+1 = 3, \dots, 8$, for γ^n sampled from 3 sampling schemes.

We use $\bar{\mathcal{R}}_n(v_m)$ to denote the maximum value of $\mathcal{R}_n$ given v_m under complete ranking of $n+1$ items. This describes the maximum loss of efficiency in using the estimator $\hat{\theta}_{BT}$ when the difference in worth between the highest and lowest item is described by v_m . From Figure 3.1, we observe that as a function of v_m , $\bar{\mathcal{R}}_n(v_m)$ is inversely related to v_m . As $v_m \rightarrow 0$, $\bar{\mathcal{R}}_n(v_m)$ stabilizes. We denote the value of $\bar{\mathcal{R}}_n(v_m)$ as $v_m \rightarrow 0$ as $\bar{\mathcal{R}}_n(0)$. Given v_m , $\bar{\mathcal{R}}_n(v_m)$ increases with n . We already know that $\bar{\mathcal{R}}_n(0.25) = \mathcal{R}_n^{\text{Unif}}$ converges to $\frac{4}{3}$ as $n \rightarrow \infty$. Using an approximation for the inverse of the matrices $\mathcal{M}$, $\mathcal{Q}$ and $\mathcal{J}$ proposed by [Simons and Yao \(1998, 1999\)](#), we study an approximate quantity

of $\mathcal{R}_n$ given by

$$\tilde{\mathcal{R}}_n = \left(\prod_{i=0}^n \frac{\mathcal{Q}_{ii}}{\mathcal{J}_{ii}^2} \right)^{1/n} \left(\frac{\sum_{i=0}^n \mathcal{J}_{ii} \sum_{i=0}^n \mathcal{J}_{ii}}{\sum_{i=0}^n \mathcal{Q}_{ii}} \right)^{1/n} \left(\prod_{i=0}^n \frac{\mathcal{M}_{ii}}{\sum_{i=0}^n \mathcal{M}_{ii}} \right)^{1/n} \quad (3.21)$$

where $\mathcal{J}_{00}$, $\mathcal{Q}_{00}$ and $\mathcal{M}_{00}$ are defined by extending the definition in Equations (3.12), (3.14) and (3.16) to include the case $i = 0$ respectively. Appendix B.3.1 details the derivation of this approximation and provides some numerical justification for using $\tilde{\mathcal{R}}_n$ to approximate $\mathcal{R}_n$. In Appendix B.3.3, we prove that given v_m ,

$$\tilde{\mathcal{R}}_n \leq \frac{(1 + r_M)^2}{1 + 2r_M}, \quad n \rightarrow \infty.$$

This provides theoretical justification that $\bar{\mathcal{R}}_n(v_m)$ converges to a limit when $n \rightarrow \infty$. Existence of this limit offers assurance that the loss of efficiency in the estimator $\hat{\theta}_{BT}$ is bounded as $n \rightarrow \infty$. Unfortunately, this upper bound clearly failed to capture the empirical observation that $\bar{\mathcal{R}}_n(v_m)$ stabilizes as $v_m \rightarrow 0$. Hence, we are unable to study the behavior of $\bar{\mathcal{R}}_n(0)$. This bound is also too loose to give useful bounds on $\mathcal{R}_n$.

We summarize the observed upper limits on $\mathcal{R}_n$ by approximating $\bar{\mathcal{R}}_n(0)$ with maximum observed values seen in the simulation in Figure 3.2. We add results from ranking of 9 and 10 items by conditionally sampling at $\log(v_m) = -9$. In the range of n that we used in simulation study, the maximum possible loss of efficiency in using $\hat{\theta}_{BT}$ is no more than 27%.

We also observe from Figure 3.1 that as v_m decreases, the range of $\mathcal{R}_n$ widens. Using $\mathcal{R}_n^{\text{Unif}}$ as the benchmark, we find that $\mathcal{R}_n$ can fall below $\mathcal{R}_n^{\text{Unif}}$ when v_m approaches 0. In the next section, we examine the relationship between $\mathcal{R}_n$ and v_m further by assigning special structure to γ^n . We would like to see if we can characterize the γ^n vectors that give low and high values of $\mathcal{R}_n$ when v_m is small.

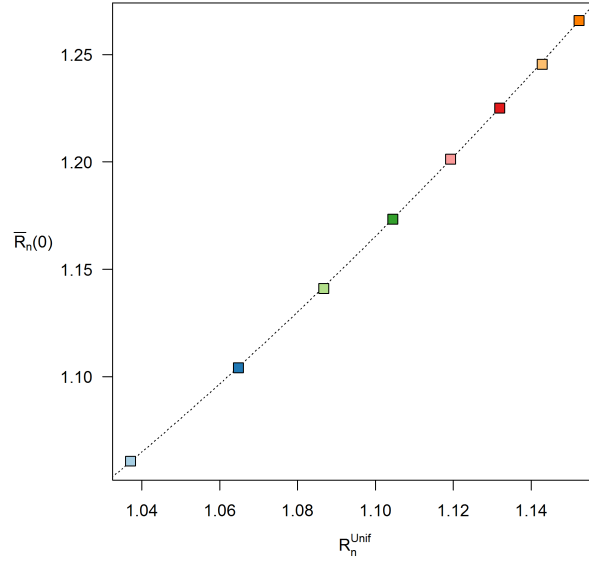


Figure 3.2: Plot of $\bar{\mathcal{R}}_n(0)$ against $\mathcal{R}_n^{\text{Unif}}$ for $n+1 = 3, \dots, 10$. Each color represent one value of n . $\bar{\mathcal{R}}_n(0)$ is obtained for $n+1 = 9, 10$ by sampling γ^n at $\log(v_m) = -9$. The value $\bar{\mathcal{R}}_n(0)$ is taken to be the maximum value of $\bar{\mathcal{R}}_n$ observed in the simulation.

3.3.2.2 NUMERICAL RESULTS BASED ON DESIGNED γ .

With a design on the spacing between worths of each item, there is a one-to-one correspondence between v_m and γ^n . We use this to examine the behavior of $\mathcal{R}_n$ as a function of v_m . Below is the the list of structures on γ^n we used. Details for deriving the γ^n under each design are given in Appendix B.4.

1. Equally-spaced γ with $\gamma_i - \gamma_{i-1} = c$ for some constant c . We label $\mathcal{R}_n$ under this case as $\mathcal{R}_n^{\text{EqG}}(v_m)$.
2. Equally-spaced θ with $\theta_i - \theta_{i-1} = c$. We label $\mathcal{R}_n$ under this case as $\mathcal{R}_n^{\text{EqT}}(v_m)$.
3. Mixture of two types of items. We assume that there are only two type of items - those worth γ_L and those worth γ_H . Let n_L be the number of items with low worth and n_H be the number of items with high worth. We consider division between low and high worth groups given by:

- (a) An extreme division with $n_L = 1$ and $n_H = n$. We label this with $\mathcal{R}_n^{\text{mixIn}}(v_m)$.
- (b) An extreme division with $n_L = n$ and $n_H = 1$. We label this with $\mathcal{R}_n^{\text{mixnI}}(v_m)$.
- (c) Half and half divide. We consider a ceiling case with $n_L = \frac{n+1}{2}$ and floor case with $n_L = \frac{n-1}{2}$. We label them with $\mathcal{R}_n^{\text{o.5c}}(v_m)$ and $\mathcal{R}_n^{\text{o.5f}}(v_m)$ respectively.

Of course, regardless of the spacing design on γ , $\mathcal{R}_n$ is always given by $\mathcal{R}_n^{\text{Unif}}$ when $v_m = 0.25$.

We are interested in describing $\mathcal{R}_n$ as v_m changes. However, we should note that when v_m is small, it is possible that the rank order data generated from θ with v_m do not satisfy the condition for existence of a unique MLE because the disparity in items' worth is too large. We offer a short discussion of this in Appendix B.5. As a reference, Table 3.2 gives the empirical probabilities of producing rank order data of size $T = 100$ that violate the sufficient conditions for uniqueness of $\hat{\theta}_{BT}$ under the different designs on spacing within the worth vector with 7 items. Observe that for several of the structures, when v_m is low, there is a high probability that the resulting rank order data can be divided into subgroups of items, where one subgroup never outrank the other.

$\log(v_m)$	EqG	EqT	mixIn	mixnI	o.5c	o.5f
-3	0.000	0.000	0.000	0.000	0.000	0.000
-5	0.008	0.000	0.176	0.022	0.007	0.012
-7	0.494	0.000	0.814	0.582	0.534	0.551
-9	0.914	0.000	0.969	0.925	0.903	0.932

Table 3.2: Empirical probabilities of producing rank order data that violate the sufficient conditions for uniqueness of MLE at $n + 1 = 7$, for different designs on spacing within γ proposed for the simulation study. The results are based on 1000 independent samples of rank order data of size $T = 100$ given γ^n .

For the simulation study, we will consider values of $\log(v_m)$ between -10 and $\log(0.25)$. The lower bound -10 covers observed values of v_m recorded in Table 3.1. We acknowledge that some settings may not be realistic since we will only ever be fitting the models on rank order data that meets the sufficient condition for uniqueness of MLE. Figure 3.3 plots $\mathcal{R}_n(v_m)$ given the designs we described

on γ^n for the $n + 1 = 7$ case. More plots for other n values are provided in Appendix B.2, where we show that the results are similar across varying n .

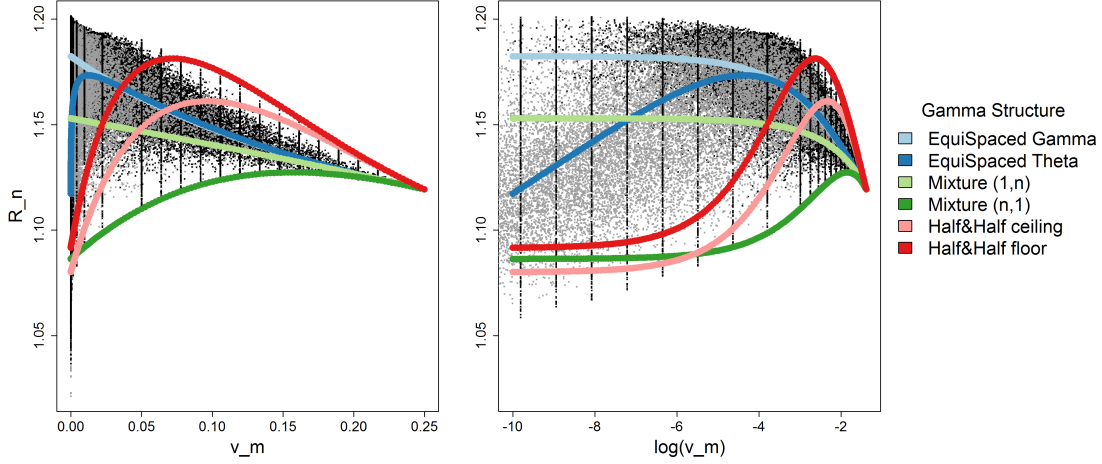


Figure 3.3: Quantity $\mathcal{R}_n$ as a function of v_m on linear and log scale of v_m for $n + 1 = 7$ under 6 different choices of design for γ^n . In black is $\mathcal{R}_n$ based on γ^n sampled from $\text{Dir}(\frac{1}{n+1}, \dots, \frac{1}{n+1})$ and γ^n sampled by conditioning on v_m for $n + 1 = 7$. In grey is $\mathcal{R}_n$ based on θ^n sample from Normal distributions.

From Figure 3.3, we observe that the relationship between $\mathcal{R}_n$ and v_m can depend heavily on the spacing and values of the elements of γ^n . In the cases with equally-spaced γ^n and two components mixture with a single item in the low-worth group, $\mathcal{R}_n$ monotonically increases as v_m decreases from 0.25. In all other four cases, $\mathcal{R}_n$ first increases as v_m decreases from 0.25, but after some threshold, $\mathcal{R}_n$ decreases and can fall below $\mathcal{R}_n^{\text{Unif}}$. Out of these four cases, the $\mathcal{R}_n$ from the three cases with different designs on the two components mixture have all stabilized as a function of v_m , but the $\mathcal{R}_n$ from equally spaced θ^n is still decreasing with v_m at $\log(v_m) = -10$. Making reference to Table 3.2, even at $\log(v_m) = -10$, equally spaced θ^n are still producing rank order data that satisfy the sufficient conditions for uniqueness of an MLE. It seems that when the worth of items are equally spaced on the θ scale, it seems theoretically possible to achieve loss of efficiency that is close to 1. From Figure 3.3 we

also observe that the relationship between $\mathcal{R}_n$ from θ^n sampled from the Normal distributions and v_m look similar to $\mathcal{R}_n^{\text{EqT}}(v_m)$.

In this section, we described the loss of efficiency in using $\hat{\theta}_{BT}$ with the n -th root of the determinant of $\text{Var}(\hat{\theta}_{BT})$ relative to the determinant of $\text{Var}(\hat{\theta}_{PL})$. Based on our simulation study, we find that the maximum possible loss of efficiency in using $\hat{\theta}_{BT}$ is positively related to n , the number of items ranked, and inversely related to v_m , a measure of the gap between highest and lowest worth items. With a total 10 items, the maximum possible loss of efficiency in using $\hat{\theta}_{BT}$ is no more than 30%. We are unable to offer exact upper bounds on this loss of efficiency for higher n , but we showed that it is finite given v_m as $n \rightarrow \infty$. We also note that a complete analysis of ranking data is not realistic for high values of n , and our simulation study can cover settings involving small n . The relationship between loss of efficiency and v_m depends heavily on how the worth of items are distributed.

We believe that our simulation study can help practitioners appreciate the efficiency cost of using $\hat{\theta}_{BT}$. Next, we draw inspiration from the paired comparison formulation of the rank order data, and use it to propose a new model diagnostic measure for fitting Plackett-Luce models.

3.4 A MODEL DIAGNOSTIC FOR PLACKETT-LUCE MODEL INSPIRED BY THE PAIRED COMPARISON DATA FORMULATION

3.4.1 A REVIEW OF MODEL DIAGNOSTICS FOR RANKING MODELS

Most research for models on rank order data have focused on setting up the model and estimating model parameters. Several authors (for example, see [Li et al. \(2016\)](#); [Cheng et al. \(2010\)](#)) have attempted to compare goodness-of-fit of different models on the same data, especially when proposing new models. These authors typically choose to compare models using predictive performances on out-of-sample data, where distance between observed ranking and fitted ranking is measure using Kendall's τ coefficient ([Kendall, 1938](#)), a non-parametric measure of distance between two rankings.

Kendall's τ compares two rankings by examining the number of concordant and discordant pairs. This is useful when comparing multiple models.

There has been minimal discussion, however, on model fit for rank order data without the comparison to competing models. The paper by [Cohen and Mallows \(1983\)](#) is one of the few and earliest works that has discussed this issue. They pointed out that the goodness-of-fit problem for ranking data can be framed as a goodness-of-fit problem on a multinomial distribution. Each of the $(n + 1)!$ rankings is a unique category and we can compare the fitted counts against observed counts across all categories. A formal test can be formulated using the likelihood ratio test for a contingency table, or using a Pearson χ^2 test. Cohen and Mallows also emphasized that this approach is manageable only for small n (Cohen and Mallows specifically recommended $n + 1 \leq 4$). When n is large, Cohen and Mallows recommended combining subsets of rankings into disjoint groups and examining goodness-of-fit on these partitions. The authors recognized that there are many possible partitionings of the $(n + 1)!$ rankings and offered two suggestions.

The first suggestion is to partition the rankings based on distance from some reference ranking. [Cohen and Mallows](#) did highlight that this partition is more sensible when applied to distances based ranking models. Under these models, a natural choice of a reference ranking is the estimated modal ranking. Also, the distance metric used in the model specification is a natural choice as a distance. However, there is still a need to decide on how to combine the rankings into groups.

The second suggestion from [Cohen and Mallows](#) is to summarize the rankings by reducing them into paired comparisons. For unique item pairs $j > i$, let p_{ij} be the model-based estimated probability that i is ranked higher than j , and T_{ij} be the observed count of the event i is ranked higher than j out of a total of T sets of ranks. Cohen and Mallows proposed the quantity

$$Z_{ij} = \frac{T_{ij} - Tp_{ij}}{\sqrt{Tp_{ij}(1 - p_{ij})}}$$

which is approximately behaves like a Normal random variable. They recommended comparing these Z values against $\binom{n+1}{2}$ standard Normal random variables. These values are not independently distributed, but Cohen and Mallows argued that the correlation is weak and hence the comparison against the standard Normal distribution is not completely unreasonable.

Another example that we have observed in previous work is to compare only the observed and expected frequency of items ranked first (D'Elia and Piccolo, 2005; Yu, 2000).

3.4.2 A NEW MODEL DIAGNOSTIC FOR PLACKETT-LUCE MODELS

Given observed ranking A of $n + 1$ items, define a set of paired comparisons Y_{ij} from A as given in Equation (3.4). For the i th item, define

$$S_i = \sum_{j=0, j \neq i}^n Y_{ij}.$$

In other words, S_i is the number of players ranked below player i in ranking A . We can re-write this as

$$S_i = (n + 1) - r(i)$$

where $r(i)$ is the rank of player i in ranking A . Also, note from before that

$$\mathbb{E}[S_i] = \mathbb{E}\left[\sum_{j=0, j \neq i}^n Y_{ij}\right] = \sum_{j=0, j \neq i}^n \frac{e^{\theta_i}}{e^{\theta_i} + e^{\theta_j}} \quad (3.22)$$

and the covariance matrix of $(S_1, \dots, S_n)'$ is precisely the matrix $\mathcal{Q}$. Hence, based on the paired comparison formulation of the rank order data under the Plackett-Luce model, we have an exact calculation for the first two moments of the rankings.

The distribution on the ranking of all $n + 1$ item is degenerate, since $\sum_{i=0}^n r(i) = \sum_{i=1}^{n+1} i =$

$(n+2)(n+1)/2$. We can choose to leave out the reference item and define the quantity

$$\delta = (\mathbf{r} - \mathbb{E}(\mathbf{r}))' \mathcal{Q}^{-1} (\mathbf{r} - \mathbb{E}(\mathbf{r}))$$

with $\mathbf{r} = (r_1, \dots, r_n)'$ and $\mathbb{E}(\mathbf{r}) = (n+1 - \mathbb{E}(S_1), \dots, n+1 - \mathbb{E}(S_n))'$. With $\mathbb{E}(\mathbf{r})$ and $\mathcal{Q}$ describing the first two moments of rank order data under the Plackett-Luce model, we can check for goodness-of-fit of the Plackett-Luce model by examining if the distribution of δ corresponds to what we expect when the data is generated from a Plackett-Luce model.

Unfortunately, δ is not a pivotal quantity. This can be easily seen from the fact that distribution of δ depends on which player has been set as the reference player. Hence, to understand the expected behavior of δ under the Plackett-Luce model, we propose to use simulation based methods. We describe below the full procedure of our proposed diagnostics for the Plackett-Luce model.

3.4.2.1 PROPOSED PROCEDURE FOR MODEL DIAGNOSTICS

Given T independent ranked observations $A_1, \dots, A_T$, we can calculate $\hat{\boldsymbol{\theta}}$ under the Plackett-Luce model using any algorithm of choice. Below are the steps to perform model diagnostics using proposed quantity δ .

1. Given $\hat{\boldsymbol{\theta}}$, evaluate $\hat{\mathbf{r}}$ and $\hat{\mathcal{Q}}$, excluding the reference item. We define $\hat{\mathbf{r}}$ as $E(\mathbf{r})$ evaluated at $\hat{\boldsymbol{\theta}}$ using Equation (3.22).
2. Calculate $\delta_t = (\mathbf{r}_t - \hat{\mathbf{r}})' \hat{\mathcal{Q}}^{-1} (\mathbf{r}_t - \hat{\mathbf{r}})$ for $t = 1, \dots, T$.
3. Obtain a reference distribution for δ_t through a parametric bootstrap (Efron, 1982). Let B be the number of bootstrapped samples to simulate, with B being chosen to be much larger than T . For $b = 1, \dots, B$, perform the following:

(a) Simulate rank data from the Plackett-Luce model with parameter $\hat{\theta}$. Call this data $\tilde{\mathcal{D}}^{(b)}$.

One option is to simulate the ranked data through the latent variable formulation based on Gumbel distributions.

(b) Fit the Plackett-Luce model on bootstrapped data $\tilde{\mathcal{D}}^{(b)}$ and obtain estimate $\tilde{\theta}^{(b)}$.

(c) Compute the b -th bootstrapped estimate of the diagnostic, $\tilde{\delta}_b$, based on $\tilde{\mathcal{D}}^{(b)}$ and $\tilde{\theta}^{(b)}$.

Use $\tilde{\delta}_{1:B}$ as the reference distribution under the null hypothesis that the observed data come from a Plackett-Luce model.

4. Compare distribution of $\delta_{1:T}$ and $\tilde{\delta}_{1:B}$. This can be done graphically using plots such as a quantile-quantile(QQ) plot, or plots of their empirical cumulative distribution functions (CDFs). We can also perform a formal statistical test such as the Kolmogorov–Smirnov test for differences in distributions.

Notice that δ is not continuous, given n and θ , as a unique value of δ corresponds to each possible ranking of all the players. We can interpret δ as a mapping from the set of permutations onto $\mathbb{R}^+$. The mapped value δ is a distance from the expected ranking under estimated parameters $\hat{\theta}$. Instead of comparing tabulations of rankings using a multinomial contingency table, we are comparing the distribution of mapped values δ by treating it as continuous. With this interpretation, we can draw a parallel between the diagnostic δ and [Cohen and Mallows](#)' first suggestion to partition rankings using distance from a modal ranking. Instead of using a single ranking, for example, the ranking based on $\hat{\theta}$, as a modal ranking, our approach uses the expected ranking under the Plackett-Luce model. This is arguably a better representation of central tendency under the Plackett-Luce model than any specific ranking, since it incorporates information on gaps between items' worth parameters. The choice of a distance measure under δ has been preset to the Mahalanobis distance ([Mahalanobis, 1936](#)). Furthermore, since we are treating mapped values δ as continuous, we remove the need for an arbitrary

decision on how to bin rankings into distinct groups. Hence, δ can be viewed an extension of the idea from [Cohen and Mallows](#) that is sensible to the Plackett-Luce model.

3.4.3 SIMULATION STUDY

3.4.3.1 LIST OF RANKING MODELS CONSIDERED

We illustrate our proposed model diagnostic for Plackett-Luce models by examining its behavior over data generated from the Plackett-Luce model and a variant of alternative models applied to rankings with 8 items in total. Below is the list of data generating processes used in this simulation study:

1. Plackett-Luce model, with $\theta_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$ for $i = 0, \dots, n$. Rank order data is simulated using latent variable approach with Gumbel distributions.
2. Mallows ranking model with Kendall's τ metric, with density function given by Equation (3.1). The modal ranking is randomly selected from the set of all permutations. When dispersion parameter ξ is exactly 0, the Mallows model is sampling permutations randomly, this is equivalent to a Plackett-Luce model with identical location parameters.
3. Plackett-Luce model with two mixture components. Modeling rank order data with mixture of sub-populations have been investigated by [Mollica and Tardella \(2014, 2020\)](#); [Li et al. \(2016\)](#). Let us refer to the source of each rank order data as a ranker; for example, consumers ranking their product preferences. The motivation of mixture model is built upon assumptions that there exist different subgroups of rankers who have different sets of preferences. For our simulation study, we assume that there are two subgroups, with a fraction ω from the second population. For the first population, we have $\theta_{1i} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, $i = 0, \dots, n$. For the second population, $\theta_{2i} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 0.01^2)$, $i = 0, \dots, n$. In other words, the first population involves dispersed θ_{1i} so that generated ranks are relatively consistent. The second popula-

tion is a group that assign rankings close to randomly given the Plackett-Luce probabilities for rankings are nearly uniform.

4. Plackett-Luce model with noise. Consider the latent variable approach of generating ranking A_t , under which each A_t is a ranking of Gumbel random variables $W_{it}, i = 0, \dots, n$ with location parameter θ_i . We define the Plackett-Luce model with a noise data generating process, where the ranking A_t is a rank order based on ranking the $\tilde{W}_{it} = W_{it} + \epsilon_{it}$, with $\theta_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$ and $\epsilon_{it} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \frac{\pi^2}{6c^2})$. As a reference, the average Kendall's τ correlation between rankings generated under the Plackett-Luce model without and with noise for $c = 1, 2, 5$ is 0.566, 0.75 and 0.897 respectively.
5. Thurstone-Mosteller-Daniels model, with $\theta_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$ for $i = 0, \dots, n$. Under this model, the latent variables associated with each player is Normally distributed with mean θ_i and variance σ^2 . The standard version of this model assumes $\sigma = 1$.

3.4.3.2 GRAPHICAL COMPARISONS OF MODEL DIAGNOSTIC

We first give an example of graphically comparing the diagnostic δ between rankings from data and the bootstrapped rankings from the fitted Plackett-Luce model. Figures 3.4 and 3.5 display the QQ plot and empirical CDF plot of comparing diagnostic δ against its reference distribution for four different rank order data simulated under different models. When the data are not generated from the Plackett-Luce model, distribution of δ may deviate from distribution of $\tilde{\delta}$. We observe that some deviations are not easily spotted on a QQ plot. For example, in Figure 3.4, the distribution of δ and $\tilde{\delta}$ under Mallows model seems to fall close to the $y = x$ reference line but examining the empirical CDF plot in Figure 3.5 tells us that in fact there is a slight shift between distributions of δ and $\tilde{\delta}$. Here, beyond using just graphical measures, a formal statistical test is helpful.

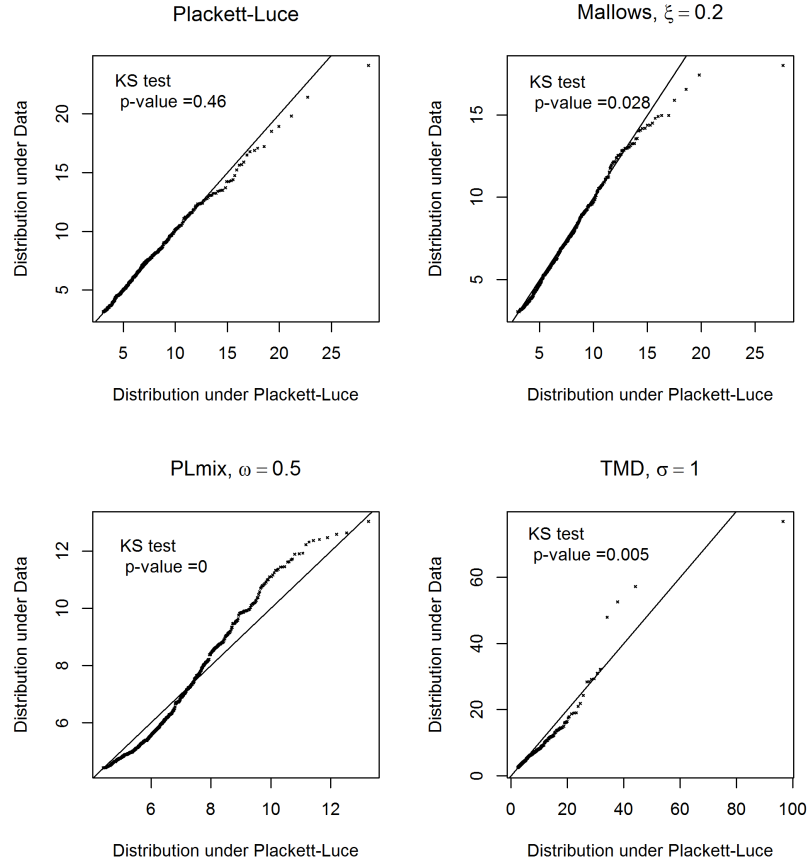


Figure 3.4: Examples of using a QQ plot to compare δ against its reference distribution for data generated under different models, sample size $T = 500$. We also include the p -value from the Kolmogorov–Smirnov test for testing the null hypothesis that distributions of δ and $\tilde{\delta}$ are identical. Upper left panel is for data generated under the Plackett-Luce model. Upper right panel is for data generating under the Mallows model with Kendall's τ metric with $\xi = 0.2$. Lower left panel is from a Plackett-Luce model with two mixture components (PLmix) with mixture weight of first component being $\omega = 0.5$. Lower right is data generated from a standard Thurstone-Mosteller-Daniels (TMD) model with $\sigma = 1$.

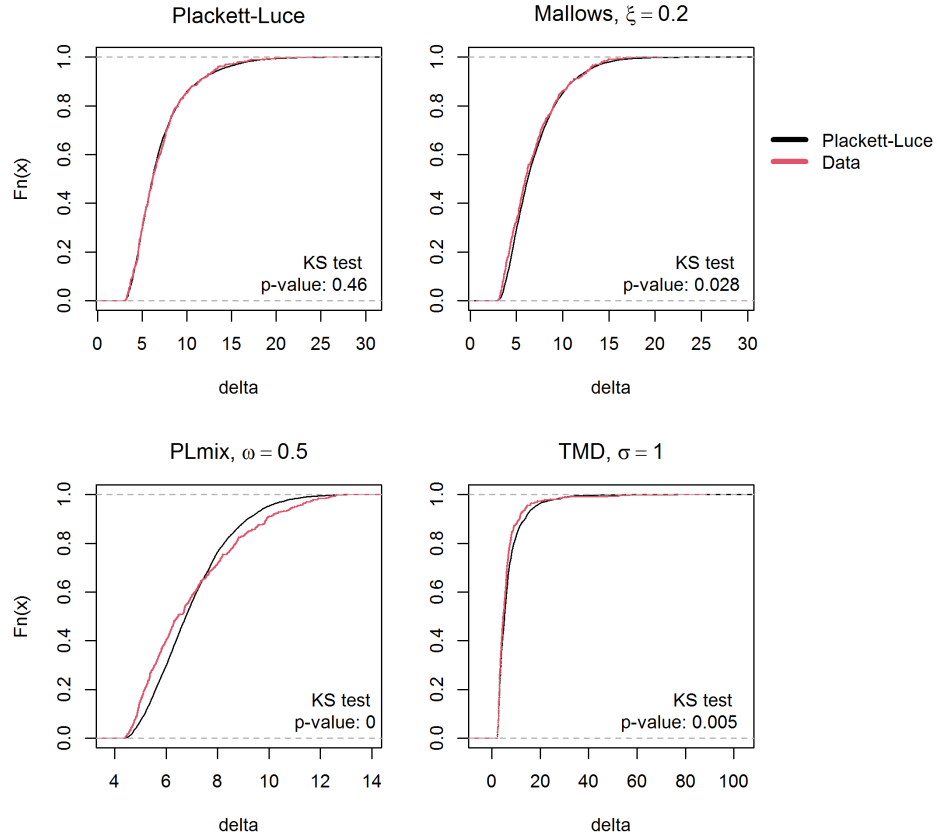


Figure 3.5: Examples of using empirical CDF of δ to compared against its reference distribution for data generated under different models, $T = 500$. Upper left panel is for data generated under the Plackett-Luce model. Upper right panel is for data generating under the Mallows model with Kendall's τ metric with $\xi = 0.2$. Lower left panel is from a Plackett-Luce model with two mixture components (PLmix) with mixture weight of first component being $\omega = 0.5$. Lower right is data generated from a standard Thurstone-Mosteller-Daniels (TMD) model with $\sigma = 1$.

3.4.3.3 POWER OF δ AS A MODEL DIAGNOSTIC MEASURE

We use simulations to examine how effective δ is at identifying deviation from the Plackett-Luce model for different ranking models listed in Section 3.4.3.1. We conduct the simulation study with 8 total items and use 500 replicates for each condition. As a reference, we compare our diagnostic measure δ against the paired-comparison diagnostic measure Z_{ij} proposed by [Cohen and Mallows \(1983\)](#). We

will refer to this diagnostic as the PC diagnostic for the rest of this section. **Cohen and Mallows** did not recommend a reference distribution for an overall test, but in the spirit of goodness-of-fit tests, we compared the test statistic $\sum_{j>i} Z_{ij}^2$ against a χ^2 distribution with $\binom{8}{2} - 8 + 1$ degree of freedom.

We first examine how δ performs at different sample sizes across different ranking models. We selected a set of parameters for each of the ranking models and selected sample sizes $T = 200, 500, 1000$. Distributions of p -values under different ranking models are plotted in Figures 3.6 and 3.7.

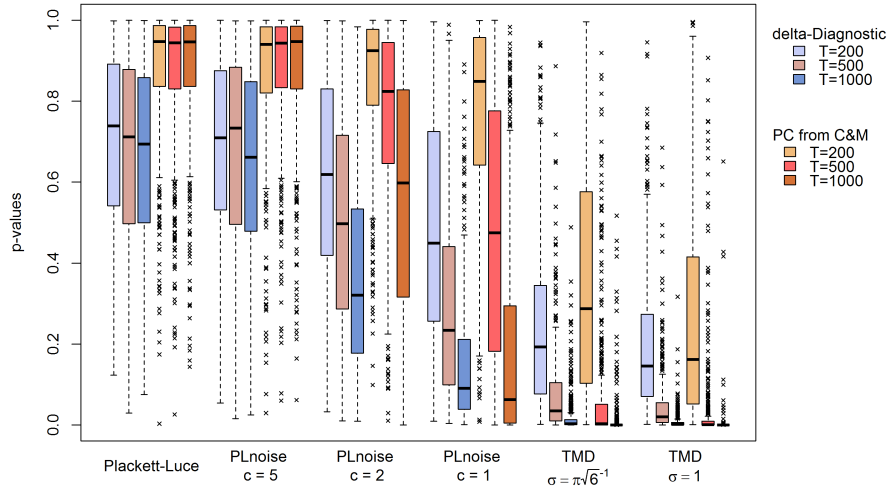


Figure 3.6: Distribution of p -values based on diagnostic δ and diagnostic based on paired comparison from **Cohen and Mallows (1983)**, for data generated under the Plackett-Luce, Plackett-Luce with noise (PLnoise) and Thurstone-Mosteller-Daniels (TMD) models. Each boxplot summarize 500 replicates.

From Figure 3.6 and 3.7, we observe that when the data are generated from the Plackett-Luce model, or when generated from a model that is very close to Plackett-Luce (Plackett-Luce model with small amount of noise with $c = 5$), both model diagnostics measures do not pick up evidence of a deviation from the Plackett-Luce model. For all other cases, the distribution of p -values shifted towards 0 as the sample size increases, suggesting an increase in evidence towards model misfit. In

this aspect, both model diagnostics measures are behaving as expected. When the data are coming from a model that is not the Plackett-Luce model, both diagnostics are able to identify deviation from Plackett-Luce model as the sample size increases.

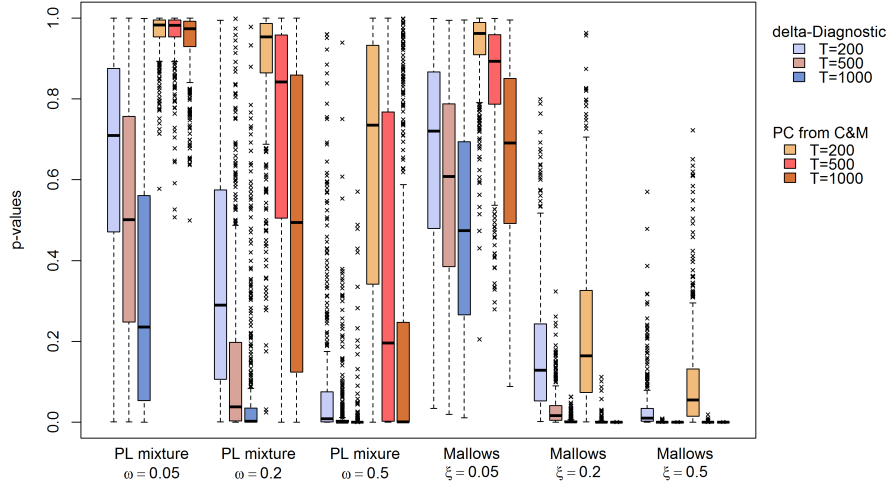


Figure 3.7: Distribution of p -values based on diagnostic δ and diagnostic based on paired comparison from [Cohen and Mallows \(1983\)](#), for data generated under the Plackett-Luce with two component mixture (PLmixture) and Mallows ranking model with Kendall's τ metric. Each boxplot summarizes 500 replicates.

Next, we compare the power of both diagnostics at $\alpha = 0.05$ for each of the non-Plackett-Luce data generating models over a sequence of parameter values. Fixing the sample size at $T = 500$, at each parameter value, we compute the proportion of p -values falling below α . Figure 3.8 gives the resulting power curves. From the top left panel of Figure 3.8, we observe that under the Plackett-Luce model with a mixture, the diagnostic δ consistently out-powers the diagnostic PC for all ω values used. We postulate that this could be an artifact of how we designed the mixture population - the second group of rankers are ranking close to randomly, hence they will not skew the distribution of counts of paired comparisons in any particular direction. The top right panel of Figure 3.8 shows that under

the Plackett-Luce model with noise, the two diagnostics are comparable.

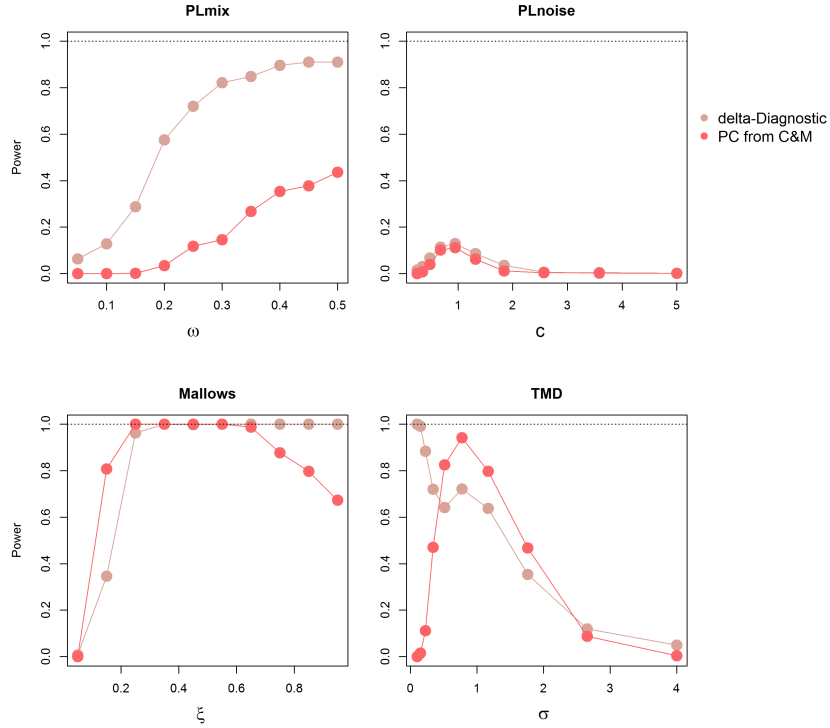


Figure 3.8: Power of model diagnostics δ and PC at $\alpha = 0.05$, for four different data generating models that are not Plackett-Luce, over a sequence of parameter values, with $T = 500$. Each power estimate is based on the proportion of p -values below $\alpha = 0.05$ over 500 replicates.

The bottom panels in Figure 3.8 illustrate the behavior of the diagnostics under the Mallows model with Kendall's τ metric and the Thurstone-Mosteller-Daniels model. We observe that the two diagnostics behave differently. For these two models, when $\xi \rightarrow 0$ and $\sigma \rightarrow \infty$, the models are behaving like uniform distributions over all rankings. When $\xi \rightarrow \infty$ and $\sigma \rightarrow 0$, the models will always return same rank order given by the model parameters. The diagnostic based on overall count of paired comparisons does not differentiate the Plackett-Luce model from the Mallows model and the Thurstone-Mosteller-Daniels model on both ends of the spectrum. In contrast, our diagnostic δ only fails to differentiate the Plackett-Luce model when the other two models are behaving like a

uniform distribution over rankings. This is sensible since there exists a set of parameter values such that the Plackett-Luce model corresponds exactly to the uniform distribution of rankings.

Overall, we conclude from the stimulation study that our diagnostic δ detects departures from an assumed model and has good power to detect such departures. Furthermore, our diagnostic is at least comparable to existing diagnostic measures.

We further investigate how the two diagnostics behave over the same data. We use the magnitude of the p -value for each model diagnostic as a proxy for strength of evidence of model misfit evaluated by the diagnostic. We measure the correlation between p -values from our diagnostic δ and the diagnostic PC using Spearman's correlation. The goal is to examine whether the two diagnostics give a consistent ranking of data in terms of how well the Plackett-Luce model fits to the data. From Table 3.3 we observe that across most data generating processes we picked, the correlation in p -values between diagnostic δ and PC are below 0.5. This hints that these two diagnostics are complementary, and can be used in tandem to identify lack of fit of the Plackett-Luce model.

T	PL	PL noise			TMD	PL mixture			Mallows		
		$c = 5$	$c = 2$	$c = 1$	$\sigma = 1$	$\omega = 0.05$	$\omega = 0.2$	$\omega = 0.5$	$\xi = 0.05$	$\xi = 0.2$	$\xi = 0.5$
200	0.060	0.054	0.196	0.333	0.415	-0.024	0.330	0.680	0.044	0.251	0.370
500	0.107	0.113	0.312	0.429	0.428	-0.008	0.571	0.812	0.177	0.215	0.446
1000	0.035	0.106	0.318	0.444	0.452	-0.051	0.682	0.834	0.222	0.198	0.397

Table 3.3: Spearman's correlation of p -values between model diagnostics δ and PC.

3.5 CONCLUSION

Observing that some practitioners may choose to analyze rank order data as paired comparisons, we were motivated to evaluate this approach. We described this approach using both a method of moments and estimating equations frameworks, which established the consistency of the estimator $\hat{\theta}_{BT}$ using paired comparisons. We also detailed the exact correction to the covariance matrix of the esti-

mator $\hat{\theta}_{BT}$ based on standard outputs of a Bradley-Terry model. Using a simulation study, we characterized the loss of efficiency in using $\hat{\theta}_{BT}$ instead of analyzing the data as rank order data under the Plackett-Luce model. Finally, we illustrated that the paired comparison formulation of rank order data gives a way to calculate first two moments of rankings. This motivated a new model diagnostic measure to assess the fit of the Plackett-Luce model. [Cohen and Mallows \(1983\)](#) proposed a goodness-of-fit diagnostic by grouping rankings using a distance measure. This idea is naturally suited for distance based ranking models. Our proposed diagnostic can be interpreted as an extension of the idea of grouping rankings using a distance measure that is suitable for the Plackett-Luce model. We demonstrated the benefits of this diagnostic in identifying evidence of misfit of a Plackett-Luce model through a simulation study.

We have focused the discussion on analysis for rank order data with complete ranking. The procedure described in Section 3.2.1 to convert rank order data into paired comparisons can also be applied to rank order data with subset rankings and top- c rankings. Suppose we have subset rankings of at most c items at a time. We believe that the loss of efficiency in using $\hat{\theta}_{BT}$ under subset ranking is less than complete ranking, since the Bradley-Terry estimator is only ignoring correlations among at most $\binom{c}{2}$ paired comparisons instead of $\binom{n+1}{2}$. We also expect the loss of efficiency under top- c ranking to lie between subset ranking of c items and complete ranking, since one set of top- c ranking converts to $\binom{c}{2} + (n+1-c)c$ paired comparisons.

When given rank order data with subset rankings, the principles of deriving our model diagnostic δ in Section 3.4.2 is still valid. However, since each ranking involves different subset of items, the calculations of the expectation and variance depends on the subset of items involved in the ranking. The computational bookkeeping involved in calculating δ may become too cumbersome. We can relieve part of the computation cost by considering the approximation of the inverse covariance matrix introduced by [Simons and Yao \(1998, 1999\)](#).

The latent variable setup of the Plackett-Luce model induces paired comparisons that are de-

scribed by the Bradley-Terry model. This extends naturally to other ordered statistics ranking models. For example, we can draw a similar relationship between the Thurstone-Mosteller-Daniels ranking model and the Thurstone-Mosteller model for paired comparison. The discussions in Section 3.2 and 3.4.2 will extend to these models. However, the exact calculations involved may not be as straight forward.

We initiated a discussion on the Plackett-Luce model, drawing on its relationship to the Bradley-Terry model. We demonstrated potential relationship between the Fisher information of a Plackett-Luce model, and the covariance matrix from a corresponding Bradley-Terry analysis. From the paired comparison formulation, we derived exact calculations for first two moments of the rank order data. The connections between the Plackett-Luce model and the Bradley-Terry model continue to be an area of potential research.



Supplemental Material of Chapter 2

A.1 ESTIMATING GLOBAL PARAMETERS

Here we provide a procedure to give reasonable estimates for global parameters h^2 and ν^2 and λ^2 , assuming that the user has access to a software that fits a Bradley-Terry model. This procedure requires data from the first 2 rating periods $(y_{ij}^{(1,2)}, n_{ij}^{(1,2)})$ for $j > i$.

A.1.1 ESTIMATING λ^2 AND ν^2

Fit a Bradley-Terry model using data from the first period $(y_{ij}^{(1)}, n_{ij}^{(1)})$, and second period $(y_{ij}^{(2)}, n_{ij}^{(2)})$. Obtain MLEs $\hat{\boldsymbol{\theta}}^{(1)}$ and $\hat{\boldsymbol{\theta}}^{(2)}$ with linear constraints $\sum_{i=1}^N \hat{\theta}_i^{(r)} = 0$. This can be done by fitting each period's Bradley-Terry model with the constraint $\theta_1 = 0$ and then center the estimates $\hat{\boldsymbol{\theta}}$.

Treating $\hat{\boldsymbol{\theta}}^{(1)}$ as the true parameter realizations from $\mathcal{N}(0, \lambda^2)$, by conjugacy we have

$$\lambda^2 | \hat{\boldsymbol{\theta}}^{(1)} \sim \text{Inv-Gamma} \left(\lambda_\alpha + N, \lambda_\beta + \frac{1}{2} \sum_{i=1}^N \left(\hat{\theta}_i^{(1)} \right)^2 \right).$$

We can estimate λ^2 by its posterior mean

$$\frac{\lambda_\beta + \frac{1}{2} \sum_{i=1}^N \left(\hat{\theta}_i^{(1)} \right)^2}{\lambda_\alpha + N - 1}.$$

Similarly, treating $\hat{\boldsymbol{\theta}}^{(1)}$ and $\hat{\boldsymbol{\theta}}^{(2)}$ as true realizations,

$$\nu^2 | \hat{\boldsymbol{\theta}}^{(1,2)} \sim \text{Inv-Gamma} \left(\nu_\alpha + N, \nu_\beta + \frac{1}{2} \sum_{i=1}^N \left(\hat{\theta}_i^{(2)} - \hat{\theta}_i^{(1)} \right)^2 \right)$$

so an estimate of ν^2 is given by

$$\frac{\nu_\beta + \frac{1}{2} \sum_{i=1}^N \left(\hat{\theta}_i^{(2)} - \hat{\theta}_i^{(1)} \right)^2}{\nu_\alpha + N - 1}.$$

A.1.2 ESTIMATING h^2

We first obtain a crude estimates for $\hat{\mathbf{k}}^{(r)}$ using the following procedure. Given $\hat{\boldsymbol{\theta}}^{(r)}$, we have

$$\eta_{ij}^{(r)} \sim \mathcal{N}(\hat{\theta}_i^{(r)} - \hat{\theta}_j^{(r)}, k_i^{2(r)} + k_j^{2(r)}).$$

Let $\text{logit}(\hat{p}_{ij}^{(r)}) = \hat{\theta}_i^{(r)} - \hat{\theta}_j^{(r)}$. By the Delta method, we have an approximate distribution on $p_{ij}^{(r)}$ given by

$$p_{ij}^{(r)} \sim \mathcal{N}\left(\hat{p}_{ij}^{(r)}, \left[\hat{p}_{ij}^{(r)}(1 - \hat{p}_{ij}^{(r)})\right]^2 (k_i^{2(r)} + k_j^{2(r)})\right).$$

We estimate $k_i^{2(r)} + k_j^{2(r)}$ using $\hat{\delta}_{ij}^2 = \left(\frac{y_{ij}^{(r)}}{n_{ij}^{(r)}} - \hat{p}_{ij}^{(r)}\right)^2 \left[\hat{p}_{ij}^{(r)}(1 - \hat{p}_{ij}^{(r)})\right]^{-2}$. Inspired by the quasi-variance approach of [Firth and De Menezes \(2004\)](#), we estimate $\hat{k}_i^{2(r)}$ by minimizing the mean square of error $\log(k_i^{2(r)} + k_j^{2(r)}) - \log(\hat{\delta}_{ij}^2)$. Then, an estimate of h^2 is given by

$$\frac{h_\beta + \frac{1}{2} \sum_{i=1}^N \left(\log(\hat{k}_i^{2(2)}) - \log(\hat{k}_i^{2(1)})\right)^2}{h_\alpha + N - 1}.$$

The procedures for getting a point estimate for ν^2 and h^2 can be extended to include data from multiple time periods. The estimates will be more reliable. However, in the spirit of initializing a new rating system, we discussed the procedure using only two time periods, which is the minimum length of observations required to give estimates of the global parameters.

A.2 FURTHER DETAILS FOR APPROXIMATION

Given $\log(\hat{k}_i^2)$, the log-posterior distribution of $\hat{\theta}_i$ is approximated by

$$\begin{aligned} \log \tilde{f}(\theta_i | \log(k_i^2), \mathbf{y}, \mathbf{n}) &= -\frac{1}{2(\nu^2 + \sigma_i^2)} (\theta_i - \mu_i)^2 \\ &+ \sum_{j \neq i} y_{ij} \frac{\theta_i - \mu_j}{g(\omega_j)} - n_{ij} \log \left(1 + e^{\frac{\theta_i - \mu_j}{g(\omega_j)}} \right). \end{aligned} \quad (\text{A.1})$$

We are interested in

$$\tilde{\theta}_i(k_i^2) = \underset{\theta_i}{\operatorname{argmax}} \log \tilde{f}(\theta_i | \log(k_i^2), \mathbf{y}, \mathbf{n}). \quad (\text{A.2})$$

The following series of approximation steps are based on the methods used in [Glickman \(1999, 2001\)](#).

Define

$$\begin{aligned} a(\theta_i) &= \sum_{j \neq i} y_{ij} \frac{\theta_i - \mu_j}{g(\omega_j)} - n_{ij} \log \left(1 + e^{\frac{\theta_i - \mu_j}{g(\omega_j)}} \right), \\ a'(\theta_i) &= \sum_{j \neq i} \frac{y_{ij}}{g(\omega_j)} - \frac{n_{ij}}{g(\omega_j)} \frac{e^{\frac{\theta_i - \mu_j}{g(\omega_j)}}}{1 + e^{\frac{\theta_i - \mu_j}{g(\omega_j)}}}, \\ a''(\theta_i) &= - \sum_{j \neq i} \frac{n_{ij}}{g(\omega_j)^2} \frac{1}{1 + e^{\frac{\theta_i - \mu_j}{g(\omega_j)}}} \frac{e^{\frac{\theta_i - \mu_j}{g(\omega_j)}}}{1 + e^{\frac{\theta_i - \mu_j}{g(\omega_j)}}}. \end{aligned}$$

Let θ_i^* be the solution of $a'(\theta_i) = 0$. Taylor approximation gives

$$\begin{aligned} a(\theta_i) &\approx a(\theta_i^*) + \frac{1}{2} a''(\theta_i^*) (\theta_i - \theta_i^*)^2 \\ f(\theta_i | \log(k_i^2), \mathbf{y}, \mathbf{n}) &\stackrel{\text{approx}}{\propto} \phi(\theta_i; \mu_i, \sigma_i^2 + \nu^2) \phi(\theta_i; \theta_i^*, \delta_i^2) \end{aligned}$$

where

$$(\delta_i^2)^{-1} = -a''(\theta_i^*).$$

The conditional distribution $f(\theta_i | \log(k_i^2), \mathbf{y}, \mathbf{n})$ is approximately Normal with mean μ_i' and variance $\sigma_i^{2'}$ given by

$$\sigma_i^{2'} = \left(\frac{1}{\sigma_i^2 + \nu^2} + \frac{1}{\delta_i^2} \right)^{-1},$$

$$\begin{aligned}\mu'_i &= \sigma_i^{2'} \left(\frac{\mu_i}{\sigma_i^2 + \nu^2} + \frac{\theta_i^*}{\delta_i^2} \right) \\ &= \mu_i + \frac{1/\delta_i^2}{1/(\sigma_i^2 + \nu^2) + 1/\delta_i^2} (\theta_i^* - \mu_i).\end{aligned}$$

Let

$$h(\theta_i) = \sum_{j \neq i} \frac{n_{ij}}{g(\omega_j)} \frac{e^{\frac{\theta_i - \mu_j}{g(\omega_j)}}}{1 + e^{\frac{\theta_i - \mu_j}{g(\omega_j)}}}.$$

Using the fact that $h(\theta_i^*) = \sum_{j \neq i} \frac{y_{ij}}{g(\omega_j)}$ and the approximation $h(\mu_i) \approx h(\theta_i^*) + h'(\theta_i^*)(\mu_i - \theta_i^*)$

we have

$$\theta_i^* - \mu_i \approx \frac{h(\theta_i^*) - h(\mu_i)}{h'(\theta_i^*)}.$$

Recognizing that $h'(\theta_i) = -a''(\theta_i)$ and $h(\theta_i^*) - h(\theta_i) = a'(\theta_i)$, we have

$$\begin{aligned}\mu'_i &= \mu_i + \frac{1/\delta_i^2}{1/(\sigma_i^2 + \nu^2) + 1/\delta_i^2} (\theta_i^* - \mu_i) \\ &\approx \mu_i + \frac{1/\delta_i^2}{1/(\sigma_i^2 + \nu^2) + 1/\delta_i^2} \frac{h(\theta_i^*) - h(\mu_i)}{h'(\theta_i^*)} \\ &= \mu_i + \frac{a'(\mu_i)}{1/(\sigma_i^2 + \nu^2) + 1/\delta_i^2} \\ &\approx \mu_i + \frac{a'(\mu_i)}{1/(\sigma_i^2 + \nu^2) + 1/\tilde{\delta}_i^2}\end{aligned}$$

where in the last step we introduce the approximation $(\delta_i^2)^{-1} = -a''(\theta_i^*) \approx -a''(\mu_i) = (\tilde{\delta}_i^2)^{-1}$.

Using the approximation based on methods from [Glickman \(1999, 2001\)](#) derived above, we have

$$\tilde{\theta}_i(k_i^2) \approx \mu_i + \frac{a'(\mu_i)}{1/(\sigma_i^2 + \nu^2) - a''(\mu_i)}.$$

B

Supplemental Materials of Chapter 3

B.1 CALCULATING FISHER INFORMATION UNDER PLACKETT-LUCE MODEL

Given ranking indices $A = (a(1), \dots, a(n+1))'$ and equivalently ranks $(r(0), \dots, r(n))$, the loglikelihood of the parameters under the Plackett-Luce model is

$$\ell_{PL}(\boldsymbol{\theta}) = \sum_{k=1}^n \log \left(\frac{\gamma_{a(k)}}{\sum_{l=k}^{n+1} \gamma_{a(l)}} \right) = \sum_{k=1}^n \log \gamma_{a(k)} - \sum_{k=1}^n \log \sum_{l=k}^{n+1} \gamma_{a(l)}$$

Notice that θ_i only appears in the expression $\sum_{l=k}^{n+1} \gamma_{a(l)}$ if $r(i) \geq k$.

$$\begin{aligned}\frac{\partial \ell_{PL}}{\partial \theta_i} &= 1_{\{r(i) < n+1\}} - \sum_{k=1}^{r(i)} \frac{e^{\theta_i}}{\sum_{l=k}^{n+1} \gamma_{a(l)}} \\ \frac{\partial^2 \ell_{PL}}{\partial \theta_i^2} &= - \sum_{k=1}^{r(i)} \frac{e^{\theta_i}}{\sum_{l=k}^{n+1} \gamma_{a(l)}} \left(1 - \frac{e^{\theta_i}}{\sum_{l=k}^{n+1} \gamma_{a(l)}} \right) \\ \frac{\partial^2 \ell_{PL}}{\partial \theta_i \partial \theta_j} &= \sum_{k=1}^{\min(r(i), r(j))} \frac{e^{\theta_i}}{\sum_{l=k}^{n+1} \gamma_{a(l)}} \frac{e^{\theta_j}}{\sum_{l=k}^{n+1} \gamma_{a(l)}}\end{aligned}$$

The second derivatives are unchanged if we shift all of θ_i by the same constant. Hence, we can rescale γ_i by a multiplicative constant such that $\sum_{i=0}^n \gamma_i = 1$.

Let $\sigma_s, s = 1, \dots, (n+1)!$, be the set of permutations of $n+1$ players. Let $r_s(i)$ be the ranking of player i under permutation σ_s , let $\sigma_s(k)$ be the index of player ranked k under permutation σ_s . We are interested in calculating the expectation of the second derivative of the log-likelihood.

$$\begin{aligned}\mathbb{E} \left[-\frac{\partial^2 \ell_{PL}}{\partial \theta_i^2} \right] &= \sum_{s=1}^{(n+1)!} \mathbb{P}(\sigma_s) \left(\sum_{k=1}^{r_s(i)} \frac{e^{\theta_i}}{\sum_{l=k}^{n+1} \gamma_{a(l)}} \left(1 - \frac{e^{\theta_i}}{\sum_{l=k}^{n+1} \gamma_{a(l)}} \right) \right) \\ &= \sum_{k=1}^n \sum_{s=1}^{(n+1)!} \mathbb{P}(\sigma_s) \left(\frac{e^{\theta_i}}{\sum_{l=k}^{n+1} \gamma_{a(l)}} \left(1 - \frac{e^{\theta_i}}{\sum_{l=k}^{n+1} \gamma_{a(l)}} \right) 1_{\{r_s(i) \geq k\}} \right)\end{aligned}$$

Let us consider the sum by values of k . Since $r_s(i) \geq 1, \forall s$,

$$\sum_{s=1}^{(n+1)!} \mathbb{P}(\sigma_s) \left(\frac{e^{\theta_i}}{\sum_{l=1}^{n+1} \gamma_{a(l)}} \left(1 - \frac{e^{\theta_i}}{\sum_{l=1}^{n+1} \gamma_{a(l)}} \right) \right) = e^{\theta_i} (1 - e^{\theta_i}) \sum_{s=1}^{(n+1)!} \mathbb{P}(\sigma_s) = \gamma_i (1 - \gamma_i)$$

The indicator $1_{\{r_s(i) \geq 2\}}$ is non-zero only if $\sigma_s(1) \neq i$ and the denominator in the sum $\sum_{l=2}^{n+1} \gamma_{a(l)}$

only depends on value of $\sigma_s(1)$.

$$\begin{aligned}
& \sum_{s=1}^{(n+1)!} \mathbb{P}(\sigma_s) \left(\frac{e^{\theta_i}}{\sum_{l=2}^{n+1} \gamma_{a(l)}} \left(1 - \frac{e^{\theta_i}}{\sum_{l=2}^{n+1} \gamma_{a(l)}} \right) 1_{\{r_s(i) \geq 2\}} \right) \\
&= \sum_{j \neq i} \frac{e^{\theta_i}}{1 - e^{\theta_j}} \left(1 - \frac{e^{\theta_i}}{1 - e^{\theta_j}} \right) \left(\sum_{s: \sigma_s(1)=j} \mathbb{P}(\sigma_s) \right) \\
&= \sum_{j \neq i} \gamma_j \frac{\gamma_i}{1 - \gamma_j} \left(1 - \frac{\gamma_i}{1 - \gamma_j} \right)
\end{aligned}$$

We can work through all the values of k by observing that the denominator only depends on a permutation of the first $k - 1$ players, and the first $k - 1$ players must exclude player i for the summation term to be present.

$$\begin{aligned}
\mathbb{E} \left[-\frac{\partial^2 \ell_{PL}}{\partial \theta_i^2} \right] &= \gamma_i(1 - \gamma_i) + \sum_{(s_1)} \gamma_{s_1} \frac{\gamma_i}{1 - \gamma_{s_1}} \left(1 - \frac{\gamma_i}{1 - \gamma_{s_1}} \right) \\
&+ \sum_{(s_1, s_2)} \gamma_{s_1} \frac{\gamma_{s_2}}{1 - \gamma_{s_1}} \frac{\gamma_i}{1 - \gamma_{s_1} - \gamma_{s_2}} \left(1 - \frac{\gamma_i}{1 - \gamma_{s_1} - \gamma_{s_2}} \right) \\
&\vdots \\
&+ \sum_{(s_1, \dots, s_{n-1})} \left(\gamma_{s_1} \prod_{l=2}^{n-2} \frac{\gamma_{s_{l+1}}}{1 - \sum_{m=1}^l \gamma_{s_m}} \right) \frac{\gamma_i}{1 - \sum_{l=1}^{n-1} \gamma_{s_l}} \left(1 - \frac{\gamma_i}{1 - \sum_{l=1}^{n-1} \gamma_{s_l}} \right)
\end{aligned} \tag{B.1}$$

where $(s_1, \dots, s_k)$ is the index of players at position 1 to k when choosing k players sequentially from all players excluding player i , and the summation is over all possible sets of indices.

Similarly, we can work through the derivation of $\mathbb{E} \left[\frac{\partial^2 \ell_{PL}}{\partial \theta_i \partial \theta_j} \right]$ where we have

$$\begin{aligned}
\mathbb{E} \left[\frac{\partial^2 \ell_{PL}}{\partial \theta_i \partial \theta_j} \right] &= \gamma_i \gamma_j + \sum_{(s_1)} \gamma_{s_1} \frac{\gamma_i}{1 - \gamma_{s_1}} \frac{\gamma_j}{1 - \gamma_{s_1}} \\
&\quad + \sum_{(s_1, s_2)} \gamma_{s_1} \frac{\gamma_{s_2}}{1 - \gamma_{s_1}} \frac{\gamma_i}{1 - \gamma_{s_1} - \gamma_{s_2}} \frac{\gamma_j}{1 - \gamma_{s_1} - \gamma_{s_2}} \\
&\quad \vdots \\
&\quad + \sum_{(s_1, \dots, s_{n-1})} \left(\gamma_{s_1} \prod_{l=2}^{n-2} \frac{\gamma_{s_{l+1}}}{1 - \sum_{m=1}^l \gamma_{s_m}} \right) \frac{\gamma_i}{1 - \sum_{l=1}^{n-1} \gamma_{s_l}} \frac{\gamma_j}{1 - \sum_{l=1}^{n-1} \gamma_{s_l}}
\end{aligned} \tag{B.2}$$

where $(s_1, \dots, s_k)$ is the index of items at position 1 to k when choosing k items sequentially from all players excluding item i, j , and the summation is over all possible sets of indices.

Comparing Equations (B.1) and (B.2), we can also conclude that

$$\mathbb{E} \left[-\frac{\partial^2 \ell_{PL}}{\partial \theta_i^2} \right] = \sum_{j=0, j \neq i}^n \mathbb{E} \left[\frac{\partial^2 \ell_{PL}}{\partial \theta_i \partial \theta_j} \right].$$

This completes derivation of the expression of matrix $\mathcal{M}$.

B.2 ADDITIONAL PLOTS FROM THE SIMULATION STUDY

Results from Section 3.3.2.2 for $n + 1 = 3, 5$ and 8. In black is $\mathcal{R}_n$ based on γ^n sampled from $\text{Dir}(\frac{1}{n+1}, \dots, \frac{1}{n+1})$ and γ^n sampled by conditioning on v_m for $n + 1 = 7$. In grey is $\mathcal{R}_n$ based on θ^n sampled from Normal distributions.

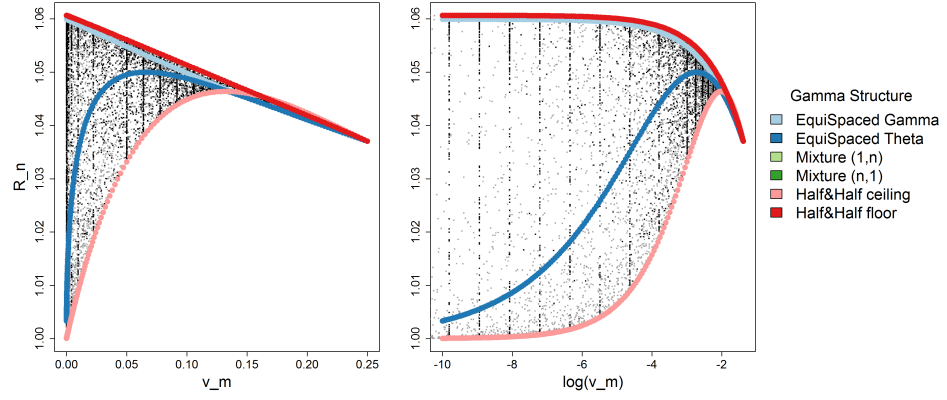


Figure B.1: Quantity $\mathcal{R}_n$ as a function of v_m on linear and log scale of v_m for $n+1=3$.

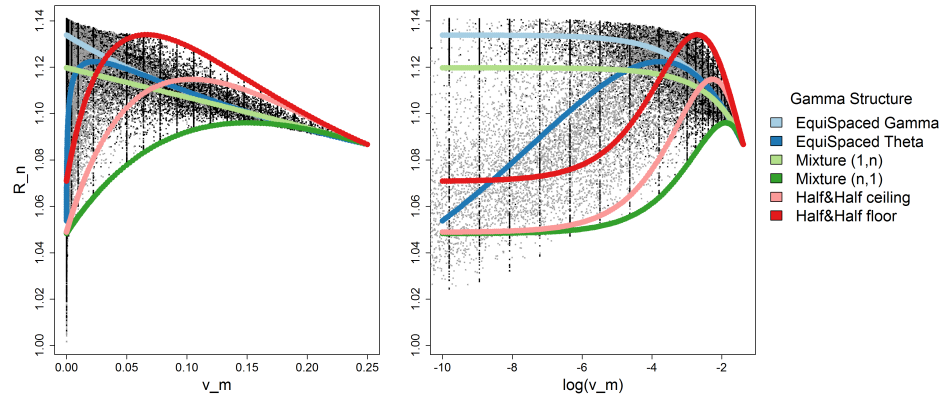


Figure B.2: Quantity $\mathcal{R}_n$ as a function of v_m on linear and log scale of v_m for $n+1=5$.

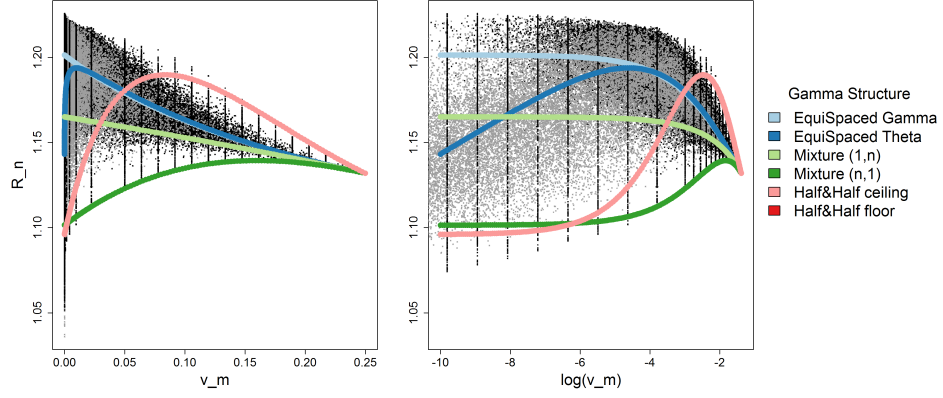


Figure B.3: Quantity $\mathcal{R}_n$ as a function of v_m on linear and log scale of v_m for $n + 1 = 8$.

B.3 ANALYZING $\mathcal{R}_n$ APPROXIMATELY

B.3.1 APPROXIMATING $\mathcal{R}_n$

Given $n \times n$ symmetric positive definite matrix $\mathcal{T} = (\tau_{ij})$ with negative off-diagonal elements and positive row sums, define an array of positive numbers $\{u_{ij}\}_{i,j=1}^n$ with

$$u_{ij} = -\tau_{ij}, \quad i \neq j$$

and

$$u_{ii} = \sum_{k=1}^n \tau_{ik}.$$

We have $u_{ij} > 0$, $u_{ij} = u_{ji}$ and $\tau_{ii} = \sum_{k=1}^n u_{ik}$. With notation $u_m = \min_{i,j} u_{ij}$, $u_M = \max_{i,j} u_{ij}$ and $\tau_{..} = \sum_{i,j=1}^n \tau_{ij} = \sum u_{kk} > 0$, $\|\mathcal{A}\| = \max_{i,j} |a_{ij}|$ for a general matrix $\mathcal{A}$.

[Simons and Yao \(1998, 1999\)](#) determine a symmetric positive definite matrix $\mathcal{S}$ with

$$s_{ij} = \frac{\delta_{i,j}}{\tau_{ii}} + \frac{1}{\tau_{..}}$$

where $\delta_{i,j}$ is the Kronecker delta function (equals 1 if $i = j$ and 0 otherwise) such that

$$\|\mathcal{T}^{-1} - \mathcal{S}\| \leq \frac{C(u_m, u_M)}{n^2}$$

where $C(u_m, u_M) = \left(1 + \frac{u_M}{u_m}\right) \frac{u_M}{u_m^2}$. So $\mathcal{S}$ is a close approximation for $\mathcal{T}^{-1}$ when n is large.

DERIVING $\tilde{\mathcal{R}}_n$ Matrices $\mathcal{M}$, $\mathcal{Q}$ and $\mathcal{J}$ are symmetric positive definite matrices with negative off-diagonal elements. Using results from [Simons and Yao \(1998, 1999\)](#), with $\mathcal{M}$ as an example, the approximation for inverse of the matrix is given by

$$\tilde{\mathcal{M}}^{-1} = \begin{pmatrix} \mathcal{M}_{11}^{-1} & & \\ & \ddots & \\ & & \mathcal{M}_{nn}^{-1} \end{pmatrix} + \frac{1}{\sum_{i=1}^n \sum_{j=1}^n \mathcal{M}_{ij}} \mathbf{1}_n \mathbf{1}_n^T$$

By definition of $\mathcal{M}$,

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^n \mathcal{M}_{ij} &= \sum_{i=1}^n \mathcal{M}_{ii} + \sum_{i=1}^n \sum_{j=1, j \neq i}^n \mathcal{M}_{ij} \\ &= \sum_{i=1}^n \sum_{j=0, j \neq i}^n \Delta_{ij} - \sum_{i=1}^n \sum_{j=1, j \neq i}^n \Delta_{ij} \\ &= \sum_{j=1}^n \Delta_{j0} \\ &= \sum_{j=1}^n \Delta_{0j} = \mathcal{M}_{00} \end{aligned}$$

where $\mathcal{M}_{00}$ is defined by extending the definition in Equation (3.16) to the case of $i = 0$. It follows that

$$\begin{aligned}\det(\tilde{\mathcal{M}}^{-1}) &= \left(\prod_{i=1}^n \frac{1}{\mathcal{M}_{ii}} \right) \left(1 + \frac{\sum_{i=1}^n \mathcal{M}_{ii}}{\mathcal{M}_{00}} \right) \\ &= \prod_{i=0}^n \frac{1}{\mathcal{M}_{ii}} \left(\sum_{i=0}^n \mathcal{M}_{ii} \right).\end{aligned}$$

Using similar methods, we can also show that $\sum_{i=1}^n \sum_{j=1}^n \mathcal{Q}_{ij} = \mathcal{Q}_{00}$ and $\sum_{i=1}^n \sum_{j=1}^n \mathcal{J}_{ij} = \mathcal{J}_{00}$.

In fact, this conclusion can also be argued through the definition of $\mathcal{J}$ and $\mathcal{Q}$ as covariance matrices.

We therefore have

$$\begin{aligned}\det(\tilde{\mathcal{Q}}^{-1}) &= \prod_{i=0}^n \frac{1}{\mathcal{Q}_{ii}} \left(\sum_{i=0}^n \mathcal{Q}_{ii} \right) \\ \det(\tilde{\mathcal{J}}^{-1}) &= \prod_{i=0}^n \frac{1}{\mathcal{J}_{ii}} \left(\sum_{i=0}^n \mathcal{J}_{ii} \right)\end{aligned}$$

This expression for approximating the determinant is also reasonable in the sense that it does not depend on the choice of the reference item in estimation. It follows that an approximation for

$$\mathcal{R}_n = \left(\frac{\det(\mathcal{J}^{-1} \mathcal{Q} \mathcal{J}^{-1})}{\det(\mathcal{M}^{-1})} \right)^{\frac{1}{n}}$$

is given by

$$\begin{aligned}\tilde{R}_n &= \left(\frac{\det(\tilde{\mathcal{J}}^{-1}) \det(\tilde{\mathcal{J}}^{-1})}{\det(\tilde{\mathcal{Q}}^{-1}) \det(\tilde{\mathcal{M}}^{-1})} \right)^{\frac{1}{n}} \\ &= \left(\left(\prod_{i=0}^n \frac{\mathcal{Q}_{ii} \mathcal{M}_{ii}}{\mathcal{J}_{ii}^2} \right) \left(\frac{\sum_{i=0}^n \mathcal{J}_{ii} \sum_{i=0}^n \mathcal{J}_{ii}}{\sum_{i=0}^n \mathcal{Q}_{ii} \sum_{i=0}^n \mathcal{M}_{ii}} \right) \right)^{\frac{1}{n}} \\ &= \left(\prod_{i=0}^n \frac{\mathcal{Q}_{ii}}{\mathcal{J}_{ii}^2} \right)^{1/n} \left(\frac{\sum_{i=0}^n \mathcal{J}_{ii} \sum_{i=0}^n \mathcal{J}_{ii}}{\sum_{i=0}^n \mathcal{Q}_{ii}} \right)^{1/n} \left(\prod_{i=0}^n \frac{\mathcal{M}_{ii}}{\sum_{i=0}^n \mathcal{M}_{ii}} \right)^{1/n}.\end{aligned}$$

B.3.2 NUMERICAL EVIDENCE FOR STUDYING $\tilde{\mathcal{R}}_n$

The approximation for the matrix inverse by [Simons and Yao \(1998, 1999\)](#) has an error bound inversely related to the square of n . However, even when n is low, like in our simulations study, we find that $\tilde{\mathcal{R}}_n$ behaves well as an approximation for $\mathcal{R}_n$.

Figure B.4 plots $\tilde{\mathcal{R}}_n$ against $\mathcal{R}_n$ at $n + 1 = 7$. Even at small values of n , most of the points fall around the $y = x$ line. The estimate $\tilde{\mathcal{R}}_n$ tends to over-approximate $\mathcal{R}_n$ at high values of $\mathcal{R}_n$. We are using $\tilde{\mathcal{R}}_n$ to derive an upper bound for $\mathcal{R}_n$. Over-approximation is not a concern.

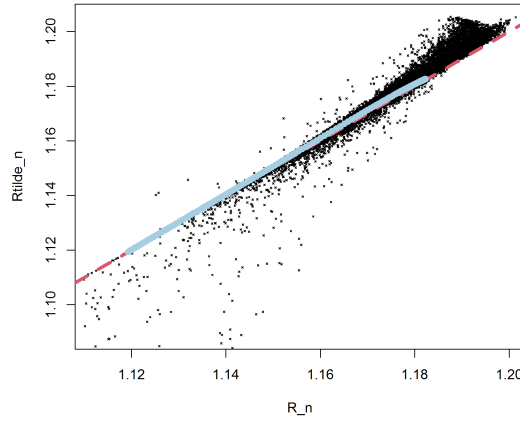


Figure B.4: $\tilde{\mathcal{R}}_n$ against $\mathcal{R}_n$ with $n + 1 = 7$, for γ^n sampled from $\text{Dir}(\frac{1}{n+1}, \dots, \frac{1}{n+1})$ and sampled by conditioning on v_m . $\tilde{\mathcal{R}}_n^{\text{EqG}}$ and $\mathcal{R}_n^{\text{EqG}}$ given in light blue. Red dotted line gives the $y = x$ line.

Figure B.5 replicates Figure 3.3 but uses $\tilde{\mathcal{R}}_n$ on the y -axis. Comparing against Figure 3.3, we observe that the behavior of $\tilde{\mathcal{R}}_n(v_m)$ over v_m is similar to the behavior of $\mathcal{R}_n(v_m)$. Under different structures for γ^n , the biggest difference between $\tilde{\mathcal{R}}_n$ and $\mathcal{R}_n$ occur when the two types of items are mixed half and half. $\tilde{\mathcal{R}}_n^{0.5c}(0)$ and $\tilde{\mathcal{R}}_n^{0.5f}(0)$ is smaller than $\mathcal{R}_n^{0.5c}(0)$ and $\mathcal{R}_n^{0.5f}(0)$.

Overall, we find that even with small n , the numerical evidence suggests that it is reasonable to study $\tilde{\mathcal{R}}_n$ as an approximation to $\mathcal{R}_n$.

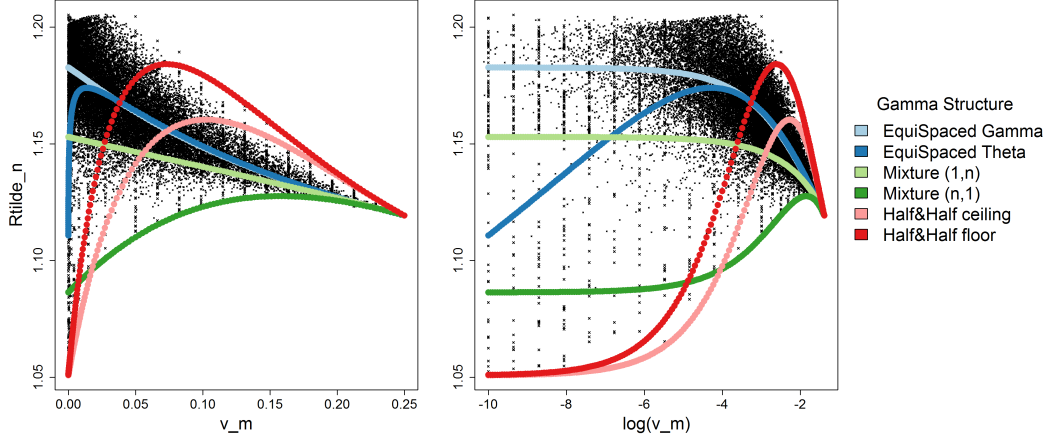


Figure B.5: Quantity $\tilde{R}_n$ as a function of v_m on linear and log scale of v_m for $n + 1 = 7$ under 6 different choices of structure for γ^n . In black is $\mathcal{R}_n$ based on γ^n sampled from $\text{Dir}(\frac{1}{n+1}, \dots, \frac{1}{n+1})$ and γ^n sampled by conditioning on v_m for $n + 1 = 7$.

B.3.3 BOUNDING $\tilde{R}_n$ AS $n \rightarrow \infty$, GIVEN v_m

In this section, we study the upper bound of $\tilde{R}_n$ by fixing v_m , the maximum gap in worth between two items and letting $n \rightarrow \infty$. We will show that fixing v_m , $\tilde{R}_n$ is finite as $n \rightarrow \infty$.

Using the fact that $\sum_{i=0}^n \mathcal{M}_{ii} \leq n$ (derivation in Appendix B.3.4) and the inequality of arithmetic and geometric means (AM-GM inequality), we have

$$\begin{aligned}
 \left(\prod_{i=0}^n \frac{\mathcal{M}_{ii}}{\sum_{i=0}^n \mathcal{M}_{ii}} \right)^{1/n} &\leq \left(\frac{\left(\frac{1}{n+1} \sum_{i=0}^n \mathcal{M}_{ii} \right)^{n+1}}{\sum_{i=0}^n \mathcal{M}_{ii}} \right)^{1/n} \\
 &= \left(\frac{1}{n+1} \right)^{\frac{n+1}{n}} \sum_{i=0}^n \mathcal{M}_{ii} \\
 &\leq n \left(\frac{1}{n+1} \right)^{\frac{n+1}{n}} \\
 &\rightarrow 1, \quad n \rightarrow \infty
 \end{aligned}$$

Using Equation (3.19) and (3.20), we have inequalities

$$J_{ii} \leq \frac{n}{4}$$

and

$$\mathcal{Q}_{ii} \geq nv_m + n(n-1)v_m^2 (1+r_M^{-1})^2 \frac{1}{1+2r_M^{-1}}.$$

Detailed derivation of lower bound for $\mathcal{Q}_{ii}$ is given in Appendix B.3.5. With this, we have

$$\begin{aligned} \left(\frac{\sum_{i=0}^n \mathcal{J}_{ii} \sum_{i=0}^n \mathcal{J}_{ii}}{\sum_{i=0}^n \mathcal{Q}_{ii}} \right)^{1/n} &\leq \left(\frac{((n+1)\frac{n}{4})^2}{(n+1)nv_m + (n+1)n(n-1)v_m^2 (1+r_M^{-1})^2 \frac{1}{1+2r_M^{-1}}} \right)^{1/n} \\ &\rightarrow n^{1/n} = 1, n \rightarrow \infty \end{aligned}$$

We also have

$$\begin{aligned} \mathcal{Q}_{ii} &\leq \sum_{j=0, j \neq i}^n v_{ij} + \frac{(1+r_M)^2}{1+2r_M} \sum_{j=0, j \neq i}^n \sum_{k \neq i, j}^n v_{ij} v_{ik} \\ &= \sum_{j=0, j \neq i}^n v_{ij} + \frac{(1+r_M)^2}{1+2r_M} \left(\left(\sum_{j=0, j \neq i}^n v_{ij} \right)^2 - \sum_{j=0, j \neq i}^n v_{ij}^2 \right) \end{aligned}$$

Using the AM-GM inequality again,

$$\begin{aligned} \left(\prod_{i=0}^n \frac{\mathcal{Q}_{ii}}{\mathcal{J}_{ii}^2} \right)^{\frac{1}{n+1}} &\leq \frac{1}{n+1} \sum_{i=0}^n \frac{\mathcal{Q}_{ii}}{\mathcal{J}_{ii}^2} \\ &= \frac{1}{n+1} \sum_{i=0}^n \frac{\sum_{j=0, j \neq i}^n \left[v_{ij} + \sum_{k=0, k \neq i, j}^n \frac{(\gamma_i + \gamma_j)(\gamma_i + \gamma_k)}{(\gamma_i + \gamma_j + \gamma_k)\gamma_i} v_{ij} v_{ik} \right]}{\left(\sum_{j=0, j \neq i}^n v_{ij} \right)^2} \end{aligned}$$

$$\begin{aligned}
&\leq \frac{1}{n+1} \sum_{i=0}^n \frac{1}{\sum_{j \neq i} v_{ij}} + \frac{(1+r_M)^2}{1+2r_M} \left(1 - \sum_{j=0, j \neq i}^n \left(\frac{v_{ij}}{\sum_{k \neq i} v_{ik}} \right)^2 \right) \\
&\leq \frac{1}{nv_m} + \frac{(1+r_M)^2}{1+2r_M} \\
&\rightarrow \frac{(1+r_M)^2}{1+2r_M}, n \rightarrow \infty.
\end{aligned}$$

Combining the three statements, given v_m ,

$$\begin{aligned}
\tilde{\mathcal{R}}_n &= \left(\prod_{i=0}^n \frac{\mathcal{Q}_{ii}}{\mathcal{J}_{ii}^2} \right)^{1/n} \left(\frac{\sum_{i=0}^n \mathcal{J}_{ii} \sum_{i=0}^n \mathcal{J}_{ii}}{\sum_{i=0}^n \mathcal{Q}_{ii}} \right)^{1/n} \left(\prod_{i=0}^n \frac{\mathcal{M}_{ii}}{\sum_{i=0}^n \mathcal{M}_{ii}} \right)^{1/n} \\
&\leq 1 \cdot 1 \cdot \frac{(1+r_M)^2}{1+2r_M}, n \rightarrow \infty.
\end{aligned}$$

Here we derived that given v_m , $\tilde{\mathcal{R}}_n$ is finite as $n \rightarrow \infty$. Unfortunately, this bound is too loose to be numerically useful. In fact, this upper bound fails to reflect the fact that $\tilde{\mathcal{R}}_n(v_m)$ stabilizes as $v_m \rightarrow 0$, where we use $\tilde{\mathcal{R}}_n(v_m)$ to denote maximum value of $\tilde{\mathcal{R}}_n$ given v_m . The main reason we failed to capture this behavior is that we did not incorporate the fact that $\mathcal{M}_{ii}$ is inversely related to $\mathcal{Q}_{ii}$ in our bound.

B.3.4 BOUNDING DIAGONAL ENTRIES OF $\mathcal{M}$

We offer two perspectives. The first bounds $\mathcal{M}_{ii}$ from above individually, but relies on sufficient condition for uniqueness of MLE. The second directly bounds $\sum_{i=0}^n \mathcal{M}_{ii}$ from above.

BOUNDING $\mathcal{M}_{ii}$ Given T independent ranking data, let $r_t(i)$ be the rank of item i in ranking t . Suppressing subscript PL for now, under the Plackett-Luce model, when the sufficient conditions for

uniqueness of MLE are met, we must have $\hat{\boldsymbol{\theta}}$ satisfying

$$0 = \sum_{t=1}^T 1_{\{r_t(i) < n+1\}} - \sum_{t=1}^T \sum_{k=1}^{r_t(i)} \frac{e^{\hat{\theta}_i}}{\sum_{l=k}^{n+1} \hat{\gamma}_{a(l)}}$$

for $i = 1, \dots, n$. Dividing both side by T , with consistency of the MLE, when $T \rightarrow \infty$, we have

$$\mathbb{E} [1_{\{r(i) < n+1\}}] = \mathbb{E} \left[\sum_{k=1}^{r(i)} \frac{e^{\theta_i}}{\sum_{l=k}^{n+1} \gamma_{a(l)}} \right]$$

where distribution of $r(i)$ is given by Plackett-Luce model with parameter $\boldsymbol{\theta}$. Using the same trick as Appendix B.1,

$$\begin{aligned} \mathbb{E} \left[\sum_{k=1}^{r(i)} \frac{e^{\theta_i}}{\sum_{l=k}^{n+1} \gamma_{a(l)}} \right] &= \gamma_i + \sum_{(s_1)} \gamma_{s_1} \frac{\gamma_i}{1 - \gamma_{s_1}} \\ &\quad + \sum_{(s_1, s_2)} \gamma_{s_1} \frac{\gamma_{s_2}}{1 - \gamma_{s_1}} \frac{\gamma_i}{1 - \gamma_{s_1} - \gamma_{s_2}} \\ &\quad \vdots \\ &\quad + \sum_{(s_1, \dots, s_{n-1})} \left(\gamma_{s_1} \prod_{l=2}^{n-2} \frac{\gamma_{s_{l+1}}}{1 - \sum_{m=1}^l \gamma_{s_m}} \right) \frac{\gamma_i}{1 - \sum_{l=1}^{n-1} \gamma_{s_l}} \end{aligned}$$

where $(s_1, \dots, s_k)$ is the index of players at position 1 to k when choosing k players sequentially from all players excluding player i , and the summation is over all possible sets of indices. We also have

$$\mathcal{M}_{ii} \leq \mathbb{E} \left[\sum_{k=1}^{r(i)} \frac{e^{\theta_i}}{\sum_{l=k}^{n+1} \gamma_{a(l)}} \right]$$

since the l th term in the expression of $\mathcal{M}_{ii}$ corresponds to the l th term $\mathbb{E} \left[\sum_{k=1}^{r(i)} \frac{e^{\theta_i}}{\sum_{l=k}^{n+1} \gamma_{a(l)}} \right]$ multiplied by a number between 0 and 1, and each term of the expression is positive. Hence we have

$$\mathcal{M}_{ii} \leq \mathbb{P}(r(i) < n+1) = 1 - \mathbb{P}(r(i) = n+1).$$

We also have

$$\sum_{i=0}^n \mathcal{M}_{ii} \leq \sum_{i=0}^n (1 - \mathbb{P}(r(i) = n+1)) = n+1 - \sum_{i=0}^n \mathbb{P}(r(i) = n+1) = n.$$

DIRECT BOUND ON $\sum_{i=0}^n \mathcal{M}_{ii}$ Following the definition of $\mathcal{M}_{ii}$ given by

$$\mathcal{M}_{ii} = \sum_{j=0, j \neq i}^n \Delta_{ij} \quad i = 1, \dots, n$$

and extending it to the case $i = 0$. We have

$$\sum_{i=0}^n \mathcal{M}_{ii} = \sum_{i=0}^n \sum_{j=0, j \neq i}^n \Delta_{ij}$$

where Δ_{ij} consists of n terms. We work through the summation one by one. For the first term in Δ_{ij} ,

$$\sum_{i=0}^n \sum_{j=0, j \neq i}^n \gamma_i \gamma_j = \left(\sum_{i=0}^n \gamma_i \right)^2 - \sum_{i=0}^n \gamma_i^2 \leq 1$$

by the constraint $\sum_{i=0}^n \gamma_i = 1$. Similarly, the summation over the second term in Δ_{ij} is

$$\sum_{i=0}^n \sum_{j=0, j \neq i}^n \sum_{k \neq i, j} \frac{\gamma_k}{1 - \gamma_k} \gamma_i \gamma_j = \sum_{k=0}^n \frac{\gamma_k}{1 - \gamma_k} \left(\sum_{i \neq k} \sum_{j \neq i, k} \gamma_i \gamma_j \right)$$

$$\begin{aligned}
&= \sum_{k=0}^n \frac{\gamma_k}{1-\gamma_k} \left(\left(\sum_{i \neq k} \gamma_i \right)^2 - \sum_{i \neq k} \gamma_i^2 \right) \\
&\leq \sum_{k=0}^n \frac{\gamma_k}{1-\gamma_k} (1-\gamma_k)^2 \\
&= \sum_{k=0}^n \gamma_k - \sum_{k=0}^n \gamma_k^2 \leq 1.
\end{aligned}$$

This can be repeated for each of the terms in Δ_{ij} , and every term can be bounded from above by 1.

Since there are n terms, we have

$$\sum_{i=0}^n \mathcal{M}_{ii} \leq n.$$

B.3.5 BOUNDING DIAGONAL ENTRIES OF $\mathcal{Q}$

In this section we give upper and lower bounds to $\mathcal{Q}_{ii}$ given v_m . With $v_m = \min_{i,j} v_{ij} = \frac{r_M}{(1+r_M)^2}$, $r_M = \max_{i,j} \frac{\gamma_i}{\gamma_j}$ and $r_M^{-1} = \min_{i,j} \frac{\gamma_i}{\gamma_j}$. The function

$$f(x, z) = \frac{(1+z)(1+x)}{(1+z+x)}$$

is an increasing function in x and z for positive x and z . Hence, the upper and lower bound of the function is achieved at upper and lower bound of x and z .

For any distinct triplet i, j and k we have

$$\begin{aligned}
\frac{(\gamma_i + \gamma_j)(\gamma_i + \gamma_k)}{\gamma_i(\gamma_i + \gamma_j + \gamma_k)} &= \left(1 + \frac{\gamma_j}{\gamma_i}\right) \left(1 + \frac{\gamma_k}{\gamma_i}\right) \frac{1}{1 + \frac{\gamma_j}{\gamma_i} + \frac{\gamma_k}{\gamma_i}} \\
\left(1 + \frac{\gamma_j}{\gamma_i}\right) \left(1 + \frac{\gamma_k}{\gamma_i}\right) \frac{1}{1 + \frac{\gamma_j}{\gamma_i} + \frac{\gamma_k}{\gamma_i}} &\leq (1 + r_M) (1 + r_M) \frac{1}{1 + r_M + r_M} \\
\left(1 + \frac{\gamma_j}{\gamma_i}\right) \left(1 + \frac{\gamma_k}{\gamma_i}\right) \frac{1}{1 + \frac{\gamma_j}{\gamma_i} + \frac{\gamma_k}{\gamma_i}} &\geq (1 + r_M^{-1}) (1 + r_M^{-1}) \frac{1}{1 + r_M^{-1} + r_M^{-1}}
\end{aligned}$$

Substituting the lower bound into the expression for $\mathcal{Q}_{ii}$, we get

$$\begin{aligned}\mathcal{Q}_{ii} &= \sum_{j=0, j \neq i}^n \left[v_{ij} + \sum_{k=0, k \neq i, j}^n \frac{(\gamma_i + \gamma_j)(\gamma_i + \gamma_k)}{(\gamma_i + \gamma_j + \gamma_k)\gamma_i} v_{ij}v_{ik} \right] \\ &\geq nv_m + n(n-1)v_m^2 (1 + r_M^{-1})^2 \frac{1}{1 + 2r_M^{-1}}.\end{aligned}$$

B.4 FURTHER DETAILS FOR DESIGN ON SPACING OF ITEMS' WORTH

Here are details of deriving each γ vector with specific design on spacing with γ^n . For bookkeeping purposes, in this section, we will index the items in ascending order of their worth. In other words, $\gamma_0 \leq \dots \leq \gamma_n$. We can also write $v_m = v_{0n}$ and $r_M = \frac{\gamma_n}{\gamma_0}$.

1. Equi-spaced γ . Here, we let $\gamma_i - \gamma_{i-1} = c$. Given $r_M = \frac{\gamma_n}{\gamma_0}$,

$$\begin{aligned}c &= \frac{\gamma_n - \gamma_0}{n} = \frac{(r_M - 1)\gamma_0}{n}, \\ \sum_{i=0}^n \gamma_i &= (n+1)\gamma_0 + \sum_{i=0}^n ic = 1 \implies \gamma_0 = \frac{2}{(n+1)(r_M + 1)}.\end{aligned}$$

2. Equi-spaced θ . Let $\theta_i - \theta_{i-1} = c$.

$$\begin{aligned}\frac{\gamma_n}{\gamma_0} &= (e^c)^n \implies c = \frac{1}{n} \log r_M, \\ \sum_{i=0}^n \gamma_i &= \sum_{i=0}^n (e^c)^i \gamma_0 = \gamma_0 \frac{e^{nc} - 1}{e^c - 1} = 1 \implies \gamma_0 = \frac{r_M^{1/n} - 1}{r_M - 1}.\end{aligned}$$

3. Mixture of two types of items. We assume that there are only two type of items - those worth

γ_0 and those worth γ_n . With n_L and n_H items in low and high worth groups respectively,

$$n_L\gamma_0 + n_H r_M \gamma_0 = 1 \implies \gamma_0 = \frac{1}{n_L + n_H r_M}.$$

B.5 REASONABLE VALUES OF v_m

The sufficient condition for uniqueness of MLE under Plackett-Luce model is that there is no partition of the items such that one partition is always ranked lower than the other partition. We attempt some calculations to understand how likely it is to generate data without a unique MLE when v_m is low, with assumptions on spacing design of γ .

Using the latent variable setup of the Plackett-Luce model, we have $W_i \sim \text{Gumbel}(\theta_i)$ with location parameter θ_i . It follows that $\exp(-(W_i - \theta_i)) \sim \text{Exp}(1)$. Let $V_i = \exp(-W_i)$, $\gamma_i V_i \sim \text{Exp}(1)$. A ranking on W_i is equivalent to a ranking on V_i in the opposite direction.

Given any partition $\mathcal{S}$ and $\mathcal{S}'$, the probability of every player in $\mathcal{S}$ being ranked higher than every player in $\mathcal{S}'$ is equivalent to $v_i \leq v_j, \forall i \in \mathcal{S}, j \in \mathcal{S}'$. Let $M = \min_{j \in \mathcal{S}'} V_j$, we know that $M \sim \text{Exp}(\sum_{j \in \mathcal{S}'} \gamma_j)$. Thus

$$\mathbb{P}(v_i \leq v_j, \forall i \in \mathcal{S}, j \in \mathcal{S}') = \mathbb{E}_M \left[\prod_{i \in \mathcal{S}} \mathbb{P}(V_i \leq M) \right] = \mathbb{E}_M \left[\prod_{i \in \mathcal{S}} (1 - e^{-\gamma_i M}) \right].$$

The probability of any partition being separated in ranking is at least as great as the probability of some partitions being separated in ranking. With a mixture of two types of items, it suffices to examine partitions of high worth items and low worth items. With equi-spaced γ and θ , we should at least examine the n partitions given by separating items 0 to i and $i + 1$ to n for $i = 0, \dots, n - 1$, where items are indexed in ascending order of their worth.

MIXTURE OF TWO TYPES OF ITEMS WITH $n_L = n$ We illustrate with the case when there are two types of items, those with low worth and those with high worth. For simplicity, we consider the case where the $n_L = n$ and $n_H = 1$. In this case, the closed-form solution is available. Let s_0 be the number of players in the low worth group, and s_n be the number of players in the high worth group. For a single ranking, the probability of all players in the low worth group being ranked lower than every players in high worth group is

$$f = \mathbb{E}_M \left[\left(1 - e^{-\gamma_n M}\right)^{s_n} \right]$$

where $M \sim \text{Exp}(s_0 \gamma_0)$.

The probability of this occurs over T independent rankings is f^T . Suppose we control $f^T < 0.01$, in other words, there is more than 99% chance of retrieving ranking data with low worth group being ranked higher than the high worth group at least once.

When $n_L = n$, we have $s_n = 1, s_0 = n$,

$$f = 1 - \frac{n}{n + r_M}$$

To control $f^T \leq p_{sep}$ given T independent rankings, we have

$$\begin{aligned} f^T \leq p_{sep} &\Leftrightarrow f \leq p_{sep}^{1/T} \\ &\Leftrightarrow \frac{n}{n + r_M} \geq 1 - p_{sep}^{1/T} \\ &\Leftrightarrow r_M \leq n \frac{p_{sep}^{1/T}}{1 - p_{sep}^{1/T}} \end{aligned}$$

As n increases, the acceptable range of r_M increases too. To get a rough sense on r_M , for $p_{sep} = 0.01$, with 7 players and $T = 100$, the maximum r_M we can accept is ≈ 148 , which gives $\log(v_m) \approx -5.01$.

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